Dell PowerFlex 4.0.x

Release Notes

August 2023
Rev. 7.0

DELL Technologies

Notes, cautions, and warnings

NOTE: A NOTE indicates important information that helps you make better use of your product.

CAUTION: A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.

WARNING: A WARNING indicates a potential for property damage, personal injury, or death.

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1 Introduction

This document provides information about the PowerFlex version 4.0.x release, including:

- Information about new features delivered in this release for PowerFlex and PowerFlex Manager
- Known issues and steps to work around them

The PowerFlex and PowerFlex Manager version 4.0.x release supports:

Product	Supported firmware and software
PowerFlex rack	Version 4.0.2 adds support for Release Certification Matrix (RCM) version 3.7.2.0. Version 4.0.1 adds support for Release Certification Matrix (RCM) version 3.7.1.0. Version 4.0 adds support for Release Certification Matrix (RCM) version 3.7.0.0. For the supported certified versions of software and firmware components, see the Dell Release Certification Matrix (RCM) Portal. To access the RCM portal go to the RCM Portal.
PowerFlex appliance	Version 4.0.2 adds support for Intelligent Catalog (IC) version 40.372.00. Version 4.0.1 adds support for Intelligent Catalog (IC) version 40.371.00. Version 4.0 adds support for Intelligent Catalog (IC) version 40.370.00. See the *Dell PowerFlex Appliance with PowerFlex 4.x Support Matrix* for supported firmware, software, and components.
Software-only, PowerFlex custom node, VxFlex Ready Node	Version 4.0 adds support for PowerFlex_Software_4.0.0.0_1211. This is a software-only catalog that provides only the packages needed for software-only deployments. It does not include ISOs or SVM packages.

The target audience for this document includes customers and Dell Technologies teams working with customers and certified partners.

NOTE: All PowerFlex 4.0.x documentation is now integrated into the PowerFlex 4.5.x documentation. For more information about PowerFlex, see the PowerFlex 4.5.x technical documentation.

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Revision history

Date	Document revision	Description of changes
August 2023	7.0	Updates for version 4.0.2
April 2023	6.0	Updated the *Install and upgrade* topic
March 2023	5.0	Editorial updates
January 2023	4.0	Updates for PowerFlex version 4.0.1
September 2022	3.0	Added VXFM-30017 to the list of fixed issues
September 2022	2.0	Updates for PowerFlex version 4.0.0.1
August 2022	1.0	Initial release

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PowerFlex version 4.0.2

What's new in release 4.0.2

PowerFlex release 4.0.2 introduces the following new features:

Feature	Details
Feature enhancements	Improved alerts for PowerFlex management nodes.
Support for security vulnerabilities	PowerFlex Manager adds support for the latest software packages and libraries to resolve security vulnerabilities.
Support for CloudLink 7.1.6 and 7.1.7	PowerFlex and PowerFlex Manager add support for CloudLink 7.1.6 and 7.1.7.
Miscellaneous issue fixes	This release includes fixes to issues discovered in earlier releases of PowerFlex and PowerFlex Manager.
Documentation	For more information about this release, see the following documents: ● *Dell PowerFlex 4.0.x Technical Overview* ● *Dell PowerFlex 4.0.x Administration Guide* ● *Dell PowerFlex 4.0.x Install and Upgrade Guide* ● *Dell PowerFlex Manager 4.0.x Online Help* ● *Dell PowerFlex 4.0.x CLI Guide* ● *Dell PowerFlex* API on the Dell Technologies Developer portal

Install and upgrade

You can install and configure PowerFlex version 4.0.x, or you can upgrade from PowerFlex version 3.6.x to version 4.0.x.

See the *Dell PowerFlex 4.0.x Install and Upgrade Guide* for detailed instructions on installing or upgrading.

Licensing

PowerFlex use requires a license. Contact your Dell account representative to purchase a license for PowerFlex.

Upgrade

The following table lists the versions from which the upgrade can be performed:

PowerFlex/ VxFlex OS / ScaleIO base version	Upgrade path	Supporting documentation
v4.0.x	If you are on 4.0.0x you must upgrade to 4.0.1 first and then upgrade to 4.0.2	
v3.6.0.x	One-step upgrade flow: 1. Upgrade to v4.0.x	*Dell PowerFlex 4.0.x Install and Upgrade Guide*
v3.0, v3.0.0.x, v3.0.1.x, v3.5.x	Two-step upgrade flow: 1. Upgrade to v3.6.0.x 2. Upgrade to v4.0.x	*Upgrade PowerFlex to v3.6 Guide*
v2.0.1.4, v2.6.x	Three-step upgrade flow:	*Upgrade VxFlex OS to v3.0.x Guide*

PowerFlex/ VxFlex OS / ScaleIO base version	Upgrade path	Supporting documentation
	1. Upgrade to latest version of v3.0.1.x 2. Upgrade to v3.6.0.x 3. Upgrade to v4.0.x	*Upgrade PowerFlex to v3.6 Guide*

In release 4.0.1, PowerFlex and PowerFlex Manager add support for CloudLink 7.1.5.

CloudLink 7.1.5 changes the disk mapping logic to use /dev/sdX, which is different from CloudLink 7.1.2 to 7.1.4. This version removes support for /dev/disk/by-id from CloudLink, thereby reverting to a /dev/sdX device mapping.

PowerFlex 3.6.0.6 supports migration and upgrade flows from CloudLink 7.1 to CloudLink 7.1.5. PowerFlex 3.6.0.6 has added code changes to identify device mapping changes under /dev/mapper.

Customers that have migrated to CloudLink 7.1.3 and adopted /dev/disk/by-id logic as documented in *Dell PowerFlex v3.6.x CloudLink for SDS Devices*, should open a support call to get assistance to revert back to the /dev/sdX device mapping to adopt the CloudLink 7.1.5. Customers that stayed on earlier CloudLink releases (CloudLink 7.1.1 or lower) can adopt the CloudLink 7.1.5 using their existing device mapping.

Due to key updates, adoption of CloudLink 7.1.5 is recommended for use with PowerFlex for data at rest encryption.

These are the supported upgrade paths for CloudLink:

PowerFlex version	CloudLink version	Description of change
3.6.0.6 and above	7.1.5	Existing CloudLink install base with the original device path logic: /dev/sdX.
3.6.0.2 , 3.6.0.3 , 3.6.0.4 , 3.6.0.5	7.1.3	CloudLink 7.1.3 enables a new device path logic for encrypted devices, using /dev/disk/by-id/wwn-*<unique_id>*.
3.0.1.x all versions 3.5.x.x, all versions 3.6 3.6.0.1	7.1 7.0	These CloudLink versions maintain legacy support for the existing CloudLink install base with the original device path logic: /dev/sdX.

Here is the expected upgrade flow for systems with CloudLink:

- Upgrade PowerFlex to 3.6.0.6.
- For CloudLink 7.1 or 7.0, upgrade to CloudLink 7.1.5.
- For CloudLink 7.1.2 or 7.1.3 or 7.1.4, upgrade to CloudLink 7.1.5 with support assistance.

Supported operating systems

For a list of supported operating systems, see the **Storage** section of the *Dell Technologies Simple Support Matrices* at https://elabnavigator.dell.com/eln/modernHomeSSM.

SDC compatibility

By design, the PowerFlex SDC in this release is compatible with newer and previously released systems that are currently supported (for example, VxFlex OS v3.0.1). In addition, the PowerFlex system (backend) in this release is backward and forward compatible with previously released SDC versions that are currently supported.

Some limitations may apply to mixed-version systems, such as:

- Replication is not supported on SDC releases earlier than PowerFlex v3.5.
- The CHAP component authentication feature cannot be enabled unless every MDM, SDC and SDS in the system is at least version 3.5, and conversely, if the feature is enabled, no pre-v3.5 SDCs can be added.
- Fine Granularity Storage Pool-based volumes cannot be mapped to a ScaleIO/VxFlex OS v2.x SDC.

- Snapshot Policy management is not available with SDC releases earlier than VxFlex OS v3.0.

Support deprecation

This section lists features that are no longer supported in PowerFlex 4.0.x.

RFCache

The RFCache feature was removed. Any customer using RFCache with the PowerFlex 3.6.x.x release must first remove the RFCache configuration and packages prior to initiating an upgrade.

VASA support

VASA support is not available in PowerFlex 4.0. VASA support is a future roadmap item.

Customers needing VASA support should continue using PowerFlex 3.6.x.x with the latest patch.

Known issues in 4.0.2

The Dell engineering team is working to resolve known issues.

Before getting started with PowerFlex and PowerFlex Manager, review the following known issues in version 4.0.2:

CLI

Issue ID	Symptom	Recommended action
SCI-67737	If the `generate_login_certificate scli` command is run without parameters, it fails.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.

MDM

Issue ID	Symptom	Recommended action
SCI-65059	IP addresses are not rebalanced when MDM disconnects from SDT.	If an SDT is unreachable, the SDT_NOT_CONNECTED_TO_HOST alert is raised. You can use connectivity monitoring utilities (such as ping) to monitor the network health.

SDS

Issue ID	Symptom	Recommended action
SCI-67744	All SDS devices get into an error state during and after server reboot. This can happen if you run	Do not use `-f` with the `reboot now` command.

Issue ID	Symptom	Recommended action
	NVMe IO for a long time, then run `reboot -f now`.	
SCI-67359	If a device-at-fault event occurs while an SDS is in protected maintenance mode (PMM), some capacity may become un-writeable, resulting in IO errors until the PMM is aborted. The device-at-fault event occurs when a device is readable but not writeable.	Remove from PMM.

SDT

Issue ID	Symptom	Recommended action
SCI-67009	Sometimes when the number of hosts and active connections is close to using the maximal number of NVMe controllers permitted by the NVMe standard, adding a new SDT to the system may fail.	Disconnect one or more hosts from the system to allow the system to reclaim NVMe controller IDs. While addition of SDT is blocked, the system can still function properly and serve all currently connected hosts.
SCI-66722	It is possible to remove 2 SDTs that are the only paths of a volume. When removing 2 SDTs, after removing the first one, the MDM will replace it in the host connectivity goal with a port from the other SDT and will update all paths that used it with system-ports from other SDTs. (Some might get a path from the new port in the goal). When removing the second one, the MDM does the same as in the first removal. However, now some mapping that got the first new port in the goal after the first removal might also get the second new port in the goal. Those mappings stay without connection.	Dell Technologies recommends not removing or entering into maintenance mode a number of SDTs that is greater than or equal to the number of paths per volume, without reconnecting the hosts.
SCI-66645	Incorrect alert definition for SDT_MULTIPLE_DISCONNECTIONS_FROM_STORAGE_BACKEND_COMPONENTS. After bringing an interface down on an SDS, SDTs disconnect from it as expected and an alert is generated and processed by PowerFlex Manager. Upon recovery (NIC up), the events confirm that the SDTs and SDCs are connected back to that interface. However, this alert stays in the user interface. Sometimes, SDT alerts do not get properly cleared.	None.
SCI-66285	SDT performance drops approximately 30% when SDT processes span 2 numas. Deploying SDT and SDR on the same server can lead to reduced performance.	Avoid deploying SDT and SDR on the same server.
SCI-64187	NVMe namespace paths may change following disconnect and reconnect to PowerFlex NVMe Target, such that volumes may be used by different paths. Use by-id paths to maintain a stable path to devices across disconnect operations.	Use /dev/disk/by-id/<unique device path>.

SDR

Issue ID	Symptom	Recommended action
SCI-70298	If you use a CSV file to configure the PowerFlex cluster, it may fail to add an SDR. The failure produces an error that says "There is no available replication capacity". The cause of the failure is that there was not enough journal capacity to add the SDR.	Adjust the journal capacity.

PowerFlex Manager

Issue ID	Symptom	Recommended action
VXFM-35665	Nodes used in Resource Group deployment are not put into "Ready" state post Service removal from the appliance.	1. Remove Node from inventory and rediscover it again. It puts the nodes to "Ready State" then. You can then proceed with BF. 2. Manually update the DB status in the backend from "Deployed" to "Ready" by the command, `update device_inventory set state = 'READY' where service_tag = '<device service tag>';`
VXFM-35668	During Gateway NDU for Storage Only service, SDS or MDM nodes are not displayed under "Ready for Upgrade".	Update Resource Group Details on the Storage Only service.
VXFM-35527	When a Node is scaled up using the scaled-up CentOS as a duplicate, the SVM OS Type under SVM OS Settings is duplicate.	Set both SLES and CentOS setting to desired OS.
VXFM-35157	After a one-time migration, the MDS gateway moved to a critical state with the error "MDM_CONNECTION_LOST, in the MDM cluster / PowerFlex gateway".	None.
VXFM-35132	SDC installation for ESXi base installation fails when there are multiple ingress and ESXi management is not the first ingress defined on the cluster.	None. **NOTE:** Ingress rules exist in the configuration file examples.
VXFM-35127	A resource group remains in lifecycle mode after a VCLS migration without any explanation.	Update the resource group details.
VXFM-35046	PowerFlex Manager Plugin apps lost Database connectivity.	Delete the platformoperator pods (`kubectl delete -A pod -l="app.kubernetes.io/name=platform-operator"`).
VXFM-34891	Incorrect Load Balancer IP & "Failed" status appears in SCG when using OOB network for PowerFlex Manager Plugin connection to SCG	None.
VXFM-34878	On completion of a successful NDU on the Resource group, the nodes are compliant but the Gateway moves to critical state and alert messages are displayed.	Restart the Gateway pod.
VXFM-34612	The 4.0.2 software catalog does not have SLES 15 SP4 in it, so upgrades performed through the user interface for that OS do not work.	None.

Issue ID	Symptom	Recommended action
VXFM-34445	Cloudiq-collector writes and sends config json file to CloudIQ	None.
VXFM-34444	The Cloudiq-collector in Capacity file contains details about only 1000 vTrees whereas the large Scale system has 131000 vTrees.	None.
VXFM-34284	Many pods fail following an increase in inventory and, some containers restart at regular intervals even after a long time since the failure.	Ensure that for large scale deployments, the RAM size is 48GB.
VXFM-34048	Need to power cycle nodes after disabling PD for disruptive update	None.
VXFM-31338 and VXFM-31116	A user with the Snapshot Manager or Replication Manager user role is not able to create, map, or unmap snapshots.	Use another role such as Super User or System Admin to manage snapshots.
VXFM-31193	After a non-disruptive upgrade completes on a storage-only resource group, the Tie Breaker node compliance state is in 'Pending Upgrade', even though the upgrade completed successfully. The state is reported incorrectly in the user interface.	Remove and re-add the resource group.
VXFM-31191	A restore to an existing cluster may fail if the CIFS password begins with a special character, such as a dollar sign ($).	Use a CIFS share that has a password that does not start with a special character.
VXFM-31185	If you attempt to perform a CSV deployment in the user interface without loading the MDM RPM, the deployment fails with an unhelpful message. The error in the event of the failed command should indicate that the MDM RPM is missing.	Load the MDM RPM and try again.
VXFM-31179	If you try to delete a hyperconverged resource group, the teardown operation does not delete the vCenter cluster. Also, the wizard for deleting the resource group does not show the checkbox that allows you to delete the cluster and remove it from vCenter.	Manually delete the vCenter cluster and re-inventory the vCenter resource.
VXFM-31093	While registering **SupportAssist** from the **Settings** page, PowerFlex Manager enables two policies by default, once the registration is successful. Instead of enabling the policy that was already there, PowerFlex Manager creates a second one.	Delete the duplicate entry if you don't want to use it.
VXFM-30889	When removing a resource group, PowerFlex Manager does not give you the option to choose whether resources should be left in a managed or unmanaged state. Instead, PowerFlex Manager automatically leaves the nodes in a managed state.	Go to the **Resources** page to remove the nodes, if you like.
VXFM-30855	The MDM events extractor now supports multiple IP addresses for each MDM. If the system is configured with two or more management IP addresses per MDM, the extractor tries to connect to each one. If the connection to multiple interfaces is lost, the extractor does not recognize the disconnect and thinks that it is still connected, and does not show an alert indicating that all are down.	None.

Issue ID	Symptom	Recommended action
	If only one connection goes down, you do see an alert.	
VXFM-30806	When an LDAP user is assigned to an LDAP group, and the LDAP group is configured with a user role, a REST login command may fail with an error stating "no role assigned to user". This can happen if you delete the LDAP user, and try to login with that user.	Define groups before adding explicit users. Also, If necessary, keep another local or federated user that can be used to log in.
VXFM-30792	On the **Getting Started** page, If you try to deploy a CSV installation file that has errors, the **Upload Installation CSV** button is disabled after the error message appears. This button should be enabled to allow you to upload a fixed file.	Close the wizard and open it again. Then, upload the fixed file.
VXFM-30663	Under rare circumstances, some snapshots may be missing from the file system GUI.	Use the export mountpoint to view the missing snapshots.
VXFM-30624	After performing a new resource group deployment, PowerFlex Manager shows a node with an update failed status, but the deployment was successful. This is an incorrect status, since no update was performed.	Remove the resource group and add it again.
VXFM-29962	During an upgrade from 4.0 to 4.0.1, a standby PostgreSQL node may end up in a failed state.	None.
VXFM-29892	Remove any text from UI which supports the statement that additional networks can be added for SO resource group.	None.

VXFM-29795	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the MDM primary node was changed, the certificate wasn't approved before. Therefore, the user isn't recognized and the import can't be done.	Change LIA LDAP authentication to LIA native authentication manually, as follows:
		1. In each PowerFlex node, go to `/opt/emc/scaleio/lia/cfg`, and re-edit the file `conf.txt` so it won't include LDAP fields at all and lia_auth_mode=NATIVE_AUTHENTICATION. Also, set lia_token=Scaleio123. Then close the file and save.
		2. Login to each PowerFlex node and reboot the LIA service, as follows:
		`[root@96N-15 ~]# cd /opt/emc/scaleio/lia/bin`
		`[root@96N-15 bin]# ./delete_service.sh`
		`Removed symlink /etc/systemd/system/multi-user.target.wants/lia.service.`
		`[root@96N-15 bin]# ./create_service.sh`
		`Created symlink from /etc/systemd/system/multi-user.target.wants/lia.service to /usr/lib/systemd/system/lia.service.`
		Importing the version 3.6 PowerFlex cluster to PowerFlex Manager will then work properly.

VXFM-29710	The old REST API for logging out does not work. If you logout with `GET https://<ip>/api/logout`, the API returns a 500 error code. In this	Logout with the new API: `POST https://<ip>/rest/auth/logout`

Issue ID	Symptom	Recommended action
	release, you need to use both the "access_token" and the "refresh_token" you got when you performed the login. The old API uses only one token, so the API returns an error.	
VXFM-29705	When a PowerFlex cluster deployment fails, you may see errors from a previous run if you attempt to redeploy repeatedly in quick succession. This is a logging issue only. If you run deployments over and over quickly, the logs may bleed over into subsequent deployments.	None.
VXFM-29637	It takes PowerFlex Manager up to 40 minutes to fully load after reboot, crash, or power off.	None.
VXFM-29619	**Delete Resource Group** does not remove nodes from CloudLink Center in the case of resource groups with encryption. The nodes are still present, but they are in a **Disconnected** state.	None.
VXFM- 29602	The PowerFlex gateway still responds to the old alert query API, but this was deprecated. In older versions of Powerflex alerts were queried via the gateway using the following API: `/api/types/Alert/instances` In this release, the alerts are handled by the alert service and queried by using this API: `/rest/v1/alerts` The deprecated API should return an error, but instead it responds normally and returns incorrect data that does not include some alerts.	Do not use the deprecated API. Use the new API instead.
VXFM- 29600	The node deployment status on the **Resources** page is set to **Deployment Cancelled** state after a successful non-disruptive update. This is a temporary state and should be corrected in a few minutes.	Check again in a few minutes to see if the deployment status has been updated.
VXFM-29584	The exception only flag for IM log collection is not supported. This flag has been deprecated and is ignored.	None.
VXFM-29543	The management workflow to allow swapping primary and backup nodes has not been added to the restricted operations after a NAS server has entered into an out of service state. This situation might occur in the following scenario: 1. You make a NAS storage pool full. 2. The NAS server on the storage pool becomes read only. 3. You try to move the NAS server (this is a prohibited operation). 4. You try to perform a swap of the NAS server (this is successful). This issue has no visible impact on file service operations.	None.

Issue ID	Symptom	Recommended action
VXFM-29500	PowerFlex Manager does not provide the ability to place an SDT in maintenance mode within the user interface.	None.
VXFM-29494	Filtering events by timestamp does not work correctly with milliseconds. For example, the following API call might return events that do not match the filter: `https://10.234.222.15:31943/rest/v1/events?filter=timestamp+gt+2022-07-05T14:57:27.807Z&filter=service_instance_id+le+Value+above+limit+`-=/<>,.?|;:[]{}&order=resource_id.dsc`	None.
VXFM-29404	A hyperconverged resource group fails if an NFS datastore is already present.	Remove the NFS datastore and retry the resource group.
VXFM-29386	The **Latency** graph on the dashboard does not show any data for NVMe/TCP.	None.
VXFM-29381	The **Summary** page of the **Initial Configuration** wizard remains static for 30 seconds while PowerFlex Manager is enabling SupportAssist.	None.
VXFM-29304	Some event and alert services cannot recover from a configuration or certificate change in the AMQ service. All services may lose connection to AMQ and reconnection attempts may fail.	Restart the pod of services that did not disconnect.
VXFM-29294	The dashboard IO performance numbers do not correlate with IO performance numbers reported by SCLI.	None.
VXFM-29272	A hyperconverged deployment fails in a partial network configuration when DNS is not configured in the hypervisor management network.	Configure DNS.
VXFM-29234	When you create a login certificate and it later expires, an attempt to login with the expired certificate gives a misleading error message: **`scli --login --p12_path /root/.scli/admin_cert.p12 --p12_password password Error: Failed to connect to MDM 127.0.0.1:8611`** There should be some indication that the certificate is no longer viable and you need to create a new one.	None.
VXFM-29190	NVMe drive replacement fails with the error "Removal of the operating system failed". The error is confusing because it says it failed, but you can remove the drive without issue and the new drive is detected and can be added back to PowerFlex and CloudLink.	None.
VXFM-29174	The **Edit Management Software Upgrade Settings** page does not have a **Test Connection** button. However, when you click **Save**, PowerFlex Manager displays a message	None.

Issue ID	Symptom	Recommended action
	indicating whether the supporting connections required for the upgrade are valid.	
VXFM-29095	There is no way to delete a source or destination on the **Notification Policies** page.	Remove or disable all the policies involving this source or destination.
VXFM-28986	PowerFlex Manager allows you to deploy 16 nodes for a PowerFlex File (NAS) compute-only deployment. This is not supported. The limit is 16 nodes. The user interface should block NAS deployments that exceed 16 nodes.	None.
VXFM-28829	If you click **Add Template** in the **Deploy Resources** section of the **Getting Started** page, the user interface shows the **Create** wizard as expected. After selecting **Create a new template**, entering a **Template Name**, and clicking **Next**, the **Create Template** page shows up correctly. After entering all fields and clicking **Save**, the user interface does not show the template details to allow further editing of the template, as it would if you clicked **Lifecycle** > **Templates** and then clicked **Create**. However, the template is created.	Navigate to **Lifecycle** > **Templates**. You will see the template in the list under **My Templates**. You can then click on the template name to edit the template as expected.
VXFM-28715	You are unable to save changes in the **Destination Protocol Settings (SupportAssist)** window if you attempt to change the connection type from **Connect via Gateway Server** to **Connect Directly**.	De-register before you try to change from one connection method to another.
VXFM-28693	The error message displayed in PowerFlex Manager is not clear when the logged in user tries to change their password.	None.
VXFM-28291	An ambiguous error appears during node expansion if a reverse DNS lookup for hostname fails to find an IP address. The **Summary** screen for adding the node shows: `An unexpected error occurred on the system. Retry the operation later.`	None.
VXFM-28281	PowerFlex Manager allows multiple instances of an iDRAC password update job to run on the same node at the same time.	None.
VXFM-28194	If there is a larger than expected time difference (over 1 hour) between PowerFlex Manager and the PowerFlex storage nodes, an upload license action may fail with the following error: `License Enforcement - Wrong Command Parameters`. This error is displayed on the **License Management** page in PowerFlex Manager. In the PowerFlex Manager log file, the following error appears: `"error_code":62,"error_message":"License Enforcement - Wrong Command Parameters."` In the primary MDM logs `/opt/emc/scaleio/mdm/logs/trc.z.XX`, the error	Align the time between the management platform and storage nodes and upload the license again.

Issue ID	Symptom	Recommended action
	displays as follows: `Error, license activation date in the future`	
VXFM-28034	If you create volumes on the **Block** page, volume creation is cancelled when you switch away from this page. Volume creation should be continued in background. Instead, volume creation is stopped and there is no progress or completion status when the **Block** page is opened again.	Wait for the operation to finish before navigating to another page.
VXFM-27967	The **View Details** page for Cisco switches shows a blank page and no metrics.	None.
VXFM-27388	If you upload a certificate file you have uploaded before using the same name you have used before, PowerFlex Manager does not display an error message and overrides the existing certificate.	None.
VXFM-26570	An attempt to tear down a storage-only CloudLink resource group does not remove the machine groups from CloudLink Center.	None.
VXFM-26550	If you run the command **`"powerflex_cli events quwery --session_token 98329575cf390107 --limit 2 --order timestamp.asc"`** with wrong command flag **`"quwery"`**, the command fails with error `"Error: unknown flag: --limit"`. This is the incorrect error message.	Run the command with the correct command flag: **`"powerflex_cli events query --session_token 98329575cf390107 --limit 2 --order timestamp.asc"`**
VXFM-26549	When using power_cli to query events with filter "severity in (MAJOR,MINOR)" it fails with a parsing error. This error occurs because the "in" operator does not exist in the CLI.	The same query can be achieved by running the query a few times, once for each value in the "in" clause.
VXFM-25895	PowerFlex Manager is unable to configure a Dell OS10 switch for deployment.	None.
VXFM-25284	While creating a snapshot rule, you need to provide the time of day. By default, the value populated on this field is current time in EST. However, internally, the snapshot rule is created with the value in UTC. This may lead to confusion as snapshots are not getting created as expected.	None.
SCI-67737	If the `generate_login_certificate scli` command is executed without parameters, it will fail.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.

Operating system known issues and limitations

This section lists known issues and limitations for specific operating systems.

Issue number & SR number	Problem summary	Workaround
VXFV-360	The device name appears incorrectly as a non-PCI device instead of iDRAC, but the current and baseline versions are accurate (6.00.30.201 to 6.00.30.202, respectively).	Dell recommends upgrading the OME catalog to the latest version.

Issue number & SR number	Problem summary	Workaround
	PowerFlex nodes running CPLD 1.0.7 stop responding during Power On Self-Test (POST) when upgrading to certain RCM or IC packages that include BIOS 1.8.2 and iDRAC 6.00.30.202.	Before upgrading to the latest version of PowerFlex, ensure that the PowerFlex nodes are running CPLD 1.0.8 or later. For details, see https://www.dell.com/support/kbdoc/en-us/000205925.
VXFV-296	Mixing the Mellanox CX-6 100GbE NIC driver with the Broadcom PCIe/OCP 25GbE NIC driver is not supported.	Remove the `bnxt_en` driver and install the Mellanox driver.
VXFV-283	iDRAC support for "Prepare for Removal" on PowerEdge R650/R6525/R750/ nodes fails.	NVMe hot-swap on Linux operating systems is supported. Skip the Prepare for Removal step in iDRAC.
VXFV-265	During execution of the command `create-namespace` with 4k alignment on RHEL 8.4, the following error is encountered: Warning: namespace0.0 unable to validate alignment	This is a false notification. The command result is as expected, and the resulting DAX device can be used with PowerFlex acceleration devices.
VXFV-243	Upgrade of a RedHat-based operating system to RHEL 8.3 fails if the node contains NVDIMMs. This is due to a known issue on RHEL 8.3: https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/8/html-single/8.3_release_notes/index#known-issue_installer-and-image-creation	Follow the procedures described in "RHEL/CentOS upgrade to version 8.3" in the .
VXFV-224	In a RHEL 8.x-based node using Cloudlink encrypted SDS devices, after reboot the device moves to failed state. **NOTE:** This issue is much rarer on RHEL7.x-based nodes, but it might occur.	1. SSH to the relevant SDS's operating system. 2. Delete the SDS service. 3. Clear the device errors using the GUI or CLI. 4. Recreate the SDS service. 5. Verify that devices come back online. Rebuild/Rebalance will occur if instant maintenance mode or protected maintenance mode was not used. 6. If instant maintenance mode or protected maintenance mode was used, exit maintenance mode.
VXFV-157	NVMe disks are not supported on RHEL 8.0 due to the following issue: https://access.redhat.com/solutions/4280341 Failure to observe this limitation might cause RHEL-based SDSs with NVMes to experience kernel panic during a disk removal flow.	None

Fixed issues in 4.0.2

Fixed issues are defects that exist in the previous release but are resolved in the 4.0.2 release of PowerFlex and PowerFlex Manager.

Issue ID	Symptom
SCI-71000	The SDC system service in SLES15.x is renamed to **scini** to maintain consistency with other Linux distributions. However, when upgrading to the latest version (scini.service), the earlier version (sdc.service) remains in use. This conflict prevents the SDC from loading.

Issue ID	Symptom
VXFM-31230	The **Getting Started** page presents an empty page after a CSV deployment. If you click on the **Deploy with Installation File** button, then click **Next** and **Cancel**, you see an empty page if you go to **Settings** and click **Getting started**.
VXFM-31160	When you create a volume on the **Block** tab, the **Map** button does not open the **Map Volumes to Hosts** window after the volume is successfully created.
VXFM-31163	The Modify IP Role feature is not working for SDRs and SDTs.
VXFM-29291	On the **File** > **NAS Servers** or **File** > **File Systems** page, the performance details do not get updated on the side pane for a selected NAS server or file system.
VXFM-28693	The error message displayed in PowerFlex Manager is not clear when the logged in user tries to change their password.
VXFM-27411	Uploading a 4.0 license to an existing service that is still on 3.6 should show a more meaningful error message.
VXFM-29896	When reconfigure network after docker already running the install_PFMP.sh phase failed.
VXFM-29880	The block-legacy-gateway still shows under Target PowerFlex Gateway.
VXFM-29803	A service that was imported into version 3.8 of PowerFlex Manager cannot use **Update Resource Group Details** in version 4.0. This could happen if you perform a one-time migration on a 3.8 environment that includes a service that was originally imported through **Add Existing Service**. If you remove a node from the resource group after the migration, and then perform **Update Resource Group Details**, you might see a 500 exception.
VXFM-29795	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the MDM primary node was changed, the certificate wasn't approved before. Therefore, the user isn't recognized and the import can't be done.
VXFM-29790	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the version 3.6 system is configured with only LDAP authentication for MDM, the LDAP user isn't recognized, so the import can't be done.
VXFM-29659	The PowerFlex management platform (PFMP) upgrade sometimes fails on the first attempt. This can happen if you attempt an upgrade immediately after enabling SupportAssist. The button to perform the upgrade is available in the user interface, but it should not be while SupportAssist configuration is still in progress.
VXFM-29656	The Notifier service fails to handle ALM events with "send_to_srs"=false due to a key error. This results in some low-level information being sent when it should not be.
VXFM-29579	A Security Admin user can perform activities defined for the Security Admin role, as well as activities allowed for the Lifecycle Admin role.
VXFM-29576	Event and alert names are not case consistent. The names for events and alerts should follow the convention of upper snake case, with upper case words separated by underscores. However, some of the names use lower snake case, with lower case words separated by underscores.
VXFM-29496	If you run the `reset_PFMP.sh` script to remove an old install and re-install PFMP, the script removes pre-requisite software rpms: • jq • nmap • skopeo The install fails if these RPMs are not there already.
VXFM-29530	PowerFlex Manager does not allow you to modify a notification source policy.
VXFM-29469	Whenever an ALM event from PowerFlex is received, PowerFlex Manager sends a message when it is received, then again when the alert is acknowledged or disacknowledged, and again when the alert is cleared. It sends an event whenever the status changes.

Issue ID	Symptom
VXFM-29241	Registering SupportAssist from the **Initial Configuration Wizard** is not activating the **Default Policy Configuration** on the **Settings** page.
VXFM-28907	On the **Protection** > **RCGs** page, while the RCG is in the failover state, the action **Resume (from pause)** should be disabled, because the RCG is not in pause mode but in switchover/failover state.
VXFM-28844	On the **Generate Troubleshooting Bundle** page, the hover text for **Download locally** indicates that NFS is supported. However, NFS is not supported for the troubleshooting bundle. The user interface correctly shows the failure if you try to use NFS, but the hover text should not show NFS.
VXFM-28828	The **Compliance Versions** page is not refreshing to show the new state of the compliance file when **Allow Unsigned File** is selected.
VXFM-28774	In some extreme bad path scenarios, when ActiveMQ is not available, some MDM events may not reach PowerFlex Manager.
VXFM-28532	Resizing the service volumes (vCLS volumes) should be blocked by PowerFlex Manager. Currently, you are able to increase the vCLS volume size. After resizing the volumes, you may also see unexpected behaviors. For example, you may see a message saying that the service is in lifecycle mode because the vCLS volumes are on local storage.
VXFM-28433	When trying to export a template without an encryption password, you may see an error.
VXFM-28288	A NilClass exception occurs if you deploy a resource group when you don't have a PowerFlex management network and you have general purpose LAN (GPL) networks that do not have a default gateway.
VXFM-26476	When using the REST API to query event history, if there is an error in the filter field, the filter is ignored, and the unfiltered list is returned instead of an error message.
VXFM-26333	When using the REST API to query alert templates, the filters are ignored, and the full list of alert templates is returned.

Software packages

This section provides information on the contents of the PowerFlex 4.0.1 software packages.

Download location

The PowerFlex complete software package, containing most of the packages listed below, can be downloaded from the following location:

https://www.dell.com/support/home/en-us/product-support/product/scaleio/overview

The RCM for PowerFlex rack can be downloaded from the following location:

Dell Download Center

Packages

Package name	Size	Contents
PFMP bundle	~19.2 GB	PFMP2-4.0.1-249.tgz
RCM for PowerFlex rack	~24.6 GB	Flex_RCM_3_7_0_1_r30.zip
Intelligent catalog for PowerFlex appliance	~23.4 GB	Intelligent_Catalog_40_371_00_r30.zip
Software-only ZIP	~5.9 GB	Software_Only_Complete_4.0.1.0_147.zip. Includes the following: • PowerFlex_4.0.1000.147_Complete_Core_SW.zip (3.4 GB)

Package name	Size	Contents
		• PowerFlex_4.0.1000.147_SDCs_for_manual_install.zip - SDC RPMs (0.6 GB) • PowerFlex_Software_4.0.1.0_147_r1.zip - software-only catalog (1.9 GB)
Compatibility file		• cm-20221129.gpg (compatibility file)
PowerFlex Manager OVA ZIP	~24.5 GB	PowerFlex_Manager_4.0.1-249.zip: • PFMP2-4.0.1-249.ova (PFMP_Installer for installing Kubernetes cluster) (23 GB) • pfmp-k8s-20230111.ova (Kubernetes OVA running SUSE 15 SP3) (2.3 GB)

The original OS layout table from 3.6 matches this file: PowerFlex_4.0.1000.147_Complete_Core_SW.zip.

4

PowerFlex version 4.0.1

What's new in release 4.0.1

PowerFlex release 4.0.1 introduces the following new features:

Feature	Details
Broadcom adapters: second source network adapter	PowerFlex rack and PowerFlex appliance have previously been single sourced on Mellanox adapters. This feature adds a second source in case of shortages. PowerFlex Manager supports the following Broadcom adapter: ● Model 57414 for 25GB (2x25) speed The NIC installation supports: ● OCP (on-board, 25G only) ● FH (Full-Height) ● LP (Low-Profile) The second source network adapter works with PowerEdge (R650. R750, and R6525). The following firmware limits apply: ● IDRAC firmware minimum: 5.10.10.00, A00 ● Broadcom NetXtreme minimum firmware: 22.0 This feature supports hyperconverged, storage-only, compute-only, and file-based compute-only deployments and works with Dell OS 10 and Cisco switches. Deployments with the Broadcom adapters are very similar to the deployments with the Mellanox adapters.
Support for import of a 4.0 software-only system	PowerFlex Manager automates the process of importing a PowerFlex 4.0 software-only system. You can import a 4.0 software-only system in two ways: ● By using the **Initial Configuration Wizard** ● By discovering a PowerFlex system on the **Resources** page
Banner notification for software upgrades	If a new version of the management software is available, PowerFlex Manager displays a banner to notify you of the new release. The banner displays a **View Details** link and an **Actions** menu to allow you to learn more about the new release and perform the upgrade.
Support for CloudLink 7.1.5	PowerFlex and PowerFlex Manager add support for CloudLink 7.1.5. CloudLink 7.1.5 changes the disk mapping logic to use /dev/sdX, which is different from CloudLink 7.1.2 to 7.1.4. This version removes support for /dev/disk/by-id from CloudLink, thereby reverting to a /dev/sdX device mapping. Due to key updates, adoption of CloudLink 7.1.5 is recommended for use with PowerFlex for data at rest encryption.
Sample templates updated to align with new network best practices	The PowerFlex rack and PowerFlex appliance network design has been updated. The default, standard design is now two data networks, instead of four. The sample templates have been updated to match the new default.
Updated object limits for file services (NAS)	The following limits apply to file services objects: ● The maximum number of file systems for each NAS Server is 125.

Feature	Details
	• The maximum number of file systems plus the number of mounted snaps for each NAS Server is 1500, including the maximum of 125 file systems. For complete details on all product limits, see the *Dell PowerFlex 4.0.x Technical Overview*.
Miscellaneous bug fixes	This release includes fixes to issues discovered in earlier releases of PowerFlex and PowerFlex Manager.
Documentation	For more information about this release, see the following documents: • *Dell PowerFlex 4.0.x Technical Overview* • *Dell PowerFlex 4.0.x Administration Guide* • *Dell PowerFlex Manager 4.0.x Online Help* • *Dell PowerFlex 4.0.x CLI Guide* • *Dell PowerFlex* API on the Dell Technologies Developer portal

Support deprecation

This section lists features that are no longer supported in PowerFlex 4.0.x.

RFCache

The RFCache feature was removed. Any customer using RFCache with the PowerFlex 3.6.x.x release must first remove the RFCache configuration and packages prior to initiating an upgrade.

VASA support

VASA support is not available in PowerFlex 4.0. VASA support is a future roadmap item.

Customers needing VASA support should continue using PowerFlex 3.6.x.x with the latest patch.

Install and upgrade

You can install and configure PowerFlex version 4.0.x, or you can upgrade from PowerFlex version 3.6.x to version 4.0.x.

See the *Dell PowerFlex 4.0.x Install and Upgrade Guide* for detailed instructions on installing or upgrading.

Licensing

PowerFlex use requires a license. Contact your Dell account representative to purchase a license for PowerFlex.

Upgrade

The following table lists the versions from which the upgrade can be performed:

PowerFlex/ VxFlex OS / ScaleIO base version	Upgrade path	Supporting documentation
v4.0.x	If you are on 4.0.0x you must upgrade to 4.0.1 first and then upgrade to 4.0.2	
v3.6.0.x	One-step upgrade flow: 1. Upgrade to v4.0.x	*Dell PowerFlex 4.0.x Install and Upgrade Guide*

PowerFlex/ VxFlex OS / ScaleIO base version	Upgrade path	Supporting documentation
v3.0, v3.0.0.x, v3.0.1.x, v3.5.x	Two-step upgrade flow: 1. Upgrade to v3.6.0.x 2. Upgrade to v4.0.x	*Upgrade PowerFlex to v3.6 Guide*
v2.0.1.4, v2.6.x	Three-step upgrade flow: 1. Upgrade to latest version of v3.0.1.x 2. Upgrade to v3.6.0.x 3. Upgrade to v4.0.x	*Upgrade VxFlex OS to v3.0.x Guide* *Upgrade PowerFlex to v3.6 Guide*

In release 4.0.1, PowerFlex and PowerFlex Manager add support for CloudLink 7.1.5.

CloudLink 7.1.5 changes the disk mapping logic to use /dev/sdX, which is different from CloudLink 7.1.2 to 7.1.4. This version removes support for /dev/disk/by-id from CloudLink, thereby reverting to a /dev/sdX device mapping.

PowerFlex 3.6.0.6 supports migration and upgrade flows from CloudLink 7.1 to CloudLink 7.1.5. PowerFlex 3.6.0.6 has added code changes to identify device mapping changes under /dev/mapper.

Customers that have migrated to CloudLink 7.1.3 and adopted /dev/disk/by-id logic as documented in *Dell PowerFlex v3.6.x CloudLink for SDS Devices*, should open a support call to get assistance to revert back to the /dev/sdX device mapping to adopt the CloudLink 7.1.5. Customers that stayed on earlier CloudLink releases (CloudLink 7.1.1 or lower) can adopt the CloudLink 7.1.5 using their existing device mapping.

Due to key updates, adoption of CloudLink 7.1.5 is recommended for use with PowerFlex for data at rest encryption.

These are the supported upgrade paths for CloudLink:

PowerFlex version	CloudLink version	Description of change
3.6.0.6 and above	7.1.5	Existing CloudLink install base with the original device path logic: /dev/sdX.
3.6.0.2 , 3.6.0.3 , 3.6.0.4 , 3.6.0.5	7.1.3	CloudLink 7.1.3 enables a new device path logic for encrypted devices, using /dev/disk/by-id/wwn-*<unique_id>*.
3.0.1.x all versions 3.5.x.x, all versions 3.6 3.6.0.1	7.1 7.0	These CloudLink versions maintain legacy support for the existing CloudLink install base with the original device path logic: /dev/sdX.

Here is the expected upgrade flow for systems with CloudLink:

- Upgrade PowerFlex to 3.6.0.6.
- For CloudLink 7.1 or 7.0, upgrade to CloudLink 7.1.5.
- For CloudLink 7.1.2 or 7.1.3 or 7.1.4, upgrade to CloudLink 7.1.5 with support assistance.

Supported operating systems

For a list of supported operating systems, see the **Storage** section of the *Dell Technologies Simple Support Matrices* at https://elabnavigator.dell.com/eln/modernHomeSSM.

SDC compatibility

By design, the PowerFlex SDC in this release is compatible with newer and previously released systems that are currently supported (for example, VxFlex OS v3.0.1). In addition, the PowerFlex system (backend) in this release is backward and forward compatible with previously released SDC versions that are currently supported.

Some limitations may apply to mixed-version systems, such as:

- Replication is not supported on SDC releases earlier than PowerFlex v3.5.
- The CHAP component authentication feature cannot be enabled unless every MDM, SDC and SDS in the system is at least version 3.5, and conversely, if the feature is enabled, no pre-v3.5 SDCs can be added.
- Fine Granularity Storage Pool-based volumes cannot be mapped to a ScaleIO/VxFlex OS v2.x SDC.
- Snapshot Policy management is not available with SDC releases earlier than VxFlex OS v3.0.

Known issues in 4.0.1

The Dell engineering team is working to resolve known issues.

Before getting started with PowerFlex and PowerFlex Manager, review the following known issues in version 4.0.1:

CLI

Issue ID	Symptom	Recommended action
SCI-67737	If the `generate_login_certificate scli` command is executed without parameters, it will fail.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.

MDM

Issue ID	Symptom	Recommended action
SCI-65059	IP addresses are not rebalanced when MDM disconnects from SDT.	If an SDT is completely unreachable, the SDT_NOT_CONNECTED_TO_HOST alert is raised. You can use connectivity monitoring utilities (such as ping) to monitor the network health.

SDS

Issue ID	Symptom	Recommended action
SCI-67744	All SDS devices get into an error state during and after server reboot. This can happen if you run NVME IO for a long time, then run `reboot -f now`.	Do not use `-f` with the `reboot now` command.
SCI-67359	If a device-at-fault event occurs while an SDS is in protected maintenance mode (PMM), some capacity may become un-writeable, resulting in IO errors until the PMM is aborted. The device-at-fault event occurs when a device is readable but not writeable.	Remove from PMM.
SCI-66916	IPv6 is not supported in version 4.0. Therefore, a non-disruptive upgrade from 3.x to v4.0 with IPv6 addresses is not supported. A non-disruptive upgrade	None.

Issue ID	Symptom	Recommended action
	from 3.x to v4.0 when SDS uses IPv6 address will fail in the query phase.	

SDC

Issue ID	Symptom	Recommended action
SCI-71000	The SDC system service in SLES15.x is renamed to **scini** to maintain consistency with other Linux distributions. However, when upgrading to the latest version (scini.service), the earlier version (sdc.service) remains in use. This conflict prevents the SDC from loading.	Ensure to disable and remove the earlier SDC service manually and then, restart the latest scini service. Perform the following procedure: 1. systemctl disable sdc.service 2. rm /usr/lib/systemd/system/sdc.service 3. systemctl restart scini.service
SCI-67107	An SDC has two IP addresses, but the **Block** tab only shows one.	Use the CLI to see other IP addresses.
SCI-66634	An error occurs during the loading of module binary /bin/emc/scaleio/scini.ko. This can happen if a machine restarts or a power outage occurs during a package upgrade. For example, this error might occur if an iDRAC power off is initiated during a zypper upgrade command, which updates packages including the PowerFlex SDC package (EMC-ScaleIO-sdc). This error occurs very infrequently.	Upgrade the SDC package again by using this command: `rpm -U <path_to_sdc_rpm>`

SDT

Issue ID	Symptom	Recommended action
SCI-67009	In some cases where the amount of hosts and active connections is close to utilizing the maximal amount of NVMe controllers permitted by the NVMe standard, adding a new SDT to the system may fail.	Disconnect one or more hosts from the system to allow the system to reclaim NVMe controller IDs. While addition of SDT is blocked, the system can still function properly and serve all currently connected hosts.
SCI-66722	It is possible to remove 2 SDTs that are the only paths of a volume. When removing 2 SDTs, after removing the first one, the MDM will replace it in the host connectivity goal with a port from the other SDT and will update all paths that used it with system-ports from other SDTs. (Some might get a path from the new port in the goal). When removing the second one, the MDM will do the same as in the first removal. However, now some mapping that got the first new port in the goal after the first removal might also get the second new port in the goal. Those mappings will stay without connection.	Dell Technologies recommends not removing or entering into maintenance mode a number of SDTs that is greater than or equal to the number of paths per volume, without reconnecting the hosts.

Issue ID	Symptom	Recommended action
SCI-66645	Incorrect alert definition for SDT_MULTIPLE_DISCONNECTIONS_FROM_STORAGE_BACKEND_COMPONENTS. After bringing an interface down on an SDS, SDTs disconnect from it as expected and an alert is generated and processed by PowerFlex Manager. Upon recovery (NIC up), the events confirm that the SDTs and SDCs are connected back to that interface. However, this alert stays in the user interface. Sometimes, SDT alerts do not get properly cleared.	None.
SCI-66285	SDT performance drops approximately 30% when SDT processes span 2 numas. Deploying SDT and SDR on the same server can lead to reduced performance.	Avoid deploying SDT and SDR on the same server.
SCI-64187	NVMe namespace paths may change following disconnect and reconnect to PowerFlex NVMe Target, such that volumes may be used by different paths. Use by-id paths to maintain a stable path to devices across disconnect operations.	Use /dev/disk/by-id/<unique device path>.

PowerFlex Manager

Issue ID	Symptom	Recommended action
SCI-70298	If you use a CSV file to configure the PowerFlex cluster, it may fail to add an SDR. The failure produces an error that says "There is no available replication capacity". The cause of the failure is that there was not enough journal capacity to add the SDR.	Adjust the journal capacity.
VXFM-31338 and VXFM-31116	A user with the Snapshot Manager or Replication Manager user role is not able to create, map, or unmap snapshots.	Use another role such as Super User or System Admin to manage snapshots.
VXFM-31232	During a 4.0.0.1 to 4.0.1 Upgrade, NFS requires an extra load balancer (LB) IP due to changes to local external traffic policy. This extra LB IP used during the upgrade must be released/freed up after the upgrade. If you attempt to configure Support Assist (SA) on this appliance now, it will fail, because there are no spare LB IPs to configure it.	
VXFM-31230	The **Getting Started** page presents an empty page after a CSV deployment. If you click on the **Deploy with Installation File** button, then click **Next** and **Cancel**, you see an empty page if you go to **Settings** and click **Getting started**.	Refresh the browser, or logout.
VXFM-31193	After a non-disruptive upgrade completes on a storage-only resource group, the Tie Breaker node compliance state is in 'Pending Upgrade', even though the upgrade completed successfully. The state is reported incorrectly in the user interface.	Remove and re-add the resource group.

Issue ID	Symptom	Recommended action
VXFM-31191	A restore to an existing cluster may fail if the CIFS password begins with a special character, such as a dollar sign ($).	Use a CIFS share that has a password that does not start with a special character.
VXFM-31185	If you attempt to perform a CSV deployment in the user interface without loading the MDM RPM, the deployment fails with an unhelpful message. The error in the event of the failed command should indicate that the MDM RPM is missing.	Load the MDM RPM and try again.
VXFM-31179	If you try to delete a hyperconverged resource group, the teardown operation does not delete the vCenter cluster. Also, the wizard for deleting the resource group does not show the checkbox that allows you to delete the cluster and remove it from vCenter.	Manually delete the vCenter cluster and re-inventory the vCenter resource.
VXFM-31163 and VXFM-29528	The Modify IP Role feature is not working for SDRs and SDTs.	Use the REST API instead. The REST command is modifyIpRole. For example: `/api/instances/Sdt::<id>/action/modifyIpRole -d '{"ip":"<ip>", "newRole":"<role>"}'` The role can be one of the following strings: StorageOnly, HostOnly, StorageAndHost
VXFM-31160	When you create a volume on the **Block** tab, the **Map** button does not open the **Map Volumes to Hosts** window after the volume is successfully created.	Click on the **Cancel** button and try again.
VXFM-31093	While registering **SupportAssist** from the **Settings** page, PowerFlex Manager enables two policies by default, once the registration is successful. Instead of enabling the policy that was already there, PowerFlex Manager creates a second one.	Delete the duplicate entry if you don't want to use it.
VXFM-30889	When removing a resource group, PowerFlex Manager does not give you the option to choose whether resources should be left in a managed or unmanaged state. Instead, PowerFlex Manager automatically leaves the nodes in a managed state.	Go to the **Resources** page to remove the nodes, if you like.
VXFM-30855	The MDM events extractor now supports multiple IP addresses for each MDM. If the system is configured with two or more management IP addresses per MDM, the extractor tries to connect to each one. If the connection to multiple interfaces is lost, the extractor does not recognize the disconnect and thinks that it is still connected, and does not show an alert indicating that all are down. If only one connection goes down, you do see an alert.	None.

Issue ID	Symptom	Recommended action
VXFM-30806	When an LDAP user is assigned to an LDAP group, and the LDAP group is configured with a user role, a REST login command may fail with an error stating "no role assigned to user". This can happen if you delete the LDAP user, and try to login with that user.	Define groups before adding explicit users. Also, If necessary, keep another local or federated user that can be used to log in.
VXFM-30792	On the **Getting Started** page, If you try to deploy a CSV installation file that has errors, the **Upload Installation CSV** button is disabled after the error message appears. This button should be enabled to allow you to upload a fixed file.	Close the wizard and open it again. Then, upload the fixed file.
VXFM-30663	Under rare circumstances, some snapshots may be missing from the file system GUI.	Use the export mountpoint to view the missing snapshots.
VXFM-30624	After performing a new resource group deployment, PowerFlex Manager shows a node with an update failed status, but the deployment was successful. This is an incorrect status, since no update was performed.	Remove the resource group and add it again.
VXFM-29962	During an upgrade from 4.0 to 4.0.1, a standby PostgreSQL node may end up in a failed state.	None.
VXFM-29892	Remove any text from UI which supports the statement that additional networks can be added for SO resource group.	None.

VXFM-29795	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the MDM primary node was changed, the certificate wasn't approved before. Therefore, the user isn't recognized and the import can't be done.	Change LIA LDAP authentication to LIA native authentication manually, as follows:
		1. In each PowerFlex node, go to `/opt/emc/scaleio/lia/cfg`, and re-edit the file `conf.txt` so it won't include LDAP fields at all and lia_auth_mode=NATIVE_AUTHENTICATION. Also, set lia_token=Scaleio123. Then close the file and save.
		2. Login to each PowerFlex node and reboot the LIA service, as follows:
		`[root@96N-15 ~]# cd /opt/emc/scaleio/lia/bin`
		`[root@96N-15 bin]# ./delete_service.sh`
		`Removed symlink /etc/systemd/system/multi-user.target.wants/lia.service.`
		`[root@96N-15 bin]# ./create_service.sh`
		`Created symlink from /etc/systemd/system/multi-user.target.wants/lia.service to /usr/lib/systemd/system/lia.service.`
		Importing the version 3.6 PowerFlex cluster to PowerFlex Manager will then work properly.

VXFM-29710	The old REST API for logging out does not work. If you logout with `GET https://<ip>/api/logout`, the API returns a 500 error code. In this release, you need to use both the "access_token" and the "refresh_token" you got when you	Logout with the new API: `POST https://<ip>/rest/auth/logout`

Issue ID	Symptom	Recommended action
	performed the login. The old API uses only one token, so the API returns an error.	
VXFM-29705	When a PowerFlex cluster deployment fails, you may see errors from a previous run if you attempt to redeploy repeatedly in quick succession. This is a logging issue only. If you run deployments over and over quickly, the logs may bleed over into subsequent deployments.	None.
VXFM-29637	It takes PowerFlex Manager up to 40 minutes to fully load after reboot, crash, or power off.	None.
VXFM-29619	**Delete Resource Group** does not remove nodes from CloudLink Center in the case of resource groups with encryption. The nodes are still present, but they are in a **Disconnected** state.	None.
VXFM- 29602	The PowerFlex gateway still responds to the old alert query API, but this was deprecated. In older versions of Powerflex alerts were queried via the gateway using the following API: `/api/types/Alert/instances` In this release, the alerts are handled by the alert service and queried by using this API: `/rest/v1/alerts` The deprecated API should return an error, but instead it responds normally and returns incorrect data that does not include some alerts.	Do not use the deprecated API. Use the new API instead.
VXFM- 29600	The node deployment status on the **Resources** page is set to **Deployment Cancelled** state after a successful non-disruptive update. This is a temporary state and should be corrected in a few minutes.	Check again in a few minutes to see if the deployment status has been updated.
VXFM-29584	The exception only flag for IM log collection is not supported. This flag has been deprecated and is ignored.	None.
VXFM-29543	The management workflow to allow swapping primary and backup nodes has not been added to the restricted operations after a NAS server has entered into an out of service state. This situation might occur in the following scenario: 1. You make a NAS storage pool full. 2. The NAS server on the storage pool becomes read only. 3. You try to move the NAS server (this is a prohibited operation). 4. You try to perform a swap of the NAS server (this is successful). This issue has no visible impact on file service operations.	None.

Issue ID	Symptom	Recommended action
VXFM-29500	PowerFlex Manager does not provide the ability to place an SDT in maintenance mode within the user interface.	None.
VXFM-29494	Filtering events by timestamp does not work correctly with milliseconds. For example, the following API call might return events that do not match the filter: `https://10.234.222.15:31943/rest/v1/events?filter=timestamp+gt+2022-07-05T14:57:27.807Z&filter=service_instance_id+le+Value+above+limit+`-=/<>,.?|;:[]{}&order=resource_id.dsc`	None.
VXFM-29404	A hyperconverged resource group fails if an NFS datastore is already present.	Remove the NFS datastore and retry the resource group.
VXFM-29386	The **Latency** graph on the dashboard does not show any data for NVMe/TCP.	None.
VXFM-29381	The **Summary** page of the **Initial Configuration** wizard remains static for 30 seconds while PowerFlex Manager is enabling SupportAssist.	None.
VXFM-29304	Some event and alert services cannot recover from a configuration or certificate change in the AMQ service. All services may lose connection to AMQ and reconnection attempts may fail.	Restart the pod of services that did not disconnect.
VXFM-29294	The dashboard IO performance numbers do not correlate with IO performance numbers reported by SCLI.	None.
VXFM-29291	On the **File** > **NAS Servers** or **File** > **File Systems** page, the performance details do not get updated on the side pane for a selected NAS server or file system.	Click somewhere else on the page and click back again to see the performance details.
VXFM-29272	A hyperconverged deployment fails in a partial network configuration when DNS is not configured in the hypervisor management network.	Configure DNS.
VXFM-29234	When you create a login certificate and it later expires, an attempt to login with the expired certificate gives a misleading error message: **`scli --login --p12_path /root/.scli/admin_cert.p12 --p12_password password Error: Failed to connect to MDM 127.0.0.1:8611`** There should be some indication that the certificate is no longer viable and you need to create a new one.	None.
VXFM-29190	NVMe drive replacement fails with the error "Removal of the operating system failed". The error is confusing because it says it failed, but you can remove the drive without issue and the new drive is detected and can be added back to PowerFlex and CloudLink.	None.

Issue ID	Symptom	Recommended action
VXFM-29174	The **Edit Management Software Upgrade Settings** page does not have a **Test Connection** button. However, when you click **Save**, PowerFlex Manager displays a message indicating whether the supporting connections required for the upgrade are valid.	None.
VXFM-29095	There is no way to delete a source or destination on the **Notification Policies** page.	Remove or disable all the policies involving this source or destination.
VXFM-28986	PowerFlex Manager allows you to deploy 16 nodes for a PowerFlex File (NAS) compute-only deployment. This is not supported. The limit is 16 nodes. The user interface should block NAS deployments that exceed 16 nodes.	None.
VXFM-28983	When trying to add an additional node to a resource group, you may see the deployment fail with a 500 error, if the storage pool media type is set to "transitional". The "transitional" media type is not supported.	None.
VXFM-28829	If you click **Add Template** in the **Deploy Resources** section of the **Getting Started** page, the user interface shows the **Create** wizard as expected. After selecting **Create a new template**, entering a **Template Name**, and clicking **Next**, the **Create Template** page shows up correctly. After entering all fields and clicking **Save**, the user interface does not show the template details to allow further editing of the template, as it would if you clicked **Lifecycle** > **Templates** and then clicked **Create**. However, the template is created.	Navigate to **Lifecycle** > **Templates**. You will see the template in the list under **My Templates**. You can then click on the template name to edit the template as expected.
VXFM-28715	You are unable to save changes in the **Destination Protocol Settings (SupportAssist)** window if you attempt to change the connection type from **Connect via Gateway Server** to **Connect Directly**.	De-register before you try to change from one connection method to another.
VXFM-28693	The error message displayed in PowerFlex Manager is not clear when the logged in user tries to change their password.	None.
VXFM-28291	An ambiguous error appears during node expansion if a reverse DNS lookup for hostname fails to find an IP address. The **Summary** screen for adding the node shows: `An unexpected error occurred on the system. Retry the operation later.`	None.
VXFM-28281	PowerFlex Manager allows multiple instances of an iDRAC password update job to run on the same node at the same time.	None.
VXFM-28194	If there is a larger than expected time difference (over 1 hour) between PowerFlex Manager and the PowerFlex storage nodes, an upload license action may fail with the following error: `License Enforcement - Wrong Command Parameters`. This error is displayed on the **License Management** page in PowerFlex Manager.	Align the time between the management platform and storage nodes and upload the license again.

Issue ID	Symptom	Recommended action
	In the PowerFlex Manager log file, the following error appears: `"error_code":62,"error_message":"License Enforcement - Wrong Command Parameters."` In the primary MDM logs `/opt/emc/scaleio/mdm/logs/trc.z.XX`, the error displays as follows: `Error, license activation date in the future`	
VXFM-28034	If you create volumes on the **Block** page, volume creation is cancelled when you switch away from this page. Volume creation should be continued in background. Instead, volume creation is stopped and there is no progress or completion status when the **Block** page is opened again.	Wait for the operation to finish before navigating to another page.
VXFM-27967	The **View Details** page for Cisco switches shows a blank page and no metrics.	None.
VXFM-27411	Uploading a 4.0 license to an existing service that is still on 3.6 should show a more meaningful error message.	Apply the license through the CLI.
VXFM-27388	If you upload a certificate file you have uploaded before using the same name you have used before, PowerFlex Manager does not display an error message and overrides the existing certificate.	None.
VXFM-26570	An attempt to tear down a storage-only CloudLink resource group does not remove the machine groups from CloudLink Center.	None.
VXFM-26550	If you run the command **"powerflex_cli events quwery --session_token 98329575cf390107 --limit 2 --order timestamp.asc"** with wrong command flag **"quwery"**, the command fails with error `"Error: unknown flag: --limit"`. This is the incorrect error message.	Run the command with the correct command flag: **"powerflex_cli events query --session_token 98329575cf390107 --limit 2 --order timestamp.asc"**
VXFM-26549	When using power_cli to query events with filter "severity in (MAJOR,MINOR)" it fails with a parsing error. This error occurs because the "in" operator does not exist in the CLI.	The same query can be achieved by running the query a few times, once for each value in the "in" clause.
VXFM-25895	PowerFlex Manager is unable to configure a Dell OS10 switch for deployment.	None.
VXFM-25284	While creating a snapshot rule, you need to provide the time of day. By default, the value populated on this field is current time in EST. However, internally, the snapshot rule is created with the value in UTC. This may lead to confusion as snapshots are not getting created as expected.	None.
SCI-67737	If the `generate_login_certificate scli` command is executed without parameters, it will fail.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.

Known issues and limitations from PowerFlex 3.6.x.x

The following table lists known issues and limitations that exist in the PowerFlex 3.6.x.x releases that also apply to this release.

NOTE: If an issue was reported by customers, the customers' Service Request numbers appear in the "Issue number & SR number" column, and serve to correlate between customer-reported issues and the issue number.

The following table lists known issues and limitations for PowerFlex Custom Node.

Issue number & SR number	Problem summary	Workaround
SCI-62961	Disk slot information for PowerEdge nodes introduced in PowerFlex 3.6 does not work on R650/R6525/R750 nodes with HBA355.	Use iDRAC or perccli to get the disk slot information.

The following table lists known issues and limitations for the PowerFlex Gateway.

Issue number & SR number	Problem summary	Workaround
SCI-57304	PowerFlex Installer (also known as the PowerFlex Gateway) is exposed to issue HSTS Missing From HTTPS Server, CVE: https://nvd.nist.gov/vuln/detail/CVE-2017-7789	None
SCI-55819	During NDU to v3.5.1.x, in rare cases, the PowerFlex Gateway might have high CPU usage and get stuck at the upload phase.	Perform a restart of the gateway service using an SSH session to the PowerFlex Gateway: service scaleio-gateway restart Then, in the PowerFlex Installer user interface, click Retry to continue the upgrade.

The following table lists known issues and limitations for the SDC.

Issue number & SR number	Problem summary	Workaround
SCI-57752	The SDC rpm package for SLES15.X installs without any issues. However, after installation, when checking if the SDC driver's scini service is loaded, the following confusing state is returned: service scini status service: no such service scini This is a false positive status alert.	None

The following table lists known issues and limitations for the SDR.

Issue number & SR number	Problem summary	Workaround
SCI-62047	A replication failover on an RCG that is still in the initial copy phase may not present all volume members at the remote target location.	Wait for all volumes in the RCG to complete their initial copy phase before performing the failover operation for that RCG.

The following table lists known issues and limitations for the SDS.

Issue number & SR number	Problem summary	Workaround
SCI-63216	When adding NVMe devices using the by-id path similar to the following: dev/	Use the alternative name path in by-id location with nvme-eui in the path. For example:

Issue number & SR number	Problem summary	Workaround
	disk/by-id/nvme-Dell_Express_Flash_NVMe_P4610_1.6TB_SFF_BTLN950106ZB1P6AGN the MDM blocks the addition due to the error: new_device_path too long, MDM supports up to 63 characters.	/dev/disk/by-id/nvme-eui.01000000010000005cd2e448b0b75151

Known issues from PowerFlex Manager 3.8.1

The following table lists known issues from PowerFlex Manager 3.8.1 that also apply to this release.

Issue ID	Symptom	Recommended action
ESRCM-3148	Remediation is available for the Apache Log4j remote code execution vulnerability that could be exploited by malicious users to compromise the affected system. Dell recommends implementing this remediation as soon as possible in light of the critical severity of the vulnerability. The PowerFlex presentation server uses Log4j with version 2.14.1 and is impacted by this issue.	For details, see https://www.dell.com/support/kbdoc/enus/000194548/dsa-2021-272-dellpowerflex-securityupdate-for-apachelog4j-remote-codeexecutionvulnerabilitycve-2021-44228?ref=emcadvisory_000194548_Critical_null
VXFM-24115	A service status is displaying incorrectly on the **Services** page after performing service mode operation.	None.
VXFM-23822	An attempt to perform a service upgrade fails after a backup and restore from an earlier release of PowerFlex Manager. This could be caused by a change in the state of the environment after the backup is performed. For example, this would happen if the hostname of the PowerFlex gateway changed after the backup was performed, since PowerFlex Manager restores the state of objects at the time of the backup, and any subsequent service operations would fail because of the mismatch in the gateway hostname.	Ensure that the state of your system resources does not change after a backup is performed. The backup state must match the state of your production environment once the restore is performed.
VXFM-23802	The SDC nodes are not listed under **View details** on the **Resources** page for PowerFlex Gateway.	None.
VXFM-23775	Disruptive upgrade on the PowerFlex hyperconverged service fails, if upgrading from VMware ESXi 7.0.Ux to VMware ESXi 7.0 Ux. The PowerFlex hyperconverged nodes become unresponsive after deactivating the protection domain and shutting down all the VMs.	Use the non-disruptive upgrade.
VXFM-23756	After upgrading from PowerFlex Manager 3.4 to PowerFlex Manager 3.8, PowerFlex storage-only service is showing as **Healthy** instead of **Lifecycle Mode**.	You need to perform **Update Service Details** after adding the latest compatibility matrix.
VXFM-23672	The PowerFlex storage-only service shows as `Incomplete` with a banner message displaying `To add volume` even though two PowerFlex service volumes exist.	Volumes created outside of the service will not display within the service until inventory and **Update Service Details** is performed.
VXFM-23639	New OS10 switches are non-compliant with newer firmware versions as expected version in RCMs and Intelligent Catalogs may be lower and downgrades are not allowed.	None.

Issue ID	Symptom	Recommended action
VXFM-23636	If you perform an update on a node in a service, you may not be able to update a second node in the service immediately after the first node is upgraded without manually running inventory on the PowerFlex gateway in PowerFlex Manager.	Run the inventory for the PowerFlex gateway and re-initiate the node upgrade.
VXFM-23630	Unable to initiate the scaleup of networks on a PowerFlex management controller 2.0 setup.	None.
VXFM-23231	If the server profile is incorrect, PowerFlex storage-only service scaleup volume completes with an **Error** deployment state.	1. Reapply the server profile so that it displays the model name correctly and retry the service. 2. Set the correct model name for PowerFlex Manager appliance when converting from PowerFlex Ready Node to PowerFlex appliance.
VXFM-23193	Editing a service with a new OS image is not reflected in the **Services** or **Compliance and OS Repositories**. New PowerFlex nodes deployed will reflect the new OS image. Only existing PowerFlex nodes upgraded will have this issue.	1. Confirm the current OS image, go to **Services** and confirm. 2. Re-adding the service will identify the latest OS image available and reflect the selected one during import.
VXFM-23136	Reserving virtual IP address comes from the **SDC role**, if the IP address range is small.	None.
VXFM-23132	In the **Resources** page, filtering resource type **PowerFlex Gateway/Presentation Server(PS)** using the corresponding service name displays `No Devices Found`.	None.
VXFM-23124	Multiple inventory jobs are causing service failures.	Retry the service.
VXFM-23066	Cannot search for volumes starting with **+** in the search text.	Do not use **+** in the search criteria.
VXFM-23061	In resource discovery, cannot select **Next** from **Assign Users** step in **Create Node Pool** wizard due to incorrect validation error.	None.
VXFM-22804	The **Filter by Groups** feature for Active Directory is not working.	None. Scroll through the list of users.
VXFM-22573	On a PowerFlex storage-only template, when changing to the **Add Networks to this port** option from **Mirror**, the selected networks are inversed.	Re-edit and the selected folders will be available.
VXFM-21849	The **Import Networks** feature is not working when adding an existing service with VMware NSX and VMware vCenter network folders.	Do not use the **Import Networks** feature while adding an existing service. Manually define the required networks in PowerFlex Manager - PowerFlex / hypervisor management and PowerFlex data networks. It is not necessary to define the general purpose LAN networks - they can be left blank.
VXFM-21559	During a deployment, if servers are discovered before switches, the service is returned in lifecycle mode.	Manually run inventory on the switches.
VXFM-21531	CloudLink Center SNMP settings are applied without user input and are not fully functional.	After completing a change configuration, save the configuration again without making any changes.

Issue ID	Symptom	Recommended action
VXFM-21467	Deployment fails if overlapping data network subnets are used.	Set up the correct network subnets and redeploy. Ensure you have one interface per subnet and redeploy.
VXFM-21353	PowerFlex gateway upgrade does not update the compliance report on the PowerFlex nodes.	Rerun the inventory on all the PowerFlex nodes.
VXFM-21278	VMware vCLS volumes fail to migrate to the shared datastore if VMware ESXi nodes are added using an IP address, but resolve to a hostname in the DNS.	Add the PowerFlex nodes back into the vCSA under the resolved hostname.
VXFM-21256	The iDRAC service module is not in a running state during an NVMe replacement. The NVMe eject operation fails.	Restart the iDRAC network and restart the ISM service `dcism`.
VXFM-21233	The migration of vCLS VMs fails on a PowerFlex hyperconverged service if there is a mix of PowerFlex compute-only nodes and hyperconverged-only nodes in same VMware cluster but in different services that you are trying to upgrade. Migration is failing because one of the vCLS VMs belongs to the PowerFlex compute-only service, but it does not have access to the new utility datastores.	Run the VMware vCLS migration on the PowerFlex hyperconverged service. If that fails, go to PowerFlex compute-only service, and run the vCLS migration on that service.
VXFM-21232	Network scale up fails on a hyperconverged service due to port down when adding service.	1. Reseat the network cable to see if uplink can be restored. 2. Remove the service. 3. Readd the service using the **Add Existing Service** function. 4. Verify the network connectivity displays in the **Port View**.
VXFM-21231 and VXFM-21180	Partial network automation option is only intended to be used with LACP networking. However, there is no validation performed in the **Add Existing Service** wizard to prevent the use of partial network automation without LACP networking. Selecting this combination can result in various failures including: ● The service shows a blank **Teaming And Bonding Configuration** in the service's **View All Settings** dialog under **OS Settings**. ● Rebooting the PowerFlex Manager appliance will cause the service to go into an **Upgrade Service Components** error status. ● Service update operations run on the service fail. ● PowerFlex cluster upgrades fail.	Remove the service and add it again by using **Add Existing Service** with the **Full Network Automation** option. Only choose **Partial Network Automation** if your service is using LACP networking.
VXFM-21186	If you are upgrading a service, the service may be placed in lifecycle mode due to validation errors. The validation errors are displayed as a part of the reason. If the service prompts for vCLS migration also, you can perform vCLS migration on the service. In this case, the validation reasons are removed, and the banner no longer includes the reasons for lifecycle mode. You no longer see the reasons, but the service is in lifecycle mode and prevents any further actions.	None.

Issue ID	Symptom	Recommended action
VXFM-21049	The service name displayed in the **Duplicate Node Summary** screen shows special characters in an HTML encoded format.	None.
VXFM-20957	During an import of Active Directory users, if you click **View Details**, PowerFlex Manager displays a broken page.	None.
VXFM-20944	Appliance and Alert Connector state shows different timezone and time stamp.	No functional impact, reboot the appliance and timezone and time stamp align.
VXFM-20940	Dell CloudLink Center deployment fails if you are using different subnet than PowerFlex Manager appliance.	Set the subnet manually.
VXFM-20868	If a service only has a single node, the **Add Existing Service** function works, but the overall service health shows error status once the service add finishes.	None.
VXFM-20877	PowerFlex Manager may show the wrong compliance status on nodes. PowerFlex Manager shows the nodes are compliant on the **Resources** page, even though the nodes are not compliant.	Run a manual inventory, and the correct compliance status displays.
VXFM-20816	Adding an OS Image with the same repository name does not return an error.	None.
VXFM-20735	VLAN 4094 through 4096 are reserved VLANs on the Dell PowerSwitch switches.	Do not use any of the reserved VLANs.
VXFM-20686	VMware vSphere vMotion vmk is missing from the second node in an VMware ESXi compute-only node deployment.	Manually create the VMware vSphere vMotion vmk on the second node.
VXFM-20666	No warning displays when adding an existing encrypted service to a brownfield deployment.	Ensure an encryption tool or CloudLink is discovered first.
VXFM-20655	Drive replacement and MDM reconfiguration are allowed through PowerFlex Manager when a service is in an error state.	Do not attempt a drive replacement or an MDM reconfiguration if a service is in an error state.
VXFM-20593	PowerFlex Manager creates an Element Manager credential via REST API fails with wrong error message if `snmpCommunityString` is omitted.	None.
VXFM-20590	When scaling up of the network on a PowerFlex compute-only service, the new network does not show under the port view for one of the three nodes in the service.	None.
VXFM-20588	PowerFlex Manager displays **Update Resource** when current version is higher than expected version.	None.
VXFM-20571	PowerFlex Manager lets you add a static route to a subnet that already exists on the node. This fails as it is not needed - the node already has a route to that subnet.	None.
VXFM-20546	If existing port channels that do not have any interfaces are reused on Dell OS10 as part of the new service, any previous VLANs on those port channels are not cleared. If new port channels are created they will only have the correct VLANs.	None.

Issue ID	Symptom	Recommended action
VXFM-20533	PowerFlex Manager fails to set the default compliance package version when you add a compliance file and select **Make this the default version for compliance checking**.	Click **Change** to reset the default version.
VXFM-20318	A PowerFlex hyperconverged service deployment using an existing VMware cluster with different storage pools fails.	Remove the nodes and add them back with the same storage pool.
VXFM-20215	In the teardown of a PowerFlex hyperconverged service where the datacenter, cluster, and network information are all moved into their own folders in VMware vCenter, the service resources are re-inventoried, and the service is removed. However, the VMware vCenter cluster and nodes are left and not removed.	Manually remove the vCenter cluster and nodes.
VXFM-20169	MDM reconfiguration fails for certain role transitions.	You can set the role manually, if needed. Retry or rescan refreshes the roles.
VXFM-20154	If you attempt to restart a failed job by clicking **Retry**, PowerFlex Manager displays a warning page with a **View Compliance Report** link. This link is broken. You may encounter this issue after a resize volume or migrate vCLS VMs operation fails.	Click **View Compliance Report** under **Service Actions**.
VXFM-19482	PowerFlex Manager fails to configure the server facing ports due to `switchport mode trunk` returning the following error: `% Error: MODE_L2HYBRID, Configured as a lag interface ToR-B(conf-if-eth1/1/3)#`	Remove `swichport mode trunk` before running through PowerFlex Manager.
VXFM-19010	The wizard for updating resources displays incorrect error messages when trying to update the PowerFlex gateway or presentation server resources from the **Services** page.	Go to the management VM service and click **Update resource**.
VXFM-18857	During an **Add Volume** job, a service will go from the **In Progress** state to the **Completed** state, then suddenly return to **In Progress**. While the job is still **In Progress**, some of the service actions may be enabled.	Allow the job to finish before you try to use any of the service actions.
VXFM-18580	Clusters already in services on the appliance are provided as selection options when importing an existing service.	None.
VXFM-18554	During a PowerFlex hyperconverged replication service deployment, retry does not set the VMware VM version. This can cause the deployment to fail when trying to add NVDIMM devices to the VM.	Delete the VM and retry the service.
VXFM-18167	Adding an existing service wizard network mapping shows `%2f` instead of `/` in port group name.	Unsupported special characters cause a rendering issue with the port group name and should not be used. Spaces, underscores, and hyphens are allowed.
VXFM-18005	Retrying a failed PowerFlex upgrade fails to cancel previous upgrade job.	If you start the upgrade, and request PowerFlex Manager to start another upgrade, it fails.

Issue ID	Symptom	Recommended action
		If you see this failure, you may must cancel the previous upgrade.
VXFM-17760	Service sees old NIC cabling data resulting in failure.	Contact Dell Support for a workaround.
VXFM-17610	The DHCP setting in PowerFlex Manager does not allow you to disable the DHCP/PXE node if the virtual appliance is not currently on the subnet.	Do not change your network while you are working on it. If you must do this, at least disable DHCP first.
VXFM-17023	Unable to use NetApp NFS share from PowerFlex Manager.	PowerFlex Manager does not support NFS shares set up this way.
VXFM-16623	A node discovery may appear to fail with the following log message: `Failed to collect the system configuration for server`	Check for the following conditions: • Networking issues between iDRAC and PowerFlex Manager • The server is booted into the BIOS screen. In this case, jobs cannot be run.
VXFM-16261	During a restore, the PowerFlex Manager appliance gets stuck at **Starting Restoration** stage when a filename is not provided, that is, if only the path is specified.	Use the full path to the restore file.
VXFM-15653	Download of the ONIE firmware upgrade package fails, but the failure is not displayed in the logs.	None.
ESRCM-2526	VMware ESXi 7.0 requires a security patch that is available in VMware ESXi 7.0 Update 1.	For VMware ESXi 7.0, you must perform a manual workaround. For details, see https://kb.vmware.com/s/article/76372.
VXFM-15308	PowerFlex Manager collects switch performance metrics only if NXAPI is listening on both HTTPS (Port 443) and HTTP (Port 80). If either of these ports are disabled, switch performance metrics are not collected.	Be sure that ports for both HTTP and HTTPS are enabled for Cisco switches.
VXFM-15227	PowerFlex Manager allows you to publish and deploy a CloudLink template containing a password with all lowercase characters.	Be sure to include at least one uppercase letter in the password.
VXFM-15267	If you try to upload an SSL certificate on the **Appliance SSL Certificates** section of the **Settings** page, you may encounter security errors with the Certificate Signing Request (CSR). The CSR does not include the Subject Alternative Name (SAN) field. It only includes the Common Name (CN) field. This problem occurs only with the Chrome browser.	Contact Dell Support for a workaround.
VXFM-15188	When deploying a service, you may see an error message stating that a static IP address could not be reserved. The error may include empty brackets ([ ]). If you see this error, the IP addresses may not be pingable.	Check your IP addresses to be sure you have enough pingable IP addresses, and try again.
VXFM-15110	You may see incorrect warning messages when using invalid characters for a hostname in a template. If you try to publish the template, you may see a message indicating that a hostname exceeds the limit of 15 characters. The message should indicate that the hostname has invalid characters.	Be sure to use valid characters and conform to the character limit for the operating system. For Windows, the limit is 15 characters for a hostname. For ESXi and Linux, the limit is 26 characters.

Issue ID	Symptom	Recommended action
VXFM-15069	A node cannot be removed from a storage-only service because of a failure removing the switch. If the switch removal fails, PowerFlex Manager leaves the node in the service.	Update the service details or remove the node manually from the service.
VXFM-15010	Adding replication networks to a hyperconverged template breaks the service.	None.
VXFM-14898	During drive replacement, if you type **CONTINUE WITHOUT DRIVE** with a bad drive, PowerFlex Manager tries to add it back and fails.	None.
VXFM-14872	You may notice a type casing inconsistency, when creating templates with uppercase host and SVM names. VMware ESXi uses lowercase host names. There is no impact to functionality.	None.
VXFM-14773	If you try to download two RCMs simultaneously, the operation fails.	Resynchronize the RCMs or download them again separately.
VXFM-14771	If you deploy a service with only one node in it, the service may show **Critical**. This condition is caused by a warning at the component level.	None.
VXFM-14770	The inventory is not running on the PowerFlex gateway after you use service mode. If you place a node in service mode, then exit service mode, then try to enter service mode again, the PowerFlex gateway is left in an abnormal state.	Manually run the inventory or wait for the next hourly update.
VXFM-14767	PowerFlex Manager does not refresh text boxes in the **Discovery Wizard**. This display problem can happen if you select **Node** as the **Resource Type** and choose **Hostname**, then change the **Resource Type** to any other type other than **Node**.	Click the trash can icon to remove the resource, then add it again.
VXFM-14763	Mellanox ConnectX-4 LC firmware must be on 14.25.80.00 or later to upgrade to 14.27.10.16 or later. This problem is not a PowerFlex Manager issue. PowerFlex Manager does not support step upgrades.	None.
VXFM-14753	If an upgrade of the PowerFlex Manager virtual appliance fails, you may see duplicate email messages and alerts.	None.
VXFM-14724	If you try to delete an RCM that is used on a service, the operation fails with a 500 internal server error.	Remove the service that previously used the RCM, delete the RCM, and add the services back as an existing service.
VXFM-14581	When you add an existing hyperconverged or storage-only service that includes a CloudLink Center VM in a cluster, the **Inventory Summary** page does not show the **Adding** icon for the CloudLink Center VM.	None.
VXFM-14535 and VXFM-15309	If you change the password for an SVM, **View All Settings** on the **Services** page still shows the old password. This issue also breaks **Update Service Details** if you try to use a new credential.	If you need to change the OS credential to a newly-defined credential and update to use that credential on a service, you have to remove the service and readd it to select that credential.
VXFM-14496 and VXFM-14254	After upgrading from an earlier version of PowerFlex Manager, you may notice that the inventory might not be updated correctly or that	Run a manual inventory after the upgrade.

Issue ID	Symptom	Recommended action
	subsequent operations such as **Add Existing Service** may fail.	
VXFM-14494 and VXFM-14762	After changing an iDRAC password, if you select the same credentials again, you may see an error message in the **Update Password** wizard. After you see the error, you cannot proceed with the wizard.	Cancel and restart the wizard.
VXFM-14470	PowerFlex Manager does not perform validation to ensure that you have a minimum of three fault sets for a protection domain and storage pool.	Ensure that your fault set configuration conforms to best practices. Dell recommends that all nodes in a protection domain should be in the same storage pool.
VXFM-14397	PowerFlex Manager displays the wrong progress bar during the resizing of a datastore.	None.
VXFM-14377	After you remove a node from a service, PowerFlex Manager may not update the PowerFlex gateway inventory.	Run a manual inventory after removing the node.
VXFM-14028	The **Services** page does not show lifecycle mode as the deployment state for NSXT services that are in lifecycle mode.	None.
VXFM-13994	If a Dell CloudLink Center is non-compliant, PowerFlex Manager disables node selection for upgrade.	Upgrade the CloudLink Center from the **Resources** page and try the upgrade from the **Services** page.
VXFM-13744	An RCM upgrade to a major version of NSX and ESXi may fail with a VIB error. This failure could be caused by a VIB incompatibility if you are using a nonstandard Dell image. Be aware that NSX VIBs are specific to the version of ESXi that is installed on the host. If you upgrade ESXi, you must install new NSX VIBs appropriate for the new ESXi version.	Install new NSX VIBs for the new ESXi version. For more information, see Upgrade to ESXi 6.5 or 6.7 in an NSX Environment.
VXFM-12990	The state of a discovery job may show as **Running** on the **Jobs** page, even though it has completed.	Restart the PowerFlex Manager virtual appliance.
VXFM-12477	On the **Resources** page, if you select a machine with self-encrypting drives (SEDs) for a CloudLink Center, you might see two entries for each drive. One entry shows the status as "encrypted" with a type of "data_sed". The other entry shows the status as "unencrypted" with a type of "data_sds". The SED drives are encrypted, but PowerFlex Manager should not show duplicate entries.	None.
VXFM-12402	After deleting a node from a storage-only service, you may see an incorrect error message in **Recent Activity**. The message indicates a failure removing the Storage Data Client (SDC), even though an SDC is not installed on a storage-only node.	None.
VXFM-12391	If you put an SDS into Protected Maintenance Mode (PMM) manually, and then run inventory for the PowerFlex gateway within PowerFlex Manager, the service does not exit service mode. When you click Exit Service Mode, PowerFlex Manager shows an error. This error occurs because PowerFlex Manager is trying to exit	Manually take the SDS out of PMM outside of PowerFlex Manager.

Issue ID	Symptom	Recommended action
	Instant Maintenance Mode (IMM) instead of PMM.	
VXFM-12366	If you cancel a service deployment, and then retry the deployment later, the service succeeds. However, the warning banner that is displayed while PowerFlex Manager is canceling the deployment remains during and after the successful deployment.	Remove the service and add it back again.
VXFM-12096	The deployment for a storage-only service may sometimes fail to start.	Retry the service deployment.
VXFM-12074	On the **View Details** page for CloudLink, PowerFlex Manager shows a hyperlink for the **IP Address** field for the SDS. The hyperlink should not be displayed for an SDS IP address.	None.
VXFM-11998	PowerFlex Manager does not always show the new status when you add an operating system image on the **OS Image Repositories** tab.	Manually refresh the screen to see the new status.
VXFM-11977	If you delete a node that was not deployed successfully in a service, PowerFlex Manager might show the node as deployed.	To remove the failed node from the service, choose the **Remove Resource** option, rather than the **Delete Resource** option.
VXFM-11930	After upgrading from version 3.3 or 3.4, you might notice that a service that has a fault set enabled shows **Deployed** as the **Deployment State**, even though it should show **Lifecycle Mode**.	Click **Update Service Details** to show the correct state.
VXFM-11760	After updating the firmware and software on a node, or during a service mode operation, PowerFlex Manager might show the node as being in service mode, even though it is no longer in service mode. Due to a time lag, the service mode status that is shown on the service does not always match the state of the system exactly. The node eventually comes out-of-service mode, and the update completes successfully.	None.
VXFM-11758	A compute-only service with ESXi displays the **Protected Maintenance Mode** option for a firmware and software update on a PowerFlex 3.5 cluster. This type of service offers only compute resources, and does not have an SDS component. The **Protected Maintenance Mode** option should not be available. The update succeeds even if you choose this option.	None.
VXFM-11705	If you attempt to discover a Cisco switch with terminal color configured, the discovery fails.	Disable the terminal color option by running `configure no terminal color persist`.
VXFM-11242	If you click **Update Service Details**, PowerFlex Manager might display an incorrect error indicating that the `device could not be connected through SSH`. The message should state that the iDRAC was not responding, most likely because of bad cabling.	None.
VXFM-10663	PowerFlex Manager has only one firmware component id set for all OS10 switches in PowerFlex Manager. This means that PowerFlex	Do not discover the management switches.

Issue ID	Symptom	Recommended action
	Manager can only include one set of OS10 switch versions for all models of OS10 switch in a given RCM package. This is a problem for management switches.	
VXFM-3727	If you try to create a network containing IP addresses that are in use by vCenter or the PowerFlex gateway inventory, `Invalid IP address range is entered` errors are thrown. An extra error message states: `The entered IP range overlaps with the IP ranges configured on one or more configured networks`. This situation only happens if you have multiple networks with the same subnet. For example, this situation might happen if you have an SDC only network and a network that supports both SDS and SDC traffic.	Remove your vCenter or gateway from the inventory, create the network ranges, and then rediscover them.
VXFM-3517	If the PowerFlex gateway is deployed onto an operating system that does not have `systemctl` (such as SuSE 11 or Red Hat Enterprise Linux 6), the PowerFlex configuration script fails to restart the gateway after configuring it. However, it does not return any error, so ultimately the deployment fails when trying to talk to the gateway REST API. This configuration is unsupported, but PowerFlex Manager does not block the configuration or provide a clear error message.	None.

Operating system known issues and limitations

This section lists known issues and limitations for specific operating systems.

Issue number & SR number	Problem summary	Workaround
VXFV-360	The device name appears incorrectly as a non-PCI device instead of iDRAC, but the current and baseline versions are accurate (6.00.30.201 to 6.00.30.202, respectively).	Dell recommends upgrading the OME catalog to the latest version.
	PowerFlex nodes running CPLD 1.0.7 stop responding during Power On Self-Test (POST) when upgrading to certain RCM or IC packages that include BIOS 1.8.2 and iDRAC 6.00.30.202.	Before upgrading to the latest version of PowerFlex, ensure that the PowerFlex nodes are running CPLD 1.0.8 or later. For details, see https://www.dell.com/support/kbdoc/en-us/000205925.
VXFV-296	Mixing the Mellanox CX-6 100GbE NIC driver with the Broadcom PCIe/OCP 25GbE NIC driver is not supported.	Remove the `bnxt_en` driver and install the Mellanox driver.
VXFV-283	iDRAC support for "Prepare for Removal" on PowerEdge R650/R6525/R750/ nodes fails.	NVMe hot-swap on Linux operating systems is supported. Skip the Prepare for Removal step in iDRAC.
VXFV-265	During execution of the command `create-namespace` with 4k alignment on RHEL 8.4, the following error is encountered: Warning: namespace0.0 unable to validate alignment	This is a false notification. The command result is as expected, and the resulting DAX device can be used with PowerFlex acceleration devices.

Issue number & SR number	Problem summary	Workaround
VXFV-243	Upgrade of a RedHat-based operating system to RHEL 8.3 fails if the node contains NVDIMMs. This is due to a known issue on RHEL 8.3: https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/8/html-single/8.3_release_notes/index#known-issue_installer-and-image-creation	Follow the procedures described in "RHEL/CentOS upgrade to version 8.3" in the .
VXFV-224	In a RHEL 8.x-based node using Cloudlink encrypted SDS devices, after reboot the device moves to failed state. **NOTE:** This issue is much rarer on RHEL7.x-based nodes, but it might occur.	1. SSH to the relevant SDS's operating system. 2. Delete the SDS service. 3. Clear the device errors using the GUI or CLI. 4. Recreate the SDS service. 5. Verify that devices come back online. Rebuild/Rebalance will occur if instant maintenance mode or protected maintenance mode was not used. 6. If instant maintenance mode or protected maintenance mode was used, exit maintenance mode.
VXFV-157	NVMe disks are not supported on RHEL 8.0 due to the following issue: https://access.redhat.com/solutions/4280341 Failure to observe this limitation might cause RHEL-based SDSs with NVMes to experience kernel panic during a disk removal flow.	None

Fixed issues in 4.0.1

Fixed issues are defects that exist in the previous release but are resolved in the 4.0.1 release of PowerFlex and PowerFlex Manager.

Issue ID	Symptom
VXFM-29896	When reconfigure network after docker already running the install_PFMP.sh phase failed.
VXFM-29880	The block-legacy-gateway still shows under Target PowerFlex Gateway.
VXFM-29803	A service that was imported into version 3.8 of PowerFlex Manager cannot use **Update Resource Group Details** in version 4.0. This could happen if you perform a one-time migration on a 3.8 environment that includes a service that was originally imported through **Add Existing Service**. If you remove a node from the resource group after the migration, and then perform **Update Resource Group Details**, you might see a 500 exception.
VXFM-29795	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the MDM primary node was changed, the certificate wasn't approved before. Therefore, the user isn't recognized and the import can't be done.
VXFM-29790	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the version 3.6 system is configured with only LDAP authentication for MDM, the LDAP user isn't recognized, so the import can't be done.
VXFM-29659	The PowerFlex management platform (PFMP) upgrade sometimes fails on the first attempt. This can happen if you attempt an upgrade immediately after enabling SupportAssist. The button to perform the upgrade is available in the user interface, but it should not be while SupportAssist configuration is still in progress.

Issue ID	Symptom
VXFM-29656	The Notifier service fails to handle ALM events with "send_to_srs"=false due to a key error. This results in some low-level information being sent when it should not be.
VXFM-29579	A Security Admin user can perform activities defined for the Security Admin role, as well as activities allowed for the Lifecycle Admin role.
VXFM-29576	Event and alert names are not case consistent. The names for events and alerts should follow the convention of upper snake case, with upper case words separated by underscores. However, some of the names use lower snake case, with lower case words separated by underscores.
VXFM-29496	If you run the `reset_PFMP.sh` script to remove an old install and re-install PFMP, the script removes pre-requisite software rpms: • jq • nmap • skopeo The install fails if these RPMs are not there already.
VXFM-29530	PowerFlex Manager does not allow you to modify a notification source policy.
VXFM-29469	Whenever an ALM event from PowerFlex is received, PowerFlex Manager sends a message when it is received, then again when the alert is acknowledged or disacknowledged, and again when the alert is cleared. It sends an event whenever the status changes.
VXFM-29241	Registering SupportAssist from the **Initial Configuration Wizard** is not activating the **Default Policy Configuration** on the **Settings** page.
VXFM-28907	On the **Protection** > **RCGs** page, while the RCG is in the failover state, the action **Resume (from pause)** should be disabled, because the RCG is not in pause mode but in switchover/failover state.
VXFM-28844	On the **Generate Troubleshooting Bundle** page, the hover text for **Download locally** indicates that NFS is supported. However, NFS is not supported for the troubleshooting bundle. The user interface correctly shows the failure if you try to use NFS, but the hover text should not show NFS.
VXFM-28828	The **Compliance Versions** page is not refreshing to show the new state of the compliance file when **Allow Unsigned File** is selected.
VXFM-28774	In some extreme bad path scenarios, when ActiveMQ is not available, some MDM events may not reach PowerFlex Manager.
VXFM-28532	Resizing the service volumes (vCLS volumes) should be blocked by PowerFlex Manager. Currently, you are able to increase the vCLS volume size. After resizing the volumes, you may also see unexpected behaviors. For example, you may see a message saying that the service is in lifecycle mode because the vCLS volumes are on local storage.
VXFM-28433	When trying to export a template without an encryption password, you may see an error.
VXFM-28288	A NilClass exception occurs if you deploy a resource group when you don't have a PowerFlex management network and you have general purpose LAN (GPL) networks that do not have a default gateway.
VXFM-26476	When using the REST API to query event history, if there is an error in the filter field, the filter is ignored, and the unfiltered list is returned instead of an error message.
VXFM-26333	When using the REST API to query alert templates, the filters are ignored, and the full list of alert templates is returned.

Software packages

This section provides information on the contents of the PowerFlex 4.0.1 software packages.

Download location

The PowerFlex complete software package, containing most of the packages listed below, can be downloaded from the following location:

https://www.dell.com/support/home/en-us/product-support/product/scaleio/overview

The RCM for PowerFlex rack can be downloaded from the following location:

Dell Download Center

Packages

Package name	Size	Contents
PFMP bundle	~19.2 GB	PFMP2-4.0.1-249.tgz
RCM for PowerFlex rack	~24.6 GB	Flex_RCM_3_7_0_1_r30.zip
Intelligent catalog for PowerFlex appliance	~23.4 GB	Intelligent_Catalog_40_371_00_r30.zip
Software-only ZIP	~5.9 GB	Software_Only_Complete_4.0.1.0_147.zip. Includes the following: ● PowerFlex_4.0.1000.147_Complete_Core_SW.zip (3.4 GB) ● PowerFlex_4.0.1000.147_SDCs_for_manual_install.zip - SDC RPMs (0.6 GB) ● PowerFlex_Software_4.0.1.0_147_r1.zip - software-only catalog (1.9 GB)
Compatibility file		● cm-20221129.gpg (compatibility file)
PowerFlex Manager OVA ZIP	~24.5 GB	PowerFlex_Manager_4.0.1-249.zip: ● PFMP2-4.0.1-249.ova (PFMP_Installer for installing Kubernetes cluster) (23 GB) ● pfmp-k8s-20230111.ova (Kubernetes OVA running SUSE 15 SP3) (2.3 GB)

The original OS layout table from 3.6 matches this file: PowerFlex_4.0.1000.147_Complete_Core_SW.zip.

5

PowerFlex version 4.0.0.1

What's new in release 4.0.0.1

This release includes fixes to some issues identified in PowerFlex release 4.0.

Support deprecation

This section lists features that are no longer supported in PowerFlex 4.0.x.

RFCache

The RFCache feature was removed. Any customer using RFCache with the PowerFlex 3.6.x.x release must first remove the RFCache configuration and packages prior to initiating an upgrade.

VASA support

VASA support is not available in PowerFlex 4.0. VASA support is a future roadmap item.

Customers needing VASA support should continue using PowerFlex 3.6.x.x with the latest patch.

Install and upgrade

You can install and configure PowerFlex version 4.0.x, or you can upgrade from PowerFlex version 3.6.x to version 4.0.x.

PowerFlex Manager allows you to use the **Initial Configuration Wizard** to perform an upgrade to version 4.0.x. Customers with PowerFlex 3.5.x can upgrade to 3.6.x, and then use the **Initial Configuration Wizard** to perform an upgrade to PowerFlex Manager 4.0.x.

Customers with PowerFlex Manager must be running PowerFlex Manager 3.8 in order to perform the upgrade.

Customers that do not have an existing PowerFlex environment have the option to deploy a new instance of PowerFlex.

See the *Dell PowerFlex Install and Upgrade Guide* for more information.

Licensing

PowerFlex use requires a license. Please call your Dell account representative to purchase a license for PowerFlex.

Upgrade

The following table lists the versions from which the upgrade can be performed:

Table 1. Upgrade support matrix

ScaleIO / VxFlex OS / PowerFlex base version	Upgrade path	Supporting documentation
v2.0.1.4, v2.6.x	Two-step upgrade flow:	*Upgrade VxFlex OS to v3.0.x Guide*

Table 1. Upgrade support matrix (continued)

ScaleIO / VxFlex OS / PowerFlex base version	Upgrade path	Supporting documentation
	1. Upgrade to latest version of v3.0.1.x 2. Upgrade to v3.6.0.x	*Upgrade PowerFlex to v3.6 Guide*
v3.0, v3.0.0.x, v3.0.1.x, v3.5.x, v3.6, v3.6.0.1, v3.6.0.2, v3.6.0.3	Single-step upgrade from: • VxFlex OS v3.0/v3.0.0.x/v3.0.1.x • PowerFlex v3.5.x • PowerFlex v3.6/v3.6.0.1/v3.6.0.2/ v3.6.0.3 to PowerFlex v3.6.0.4	*Upgrade PowerFlex to v3.6 Guide*
v3.6.0.x	Two-step upgrade from: 1. Upgrade to latest version of v3.6.0.x 2. Upgrade to v4.0.x.	*Dell PowerFlex Install and Upgrade Guide*

Supported operating systems

The user documentation contains a list of supported operating systems. For the most current information, see the *Dell Simple Support Matrix* (ESSM) at https://elabnavigator.dell.com/eln/modernHomeSSM, **Storage** option.

SDC compatibility

By design, the PowerFlex SDC in this release is compatible with newer and previously released systems that are currently supported (for example, VxFlex OS v3.0.1). In addition, the PowerFlex system (backend) in this release is backward and forward compatible with previously released SDC versions that are currently supported.

Some limitations may apply to mixed-version systems, such as:

- Replication is not supported on SDC releases earlier than PowerFlex v3.5.
- The CHAP component authentication feature cannot be enabled unless every MDM, SDC and SDS in the system is at least version 3.5, and conversely, if the feature is enabled, no pre-v3.5 SDCs can be added.
- Fine Granularity Storage Pool-based volumes cannot be mapped to a ScaleIO/VxFlex OS v2.x SDC.
- Snapshot Policy management is not available with SDC releases earlier than VxFlex OS v3.0.

Known issues in 4.0.0.1

The Dell engineering team is working to resolve known issues.

Before getting started with PowerFlex and PowerFlex Manager, review the following known issues in version 4.0.0.1:

CLI

Issue ID	Symptom	Recommended action
SCI-67737	If the `generate_login_certificate scli` command is executed without parameters, it will fail.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.

MDM

Issue ID	Symptom	Recommended action
SCI-65059	IP addresses are not rebalanced when MDM disconnects from SDT.	If an SDT is completely unreachable, the SDT_NOT_CONNECTED_TO_HOST alert is raised. You can use connectivity monitoring utilities (such as ping) to monitor the network health.

SDS

Issue ID	Symptom	Recommended action
SCI-67744	All SDS devices get into an error state during and after server reboot. This can happen if you run NVME IO for a long time, then run `reboot -f now`.	Do not use `-f` with the `reboot now` command.
SCI-67359	If a device-at-fault event occurs while an SDS is in protected maintenance mode (PMM), some capacity may become un-writeable, resulting in IO errors until the PMM is aborted. The device-at-fault event occurs when a device is readable but not writeable.	Remove from PMM.
SCI-66916	IPv6 is not supported in version 4.0. Therefore, a non-disruptive upgrade from 3.x to v4.0 with IPv6 addresses is not supported. A non-disruptive upgrade from 3.x to v4.0 when SDS uses IPv6 address will fail in the query phase.	None.

SDC

Issue ID	Symptom	Recommended action
SCI-67107	An SDC has two IP addresses, but the **Block** tab only shows one.	Use the CLI to see other IP addresses.
SCI-66634	An error occurs during the loading of module binary /bin/emc/scaleio/scini.ko. This can happen if a machine restarts or a power outage occurs during a package upgrade. For example, this error might occur if an iDRAC power off is initiated during a zypper upgrade command, which updates packages including the PowerFlex SDC package (EMC-ScaleIO-sdc). This error occurs very infrequently.	Upgrade the SDC package again by using this command: `rpm -U <path_to_sdc_rpm>`

SDT

Issue ID	Symptom	Recommended action
SCI-67009	In some cases where the amount of hosts and active connections is close to utilizing the maximal amount of NVMe controllers permitted by the NVMe standard, adding a new SDT to the system may fail.	Disconnect one or more hosts from the system to allow the system to reclaim NVMe controller IDs. While addition of SDT is blocked, the system can still function properly and serve all currently connected hosts.
SCI-66722	It is possible to remove 2 SDTs that are the only paths of a volume. When removing 2 SDTs, after removing the first one, the MDM will replace it in the host connectivity goal with a port from the other SDT and will update all paths that used it with system-ports from other SDTs. (Some might get a path from the new port in the goal). When removing the second one, the MDM will do the same as in the first removal. However, now some mapping that got the first new port in the goal after the first removal might also get the second new port in the goal. Those mappings will stay without connection.	Dell Technologies recommends not removing or entering into maintenance mode a number of SDTs that is greater than or equal to the number of paths per volume, without reconnecting the hosts.
SCI-66645	Incorrect alert definition for SDT_MULTIPLE_DISCONNECTIONS_FROM_STORAGE_BACKEND_COMPONENTS. After bringing an interface down on an SDS, SDTs disconnect from it as expected and an alert is generated and processed by PowerFlex Manager. Upon recovery (NIC up), the events confirm that the SDTs and SDCs are connected back to that interface. However, this alert stays in the user interface. Sometimes, SDT alerts do not get properly cleared.	None.
SCI-66285	SDT performance drops approximately 30% when SDT processes span 2 numas. Deploying SDT and SDR on the same server can lead to reduced performance.	Avoid deploying SDT and SDR on the same server.
SCI-64187	NVMe namespace paths may change following disconnect and reconnect to PowerFlex NVMe Target, such that volumes may be used by different paths. Use by-id paths to maintain a stable path to devices across disconnect operations.	Use /dev/disk/by-id/<unique device path>.

PowerFlex Manager

Issue ID	Symptom	Recommended action
SCI-67737	If the `generate_login_certificate scli` command is executed without parameters, it will fail.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.
VXFM-29880	The block-legacy-gateway still shows under Target PowerFlex Gateway.	None.

Issue ID	Symptom	Recommended action
VXFM-29896	When reconfigure network after docker already running the install_PFMP.sh phase failed.	None.
VXFM-29892	Remove any text from UI which supports the statement that additional networks can be added for SO resource group.	None.
VXFM-29803	A service that was imported into version 3.8 of PowerFlex Manager cannot use **Update Resource Group Details** in version 4.0. This could happen if you perform a one-time migration on a 3.8 environment that includes a service that was originally imported through **Add Existing Service**. If you remove a node from the resource group after the migration, and then perform **Update Resource Group Details**, you might see a 500 exception.	Remove the resource group, and re-add it by using **Add Existing Resource Group**. This ensures that the resource group is recreated with the correct data for the nodes.

VXFM-29795	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the MDM primary node was changed, the certificate wasn't approved before. Therefore, the user isn't recognized and the import can't be done.	Change LIA LDAP authentication to LIA native authentication manually, as follows:
		1. In each PowerFlex node, go to `/opt/emc/scaleio/lia/cfg`, and re-edit the file `conf.txt` so it won't include LDAP fields at all and lia_auth_mode=NATIVE_AUTHENTICATION. Also, set lia_token=Scaleio123. Then close the file and save.
		2. Login to each PowerFlex node and reboot the LIA service, as follows:
		`[root@96N-15 ~]# cd /opt/emc/scaleio/lia/bin`
		`[root@96N-15 bin]# ./delete_service.sh`
		`Removed symlink /etc/systemd/system/multi-user.target.wants/lia.service.`
		`[root@96N-15 bin]# ./create_service.sh`
		`Created symlink from /etc/systemd/system/multi-user.target.wants/lia.service to /usr/lib/systemd/system/lia.service.`
		
		Importing the version 3.6 PowerFlex cluster to PowerFlex Manager will then work properly.

VXFM-29790	When importing a version 3.6 PowerFlex cluster to PowerFlex Manager, when the version 3.6 system is configured with only LDAP authentication for MDM, the LDAP user isn't recognized, so the import can't be done.	Change the authentication back to "native_and_ldap_authentication" (so, the admin user works) and use the following procedure:
		`[root@96N-15 ~]# echo "Reset Admin" > /opt/emc/scaleio/mdm/logs/MDM_SERVICE_MODE; scli --reset_admin --i_am_sure ;`
		`scli --login --username admin --password admin; scli --set_password --old_password admin --new_password Scaleio123`
		`Password reset successful`
		`Logged in. User role is SuperUser. System ID is 3de5919a425cd10f`

Issue ID	Symptom	Recommended action
		Current password must be changed. Please use --set_password command Password changed successfully [root@96N-15 ~]# scli --login --username admin --password Scaleio123 Logged in. User role is SuperUser. System ID is 3de5919a425cd10f
VXFM-29710	The old REST API for logging out does not work. If you logout with `GET https://<ip>/api/logout`, the API returns a 500 error code. In this release, you need to use both the "access_token" and the "refresh_token" you got when you performed the login. The old API uses only one token, so the API returns an error.	Logout with the new API: `POST https://<ip>/rest/auth/logout`
VXFM-29659	The PowerFlex Management Platform (PFMP) upgrade sometimes fails on the first attempt. This can happen if you attempt an upgrade immediately after enabling SupportAssist. The button to perform the upgrade is available in the user interface, but it should not be while SupportAssist configuration is still in progress.	Wait a few minutes and try the upgrade again.
VXFM-29656	The Notifier service fails to handle ALM events with "send_to_srs"=false due to a key error. This results in some low-level information being sent when it should not be.	None.
VXFM-29637	It takes PowerFlex Manager up to 40 minutes to fully load after reboot, crash, or power off.	None.
VXFM-29619	**Delete Resource Group** does not remove nodes from CloudLink Center in the case of resource groups with encryption. The nodes are still present, but they are in a **Disconnected** state.	None.
VXFM- 29602	The PowerFlex gateway still responds to the old alert query API, but this was deprecated. In older versions of Powerflex alerts were queried via the gateway using the following API: `/api/types/Alert/instances` In this release, the alerts are handled by the alert service and queried by using this API: `/rest/v1/alerts` The deprecated API should return an error, but instead it responds normally and returns incorrect data that does not include some alerts.	Do not use the deprecated API. Use the new API instead.
VXFM- 29600	The node deployment status on the **Resources** page is set to **Deployment Cancelled** state after a successful non-disruptive update. This is a temporary state and should be corrected in a few minutes.	Check again in a few minutes to see if the deployment status has been updated.

Issue ID	Symptom	Recommended action
VXFM-29579	A Security Admin user can perform activities defined for the Security Admin role, as well as activities allowed for the Lifecycle Admin role.	None.
VXFM-29576	Event and alert names are not case consistent. The names for events and alerts should follow the convention of upper snake case, with upper case words separated by underscores. However, some of the names use lower snake case, with lower case words separated by underscores.	If you are performing any type of third party integration, treat the names as case insensitive. This should be necessary only if you are parsing JSON files from the events and alerts system.
VXFM-29543	The management workflow to allow swapping primary and backup nodes has not been added to the restricted operations after a NAS server has entered into an out of service state. This situation might occur in the following scenario: 1. You make a NAS storage pool full. 2. The NAS server on the storage pool becomes read only. 3. You try to move the NAS server (this is a prohibited operation). 4. You try to perform a swap of the NAS server (this is successful). This issue has no visible impact on file service operations.	None.
VXFM-29530	PowerFlex Manager does not allow you to modify a notification source policy.	Create a new source and delete the old source.
VXFM-29501	An SDT in maintenance mode shows `Disconnected` in the SCLI output.	None.
VXFM-29500	PowerFlex Manager does not provide the ability to place an SDT in maintenance mode within the user interface.	None.
VXFM-29496	If you run the `reset_PFMP.sh` script to remove an old install and re-install PFMP, the script removes pre-requisite software rpms: • jq • nmap • skopeo The install fails if these RPMs are not there already.	Install the missing RPMs again.
VXFM-29469	Whenever an ALM event from PowerFlex is received, PowerFlex Manager sends a message when it is received, then again when the alert is acknowledged or disacknowledged, and again when the alert is cleared. It sends an event whenever the status changes.	Ignore the duplicate alert.
VXFM-29416	PowerFlex Manager does not allow you to modify a syslog policy. When you attempt to modify a syslog policy, the user interface reflects the change, but syslog still behaves the same.	Delete the existing syslog policy and create a new syslog policy instead of modifying the existing policy.

Issue ID	Symptom	Recommended action
VXFM-29404	A hyperconverged resource group fails if an NFS datastore is already present.	Remove the NFS datastore and retry the resource group.
VXFM-29386	The **Latency** graph on the dashboard does not show any data for NVMe/TCP.	None.
VXFM-29381	The **Summary** page of the **Initial Configuration** wizard remains static for 30 seconds while PowerFlex Manager is enabling SupportAssist.	None.
VXFM-29304	Some event and alert services cannot recover from a configuration or certificate change in the AMQ service. All services may lose connection to AMQ and reconnection attempts may fail.	Restart the pod of services that did not disconnect.
VXFM-29294	The dashboard IO performance numbers do not correlate with IO performance numbers reported by SCLI.	None.
VXFM-29291	On the **File** > **NAS Servers** or **File** > **File Systems** page, the performance details do not get updated on the side pane for a selected NAS server or file system.	Click somewhere else on the page and click back again to see the performance details.
VXFM-29272	A hyperconverged deployment fails in a partial network configuration when DNS is not configured in the hypervisor management network.	Configure DNS.
VXFM-29241	Registering SupportAssist from the **Initial Configuration Wizard** is not activating the **Default Policy Configuration** on the **Settings** page.	After you complete the initial configuration, go to the **Settings** page and verify that **SupportAssist** has been turned on. You may need to configure the SupportAssist policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.
VXFM-29234	When you create a login certificate and it later expires, an attempt to login with the expired certificate gives a misleading error message: **`scli --login --p12_path /root/.scli/admin_cert.p12 --p12_password password Error: Failed to connect to MDM 127.0.0.1:8611`** There should be some indication that the certificate is no longer viable and you need to create a new one.	None.
VXFM-29190	NVMe drive replacement fails with the error "Removal of the operating system failed". The error is confusing because it says it failed, but you can remove the drive without issue and the new drive is detected and can be added back to PowerFlex and CloudLink.	None.
VXFM-29174	The **Edit Management Software Upgrade Settings** page does not have a **Test Connection** button. However, when you click **Save**, PowerFlex Manager displays a message indicating whether the supporting connections required for the upgrade are valid.	None.
VXFM-29095	There is no way to delete a source or destination on the **Notification Policies** page.	Remove or disable all the policies involving this source or destination.

Issue ID	Symptom	Recommended action
VXFM-28986	PowerFlex Manager allows you to deploy 16 nodes for a PowerFlex File (NAS) compute-only deployment. This is not supported. The limit is 16 nodes. The user interface should block NAS deployments that exceed 16 nodes.	None.
VXFM-28983	When trying to add an additional node to a resource group, you may see the deployment fail with a 500 error, if the storage pool media type is set to "transitional". The "transitional" media type is not supported.	None.
VXFM-28907	On the **Protection** > **RCGs** page, while the RCG is in the failover state, the action **Resume (from pause)** should be disabled, because the RCG is not in pause mode but in switchover/failover state.	None.
VXFM-28844	On the **Generate Troubleshooting Bundle** page, the hover text for **Download locally** indicates that NFS is supported. However, NFS is not supported for the troubleshooting bundle. The user interface correctly shows the failure if you try to use NFS, but the hover text should not show NFS.	None.
VXFM-28829	If you click **Add Template** in the **Deploy Resources** section of the **Getting Started** page, the user interface shows the **Create** wizard as expected. After selecting **Create a new template**, entering a **Template Name**, and clicking **Next**, the **Create Template** page shows up correctly. After entering all fields and clicking **Save**, the user interface does not show the template details to allow further editing of the template, as it would if you clicked **Lifecycle** > **Templates** and then clicked **Create**. However, the template is created.	Navigate to **Lifecycle** > **Templates**. You will see the template in the list under **My Templates**. You can then click on the template name to edit the template as expected.
VXFM-28828	The **Compliance Versions** page is not refreshing to show the new state of the compliance file when **Allow Unsigned File** is selected.	Refresh the page or navigate away from the page and come back. The **Compliance Versions** page then shows the state correctly.
VXFM-28774	In some extreme bad path scenarios, when ActiveMQ is not available, some MDM events may not reach PowerFlex Manager.	While there is no way to recover those events into PowerFlex Manager, the MDM events are still available locally on the MDM node, by using show_events.
VXFM-28715	You are unable to save changes in the **Destination Protocol Settings (SupportAssist)** window if you attempt to change the connection type from **Connect via Gateway Server** to **Connect Directly**.	De-register before you try to change from one connection method to another.
VXFM-28693	The error message displayed in PowerFlex Manager is not clear when the logged in user tries to change their password.	None.
VXFM-28532	Resizing the service volumes (vCLS volumes) should be blocked by PowerFlex Manager. Currently, you are able to increase the vCLS volume size. After resizing the volumes, you may also see unexpected behaviors. For example, you may see a message saying that the service is in lifecycle mode because the vCLS volumes are on local storage.	None.

Issue ID	Symptom	Recommended action
VXFM-28433	When trying to export a template without an encryption password, you may see an error.	Create a password on the **Backup Settings** page and the export works as expected.
VXFM-28291	An ambiguous error appears during node expansion if a reverse DNS lookup for hostname fails to find an IP address. The **Summary** screen for adding the node shows: `An unexpected error occurred on the system. Retry the operation later.`	None.
VXFM-28288	A NilClass exception occurs if you deploy a resource group when you don't have a PowerFlex management network and you have general purpose LAN (GPL) networks that do not have a default gateway.	None.
VXFM-28281	PowerFlex Manager allows multiple instances of an iDRAC password update job to run on the same node at the same time.	None.
VXFM-28194	If there is a larger than expected time difference (over 1 hour) between PowerFlex Manager and the PowerFlex storage nodes, an upload license action may fail with the following error: `License Enforcement - Wrong Command Parameters`. This error is displayed on the **License Management** page in PowerFlex Manager. In the PowerFlex Manager log file, the following error appears: `"error_code":62,"error_message":"License Enforcement - Wrong Command Parameters."` In the primary MDM logs `/opt/emc/scaleio/mdm/logs/trc.z.XX`, the error displays as follows: `Error, license activation date in the future`	Align the time between the management platform and storage nodes and upload the license again.
VXFM-28034	If you create volumes on the **Block** page, volume creation is cancelled when you switch away from this page. Volume creation should be continued in background. Instead, volume creation is stopped and there is no progress or completion status when the **Block** page is opened again.	Wait for the operation to finish before navigating to another page.
VXFM-27967	The **View Details** page for Cisco switches shows a blank page and no metrics.	None.
VXFM-27411	Uploading a 4.0 license to an existing service that is still on 3.6 should show a more meaningful error message.	Apply the license through the CLI.
VXFM-27388	If you upload a certificate file you have uploaded before using the same name you have used before, PowerFlex Manager does not display an error message and overrides the existing certificate.	None.
VXFM-26570	An attempt to tear down a storage-only CloudLink resource group does not remove the machine groups from CloudLink Center.	None.

Issue ID	Symptom	Recommended action
VXFM-26550	If you run the command **"powerflex_cli events quwery --session_token 98329575cf390107 --limit 2 --order timestamp.asc"** with wrong command flag **"quwery"**, the command fails with error `"Error: unknown flag: --limit"`. This is the incorrect error message.	Run the command with the correct command flag: **"powerflex_cli events query --session_token 98329575cf390107 --limit 2 --order timestamp.asc"**
VXFM-26549	When using power_cli to query events with filter "severity in (MAJOR,MINOR)" it fails with a parsing error. This error occurs because the "in" operator does not exist in the CLI.	The same query can be achieved by running the query a few times, once for each value in the "in" clause.
VXFM-26476	When using the REST API to query event history, if there is an error in the filter field, the filter is ignored, and the unfiltered list is returned instead of an error message.	Correct your query and run it again.
VXFM-26333	When using the REST API to query alert templates, the filters are ignored, and the full list of alert templates is returned.	Forward the REST output to a file, and use file viewer and search tools to find the alert template you need.
VXFM-25895	PowerFlex Manager is unable to configure a Dell OS10 switch for deployment.	None.
VXFM-25284	While creating a snapshot rule, you need to provide the time of day. By default, the value populated on this field is current time in EST. However, internally, the snapshot rule is created with the value in UTC. This may lead to confusion as snapshots are not getting created as expected.	None.

Known issues and limitations from PowerFlex 3.6.x.x

The following table lists known issues and limitations that exist in the PowerFlex 3.6.x.x releases that also apply to this release.

NOTE: If an issue was reported by customers, the customers' Service Request numbers appear in the "Issue number & SR number" column, and serve to correlate between customer-reported issues and the issue number.

The following table lists known issues and limitations for PowerFlex Custom Node.

Issue number & SR number	Problem summary	Workaround
SCI-62961	Disk slot information for PowerEdge nodes introduced in PowerFlex 3.6 does not work on R650/R6525/R750 nodes with HBA355.	Use iDRAC or perccli to get the disk slot information.

The following table lists known issues and limitations for the PowerFlex Gateway.

Issue number & SR number	Problem summary	Workaround
SCI-57304	PowerFlex Installer (also known as the PowerFlex Gateway) is exposed to issue HSTS Missing From HTTPS Server, CVE: https://nvd.nist.gov/vuln/detail/CVE-2017-7789	None
SCI-55819	During NDU to v3.5.1.x, in rare cases, the PowerFlex Gateway might have high CPU usage and get stuck at the upload phase.	Perform a restart of the gateway service using an SSH session to the PowerFlex Gateway: service scaleio-gateway restart

Issue number & SR number	Problem summary	Workaround
		Then, in the PowerFlex Installer user interface, click Retry to continue the upgrade.

The following table lists known issues and limitations for the SDC.

Issue number & SR number	Problem summary	Workaround
SCI-57752	The SDC rpm package for SLES15.X installs without any issues. However, after installation, when checking if the SDC driver's scini service is loaded, the following confusing state is returned: service scini status service: no such service scini This is a false positive status alert.	None

The following table lists known issues and limitations for the SDR.

Issue number & SR number	Problem summary	Workaround
SCI-62047	A replication failover on an RCG that is still in the initial copy phase may not present all volume members at the remote target location.	Wait for all volumes in the RCG to complete their initial copy phase before performing the failover operation for that RCG.

The following table lists known issues and limitations for the SDS.

Issue number & SR number	Problem summary	Workaround
SCI-63216	When adding NVMe devices using the by-id path similar to the following: dev/disk/by-id/nvme-Dell_Express_Flash_NVMe_P4610_1.6TB_SFF_BTLN950106ZB1P6AGN the MDM blocks the addition due to the error: new_device_path too long, MDM supports up to 63 characters.	Use the alternative name path in by-id location with nvme-eui in the path. For example: /dev/disk/by-id/nvme-eui.01000000010000005cd2e448b0b75151

Known issues from PowerFlex Manager 3.8.1

The following table lists known issues from PowerFlex Manager 3.8.1 that also apply to this release.

Issue ID	Symptom	Recommended action
ESRCM-3148	Remediation is available for the Apache Log4j remote code execution vulnerability that could be exploited by malicious users to compromise the affected system. Dell recommends implementing this remediation as soon as possible in light of the critical severity of the vulnerability. The PowerFlex presentation server uses Log4j with version 2.14.1 and is impacted by this issue.	For details, see https://www.dell.com/support/kbdoc/enus/000194548/dsa-2021-272-dellpowerflex-securityupdate-for-apachelog4j-remote-codeexecutionvulnerabilitycve-2021-44228?ref=emcadvisory_000194548_Critical_null
VXFM-24115	A service status is displaying incorrectly on the **Services** page after performing service mode operation.	None.
VXFM-23822	An attempt to perform a service upgrade fails after a backup and restore from an earlier	Ensure that the state of your system resources does not change after a backup is performed.

Issue ID	Symptom	Recommended action
	release of PowerFlex Manager. This could be caused by a change in the state of the environment after the backup is performed. For example, this would happen if the hostname of the PowerFlex gateway changed after the backup was performed, since PowerFlex Manager restores the state of objects at the time of the backup, and any subsequent service operations would fail because of the mismatch in the gateway hostname.	The backup state must match the state of your production environment once the restore is performed.
VXFM-23802	The SDC nodes are not listed under **View details** on the **Resources** page for PowerFlex Gateway.	None.
VXFM-23775	Disruptive upgrade on the PowerFlex hyperconverged service fails, if upgrading from VMware ESXi 7.0.Ux to VMware ESXi 7.0 Ux. The PowerFlex hyperconverged nodes become unresponsive after deactivating the protection domain and shutting down all the VMs.	Use the non-disruptive upgrade.
VXFM-23756	After upgrading from PowerFlex Manager 3.4 to PowerFlex Manager 3.8, PowerFlex storage-only service is showing as **Healthy** instead of **Lifecycle Mode**.	You need to perform **Update Service Details** after adding the latest compatibility matrix.
VXFM-23672	The PowerFlex storage-only service shows as `Incomplete` with a banner message displaying `To add volume` even though two PowerFlex service volumes exist.	Volumes created outside of the service will not display within the service until inventory and **Update Service Details** is performed.
VXFM-23639	New OS10 switches are non-compliant with newer firmware versions as expected version in RCMs and Intelligent Catalogs may be lower and downgrades are not allowed.	None.
VXFM-23636	If you perform an update on a node in a service, you may not be able to update a second node in the service immediately after the first node is upgraded without manually running inventory on the PowerFlex gateway in PowerFlex Manager.	Run the inventory for the PowerFlex gateway and re-initiate the node upgrade.
VXFM-23630	Unable to initiate the scaleup of networks on a PowerFlex management controller 2.0 setup.	None.
VXFM-23231	If the server profile is incorrect, PowerFlex storage-only service scaleup volume completes with an **Error** deployment state.	1. Reapply the server profile so that it displays the model name correctly and retry the service. 2. Set the correct model name for PowerFlex Manager appliance when converting from PowerFlex Ready Node to PowerFlex appliance.
VXFM-23193	Editing a service with a new OS image is not reflected in the **Services** or **Compliance and OS Repositories**. New PowerFlex nodes deployed will reflect the new OS image. Only existing PowerFlex nodes upgraded will have this issue.	1. Confirm the current OS image, go to **Services** and confirm. 2. Re-adding the service will identify the latest OS image available and reflect the selected one during import.
VXFM-23136	Reserving virtual IP address comes from the **SDC role**, if the IP address range is small.	None.
VXFM-23132	In the **Resources** page, filtering resource type **PowerFlex Gateway/Presentation**	None.

Issue ID	Symptom	Recommended action
	Server(PS) using the corresponding service name displays `No Devices Found`.	
VXFM-23124	Multiple inventory jobs are causing service failures.	Retry the service.
VXFM-23066	Cannot search for volumes starting with **+** in the search text.	Do not use **+** in the search criteria.
VXFM-23061	In resource discovery, cannot select **Next** from **Assign Users** step in **Create Node Pool** wizard due to incorrect validation error.	None.
VXFM-22804	The **Filter by Groups** feature for Active Directory is not working.	None. Scroll through the list of users.
VXFM-22573	On a PowerFlex storage-only template, when changing to the **Add Networks to this port** option from **Mirror**, the selected networks are inversed.	Re-edit and the selected folders will be available.
VXFM-21849	The **Import Networks** feature is not working when adding an existing service with VMware NSX and VMware vCenter network folders.	Do not use the **Import Networks** feature while adding an existing service. Manually define the required networks in PowerFlex Manager - PowerFlex / hypervisor management and PowerFlex data networks. It is not necessary to define the general purpose LAN networks - they can be left blank.
VXFM-21559	During a deployment, if servers are discovered before switches, the service is returned in lifecycle mode.	Manually run inventory on the switches.
VXFM-21531	CloudLink Center SNMP settings are applied without user input and are not fully functional.	After completing a change configuration, save the configuration again without making any changes.
VXFM-21467	Deployment fails if overlapping data network subnets are used.	Set up the correct network subnets and redeploy. Ensure you have one interface per subnet and redeploy.
VXFM-21353	PowerFlex gateway upgrade does not update the compliance report on the PowerFlex nodes.	Rerun the inventory on all the PowerFlex nodes.
VXFM-21278	VMware vCLS volumes fail to migrate to the shared datastore if VMware ESXi nodes are added using an IP address, but resolve to a hostname in the DNS.	Add the PowerFlex nodes back into the vCSA under the resolved hostname.
VXFM-21256	The iDRAC service module is not in a running state during an NVMe replacement. The NVMe eject operation fails.	Restart the iDRAC network and restart the ISM service `dcism`.
VXFM-21233	The migration of vCLS VMs fails on a PowerFlex hyperconverged service if there is a mix of PowerFlex compute-only nodes and hyperconverged-only nodes in same VMware cluster but in different services that you are trying to upgrade. Migration is failing because one of the vCLS VMs belongs to the PowerFlex compute-only service, but it does not have access to the new utility datastores.	Run the VMware vCLS migration on the PowerFlex hyperconverged service. If that fails, go to PowerFlex compute-only service, and run the vCLS migration on that service.
VXFM-21232	Network scale up fails on a hyperconverged service due to port down when adding service.	1. Reseat the network cable to see if uplink can be restored.

Issue ID	Symptom	Recommended action
		2. Remove the service. 3. Readd the service using the **Add Existing Service** function. 4. Verify the network connectivity displays in the **Port View**.
VXFM-21231 and VXFM-21180	Partial network automation option is only intended to be used with LACP networking. However, there is no validation performed in the **Add Existing Service** wizard to prevent the use of partial network automation without LACP networking. Selecting this combination can result in various failures including: ● The service shows a blank **Teaming And Bonding Configuration** in the service's **View All Settings** dialog under **OS Settings**. ● Rebooting the PowerFlex Manager appliance will cause the service to go into an **Upgrade Service Components** error status. ● Service update operations run on the service fail. ● PowerFlex cluster upgrades fail.	Remove the service and add it again by using **Add Existing Service** with the **Full Network Automation** option. Only choose **Partial Network Automation** if your service is using LACP networking.
VXFM-21186	If you are upgrading a service, the service may be placed in lifecycle mode due to validation errors. The validation errors are displayed as a part of the reason. If the service prompts for vCLS migration also, you can perform vCLS migration on the service. In this case, the validation reasons are removed, and the banner no longer includes the reasons for lifecycle mode. You no longer see the reasons, but the service is in lifecycle mode and prevents any further actions.	None.
VXFM-21049	The service name displayed in the **Duplicate Node Summary** screen shows special characters in an HTML encoded format.	None.
VXFM-20957	During an import of Active Directory users, if you click **View Details**, PowerFlex Manager displays a broken page.	None.
VXFM-20944	Appliance and Alert Connector state shows different timezone and time stamp.	No functional impact, reboot the appliance and timezone and time stamp align.
VXFM-20940	Dell CloudLink Center deployment fails if you are using different subnet than PowerFlex Manager appliance.	Set the subnet manually.
VXFM-20868	If a service only has a single node, the **Add Existing Service** function works, but the overall service health shows error status once the service add finishes.	None.
VXFM-20877	PowerFlex Manager may show the wrong compliance status on nodes. PowerFlex Manager shows the nodes are compliant on the **Resources** page, even though the nodes are not compliant.	Run a manual inventory, and the correct compliance status displays.
VXFM-20816	Adding an OS Image with the same repository name does not return an error.	None.
VXFM-20735	VLAN 4094 through 4096 are reserved VLANs on the Dell PowerSwitch switches.	Do not use any of the reserved VLANs.

Issue ID	Symptom	Recommended action
VXFM-20686	VMware vSphere vMotion vmk is missing from the second node in an VMware ESXi compute-only node deployment.	Manually create the VMware vSphere vMotion vmk on the second node.
VXFM-20666	No warning displays when adding an existing encrypted service to a brownfield deployment.	Ensure an encryption tool or CloudLink is discovered first.
VXFM-20655	Drive replacement and MDM reconfiguration are allowed through PowerFlex Manager when a service is in an error state.	Do not attempt a drive replacement or an MDM reconfiguration if a service is in an error state.
VXFM-20593	PowerFlex Manager creates an Element Manager credential via REST API fails with wrong error message if `snmpCommunityString` is omitted.	None.
VXFM-20590	When scaling up of the network on a PowerFlex compute-only service, the new network does not show under the port view for one of the three nodes in the service.	None.
VXFM-20588	PowerFlex Manager displays **Update Resource** when current version is higher than expected version.	None.
VXFM-20571	PowerFlex Manager lets you add a static route to a subnet that already exists on the node. This fails as it is not needed - the node already has a route to that subnet.	None.
VXFM-20546	If existing port channels that do not have any interfaces are reused on Dell OS10 as part of the new service, any previous VLANs on those port channels are not cleared. If new port channels are created they will only have the correct VLANs.	None.
VXFM-20533	PowerFlex Manager fails to set the default compliance package version when you add a compliance file and select **Make this the default version for compliance checking**.	Click **Change** to reset the default version.
VXFM-20318	A PowerFlex hyperconverged service deployment using an existing VMware cluster with different storage pools fails.	Remove the nodes and add them back with the same storage pool.
VXFM-20215	In the teardown of a PowerFlex hyperconverged service where the datacenter, cluster, and network information are all moved into their own folders in VMware vCenter, the service resources are re-inventoried, and the service is removed. However, the VMware vCenter cluster and nodes are left and not removed.	Manually remove the vCenter cluster and nodes.
VXFM-20169	MDM reconfiguration fails for certain role transitions.	You can set the role manually, if needed. Retry or rescan refreshes the roles.
VXFM-20154	If you attempt to restart a failed job by clicking **Retry**, PowerFlex Manager displays a warning page with a **View Compliance Report** link. This link is broken. You may encounter this issue after a resize volume or migrate vCLS VMs operation fails.	Click **View Compliance Report** under **Service Actions**.
VXFM-19482	PowerFlex Manager fails to configure the server facing ports due to `switchport mode trunk` returning the following error:	Remove `swichport mode trunk` before running through PowerFlex Manager.

Issue ID	Symptom	Recommended action
	`% Error: MODE_L2HYBRID, Configured as a lag interface ToR-B(conf-if-eth1/1/3)#`	
VXFM-19010	The wizard for updating resources displays incorrect error messages when trying to update the PowerFlex gateway or presentation server resources from the **Services** page.	Go to the management VM service and click **Update resource**.
VXFM-18857	During an **Add Volume** job, a service will go from the **In Progress** state to the **Completed** state, then suddenly return to **In Progress**. While the job is still **In Progress**, some of the service actions may be enabled.	Allow the job to finish before you try to use any of the service actions.
VXFM-18580	Clusters already in services on the appliance are provided as selection options when importing an existing service.	None.
VXFM-18554	During a PowerFlex hyperconverged replication service deployment, retry does not set the VMware VM version. This can cause the deployment to fail when trying to add NVDIMM devices to the VM.	Delete the VM and retry the service.
VXFM-18167	Adding an existing service wizard network mapping shows `%2f` instead of `/` in port group name.	Unsupported special characters cause a rendering issue with the port group name and should not be used. Spaces, underscores, and hyphens are allowed.
VXFM-18005	Retrying a failed PowerFlex upgrade fails to cancel previous upgrade job.	If you start the upgrade, and request PowerFlex Manager to start another upgrade, it fails. If you see this failure, you may must cancel the previous upgrade.
VXFM-17760	Service sees old NIC cabling data resulting in failure.	Contact Dell Support for a workaround.
VXFM-17610	The DHCP setting in PowerFlex Manager does not allow you to disable the DHCP/PXE node if the virtual appliance is not currently on the subnet.	Do not change your network while you are working on it. If you must do this, at least disable DHCP first.
VXFM-17023	Unable to use NetApp NFS share from PowerFlex Manager.	PowerFlex Manager does not support NFS shares set up this way.
VXFM-16623	A node discovery may appear to fail with the following log message: `Failed to collect the system configuration for server`	Check for the following conditions: • Networking issues between iDRAC and PowerFlex Manager • The server is booted into the BIOS screen. In this case, jobs cannot be run.
VXFM-16261	During a restore, the PowerFlex Manager appliance gets stuck at **Starting Restoration** stage when a filename is not provided, that is, if only the path is specified.	Use the full path to the restore file.
VXFM-15653	Download of the ONIE firmware upgrade package fails, but the failure is not displayed in the logs.	None.
ESRCM-2526	VMware ESXi 7.0 requires a security patch that is available in VMware ESXi 7.0 Update 1.	For VMware ESXi 7.0, you must perform a manual workaround. For details, see https://kb.vmware.com/s/article/76372.

Issue ID	Symptom	Recommended action
VXFM-15308	PowerFlex Manager collects switch performance metrics only if NXAPI is listening on both HTTPS (Port 443) and HTTP (Port 80). If either of these ports are disabled, switch performance metrics are not collected.	Be sure that ports for both HTTP and HTTPS are enabled for Cisco switches.
VXFM-15227	PowerFlex Manager allows you to publish and deploy a CloudLink template containing a password with all lowercase characters.	Be sure to include at least one uppercase letter in the password.
VXFM-15267	If you try to upload an SSL certificate on the **Appliance SSL Certificates** section of the **Settings** page, you may encounter security errors with the Certificate Signing Request (CSR). The CSR does not include the Subject Alternative Name (SAN) field. It only includes the Common Name (CN) field. This problem occurs only with the Chrome browser.	Contact Dell Support for a workaround.
VXFM-15188	When deploying a service, you may see an error message stating that a static IP address could not be reserved. The error may include empty brackets ([ ]). If you see this error, the IP addresses may not be pingable.	Check your IP addresses to be sure you have enough pingable IP addresses, and try again.
VXFM-15110	You may see incorrect warning messages when using invalid characters for a hostname in a template. If you try to publish the template, you may see a message indicating that a hostname exceeds the limit of 15 characters. The message should indicate that the hostname has invalid characters.	Be sure to use valid characters and conform to the character limit for the operating system. For Windows, the limit is 15 characters for a hostname. For ESXi and Linux, the limit is 26 characters.
VXFM-15069	A node cannot be removed from a storage-only service because of a failure removing the switch. If the switch removal fails, PowerFlex Manager leaves the node in the service.	Update the service details or remove the node manually from the service.
VXFM-15010	Adding replication networks to a hyperconverged template breaks the service.	None.
VXFM-14898	During drive replacement, if you type **CONTINUE WITHOUT DRIVE** with a bad drive, PowerFlex Manager tries to add it back and fails.	None.
VXFM-14872	You may notice a type casing inconsistency, when creating templates with uppercase host and SVM names. VMware ESXi uses lowercase host names. There is no impact to functionality.	None.
VXFM-14773	If you try to download two RCMs simultaneously, the operation fails.	Resynchronize the RCMs or download them again separately.
VXFM-14771	If you deploy a service with only one node in it, the service may show **Critical**. This condition is caused by a warning at the component level.	None.
VXFM-14770	The inventory is not running on the PowerFlex gateway after you use service mode. If you place a node in service mode, then exit service mode, then try to enter service mode again, the PowerFlex gateway is left in an abnormal state.	Manually run the inventory or wait for the next hourly update.

Issue ID	Symptom	Recommended action
VXFM-14767	PowerFlex Manager does not refresh text boxes in the **Discovery Wizard**. This display problem can happen if you select **Node** as the **Resource Type** and choose **Hostname**, then change the **Resource Type** to any other type other than **Node**.	Click the trash can icon to remove the resource, then add it again.
VXFM-14763	Mellanox ConnectX-4 LC firmware must be on 14.25.80.00 or later to upgrade to 14.27.10.16 or later. This problem is not a PowerFlex Manager issue. PowerFlex Manager does not support step upgrades.	None.
VXFM-14753	If an upgrade of the PowerFlex Manager virtual appliance fails, you may see duplicate email messages and alerts.	None.
VXFM-14724	If you try to delete an RCM that is used on a service, the operation fails with a 500 internal server error.	Remove the service that previously used the RCM, delete the RCM, and add the services back as an existing service.
VXFM-14581	When you add an existing hyperconverged or storage-only service that includes a CloudLink Center VM in a cluster, the **Inventory Summary** page does not show the **Adding** icon for the CloudLink Center VM.	None.
VXFM-14535 and VXFM-15309	If you change the password for an SVM, **View All Settings** on the **Services** page still shows the old password. This issue also breaks **Update Service Details** if you try to use a new credential.	If you need to change the OS credential to a newly-defined credential and update to use that credential on a service, you have to remove the service and readd it to select that credential.
VXFM-14496 and VXFM-14254	After upgrading from an earlier version of PowerFlex Manager, you may notice that the inventory might not be updated correctly or that subsequent operations such as **Add Existing Service** may fail.	Run a manual inventory after the upgrade.
VXFM-14494 and VXFM-14762	After changing an iDRAC password, if you select the same credentials again, you may see an error message in the **Update Password** wizard. After you see the error, you cannot proceed with the wizard.	Cancel and restart the wizard.
VXFM-14470	PowerFlex Manager does not perform validation to ensure that you have a minimum of three fault sets for a protection domain and storage pool.	Ensure that your fault set configuration conforms to best practices. Dell recommends that all nodes in a protection domain should be in the same storage pool.
VXFM-14397	PowerFlex Manager displays the wrong progress bar during the resizing of a datastore.	None.
VXFM-14377	After you remove a node from a service, PowerFlex Manager may not update the PowerFlex gateway inventory.	Run a manual inventory after removing the node.
VXFM-14028	The **Services** page does not show lifecycle mode as the deployment state for NSXT services that are in lifecycle mode.	None.
VXFM-13994	If a Dell CloudLink Center is non-compliant, PowerFlex Manager disables node selection for upgrade.	Upgrade the CloudLink Center from the **Resources** page and try the upgrade from the **Services** page.

Issue ID	Symptom	Recommended action
VXFM-13744	An RCM upgrade to a major version of NSX and ESXi may fail with a VIB error. This failure could be caused by a VIB incompatibility if you are using a nonstandard Dell image. Be aware that NSX VIBs are specific to the version of ESXi that is installed on the host. If you upgrade ESXi, you must install new NSX VIBs appropriate for the new ESXi version.	Install new NSX VIBs for the new ESXi version. For more information, see Upgrade to ESXi 6.5 or 6.7 in an NSX Environment.
VXFM-12990	The state of a discovery job may show as **Running** on the **Jobs** page, even though it has completed.	Restart the PowerFlex Manager virtual appliance.
VXFM-12477	On the **Resources** page, if you select a machine with self-encrypting drives (SEDs) for a CloudLink Center, you might see two entries for each drive. One entry shows the status as "encrypted" with a type of "data_sed". The other entry shows the status as "unencrypted" with a type of "data_sds". The SED drives are encrypted, but PowerFlex Manager should not show duplicate entries.	None.
VXFM-12402	After deleting a node from a storage-only service, you may see an incorrect error message in **Recent Activity**. The message indicates a failure removing the Storage Data Client (SDC), even though an SDC is not installed on a storage-only node.	None.
VXFM-12391	If you put an SDS into Protected Maintenance Mode (PMM) manually, and then run inventory for the PowerFlex gateway within PowerFlex Manager, the service does not exit service mode. When you click Exit Service Mode, PowerFlex Manager shows an error. This error occurs because PowerFlex Manager is trying to exit Instant Maintenance Mode (IMM) instead of PMM.	Manually take the SDS out of PMM outside of PowerFlex Manager.
VXFM-12366	If you cancel a service deployment, and then retry the deployment later, the service succeeds. However, the warning banner that is displayed while PowerFlex Manager is canceling the deployment remains during and after the successful deployment.	Remove the service and add it back again.
VXFM-12096	The deployment for a storage-only service may sometimes fail to start.	Retry the service deployment.
VXFM-12074	On the **View Details** page for CloudLink, PowerFlex Manager shows a hyperlink for the **IP Address** field for the SDS. The hyperlink should not be displayed for an SDS IP address.	None.
VXFM-11998	PowerFlex Manager does not always show the new status when you add an operating system image on the **OS Image Repositories** tab.	Manually refresh the screen to see the new status.
VXFM-11977	If you delete a node that was not deployed successfully in a service, PowerFlex Manager might show the node as deployed.	To remove the failed node from the service, choose the **Remove Resource** option, rather than the **Delete Resource** option.
VXFM-11930	After upgrading from version 3.3 or 3.4, you might notice that a service that has a fault set enabled shows **Deployed** as the **Deployment**	Click **Update Service Details** to show the correct state.

Issue ID	Symptom	Recommended action
	State, even though it should show **Lifecycle Mode**.	
VXFM-11760	After updating the firmware and software on a node, or during a service mode operation, PowerFlex Manager might show the node as being in service mode, even though it is no longer in service mode. Due to a time lag, the service mode status that is shown on the service does not always match the state of the system exactly. The node eventually comes out-of-service mode, and the update completes successfully.	None.
VXFM-11758	A compute-only service with ESXi displays the **Protected Maintenance Mode** option for a firmware and software update on a PowerFlex 3.5 cluster. This type of service offers only compute resources, and does not have an SDS component. The **Protected Maintenance Mode** option should not be available. The update succeeds even if you choose this option.	None.
VXFM-11705	If you attempt to discover a Cisco switch with terminal color configured, the discovery fails.	Disable the terminal color option by running `configure no terminal color persist`.
VXFM-11242	If you click **Update Service Details**, PowerFlex Manager might display an incorrect error indicating that the `device could not be connected through SSH`. The message should state that the iDRAC was not responding, most likely because of bad cabling.	None.
VXFM-10663	PowerFlex Manager has only one firmware component id set for all OS10 switches in PowerFlex Manager. This means that PowerFlex Manager can only include one set of OS10 switch versions for all models of OS10 switch in a given RCM package. This is a problem for management switches.	Do not discover the management switches.
VXFM-3727	If you try to create a network containing IP addresses that are in use by vCenter or the PowerFlex gateway inventory, `Invalid IP address range is entered` errors are thrown. An extra error message states: `The entered IP range overlaps with the IP ranges configured on one or more configured networks`. This situation only happens if you have multiple networks with the same subnet. For example, this situation might happen if you have an SDC only network and a network that supports both SDS and SDC traffic.	Remove your vCenter or gateway from the inventory, create the network ranges, and then rediscover them.
VXFM-3517	If the PowerFlex gateway is deployed onto an operating system that does not have `systemctl` (such as SuSE 11 or Red Hat Enterprise Linux 6), the PowerFlex configuration script fails to restart the gateway after configuring it. However, it does not return any error, so ultimately the deployment fails when trying to talk to the gateway REST API. This configuration is unsupported, but PowerFlex	None.

Issue ID	Symptom	Recommended action
	Manager does not block the configuration or provide a clear error message.	

Operating system known issues and limitations

This section lists known issues and limitations for specific operating systems.

Issue number & SR number	Problem summary	Workaround
VXFV-360	The device name appears incorrectly as a non-PCI device instead of iDRAC, but the current and baseline versions are accurate (6.00.30.201 to 6.00.30.202, respectively).	Dell recommends upgrading the OME catalog to the latest version.
	PowerFlex nodes running CPLD 1.0.7 stop responding during Power On Self-Test (POST) when upgrading to certain RCM or IC packages that include BIOS 1.8.2 and iDRAC 6.00.30.202.	Before upgrading to the latest version of PowerFlex, ensure that the PowerFlex nodes are running CPLD 1.0.8 or later. For details, see https://www.dell.com/support/kbdoc/en-us/000205925.
VXFV-296	Mixing the Mellanox CX-6 100GbE NIC driver with the Broadcom PCIe/OCP 25GbE NIC driver is not supported.	Remove the `bnxt_en` driver and install the Mellanox driver.
VXFV-283	iDRAC support for "Prepare for Removal" on PowerEdge R650/R6525/R750/ nodes fails.	NVMe hot-swap on Linux operating systems is supported. Skip the Prepare for Removal step in iDRAC.
VXFV-265	During execution of the command `create-namespace` with 4k alignment on RHEL 8.4, the following error is encountered: Warning: namespace0.0 unable to validate alignment	This is a false notification. The command result is as expected, and the resulting DAX device can be used with PowerFlex acceleration devices.
VXFV-243	Upgrade of a RedHat-based operating system to RHEL 8.3 fails if the node contains NVDIMMs. This is due to a known issue on RHEL 8.3: https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/8/html-single/8.3_release_notes/index#known-issue_installer-and-image-creation	Follow the procedures described in "RHEL/CentOS upgrade to version 8.3" in the .
VXFV-224	In a RHEL 8.x-based node using Cloudlink encrypted SDS devices, after reboot the device moves to failed state. **NOTE:** This issue is much rarer on RHEL7.x-based nodes, but it might occur.	1. SSH to the relevant SDS's operating system. 2. Delete the SDS service. 3. Clear the device errors using the GUI or CLI. 4. Recreate the SDS service. 5. Verify that devices come back online. Rebuild/Rebalance will occur if instant maintenance mode or protected maintenance mode was not used. 6. If instant maintenance mode or protected maintenance mode was used, exit maintenance mode.
VXFV-157	NVMe disks are not supported on RHEL 8.0 due to the following issue: https://access.redhat.com/solutions/4280341 Failure to observe this limitation might cause RHEL-based SDSs with NVMes to experience kernel panic during a disk removal flow.	None

Fixed issues in 4.0.0.1

Fixed issues are defects that exist in the previous release but are resolved in the 4.0.0.1 release of PowerFlex and PowerFlex Manager.

Issue ID	Symptom
SCI-68411	After a backend upgrade, VMware reports corruption when creating snapshots.
SCI-68172	SDC fails to see mapped volumes.
SCI-68129	Replication is stuck during exit from disaster recovery.
SCI-67894	Data integrity issue occurs during replication functional operations.
SCI-67628	Data integrity issue occurs during disaster recovery scenario.
SDNAS-15181	NAS panicked and memory leaks occur in SSH server.
SDNAS-15057	Rolling panics in RSA_sign happen after upgrade.
VXFM-29826	Block upgrade/NDU is failing from Flex_RCM_3_7_0_0_r31 to Flex_RCM_3_7_0_1_r2.
VXFM-29756	Online help drop has been included for 4.0.0.1 hot fix.
VXFM-29706	There are remaining active paths after unmapping NVMe/TCP device for ESXi.
VXFM-29690	SDNAS does not represent SDS or devices, whereas CLI is able to access everything.
VXFM-29660	After restoring to a new cluster, the gateway inventory is failing. In addition, after creating a volume, the consumed capacity is not reflected on the License Management page.
VXFM-30017	Adding an LDAP user when the display name (=cn) is "test test" and the sAMAccountname is "test.test" does not work.

Software packages

Learn about the contents of the PowerFlex 4.0 software packages.

Download location

The PowerFlex complete software package, containing most of the packages listed below, can be downloaded from the following location:

Dell Download Center

The RCM for PowerFlex rack can be downloaded from the following location:

Dell Download Center

Packages

Package name	Size	Contents
PowerFlex Manager bundle		PFMP2-4.0.0.1-169.tgz
RCM for PowerFlex rack	25GB	Flex_RCM_3_7_0_1_r6.zip
Intelligent catalog for PowerFlex appliance	17GB	Intelligent_Catalog_40_370_01_r5.zip
Software-only ZIP	~6.5GB	Software_Only_Complete_4.0.0.1_103.zip. Includes the following: • PowerFlex_4.0.100.103_Complete_Core_SW.zip

Package name	Size	Contents
		• PowerFlex_4.0.100.103_SDCs_for_manual_install .zip (SDC RPMs) • PowerFlex_Software_4.0.0.1_103_r1.zip (software-only catalog) • cm-20220823.gpg (compatibility file)
PowerFlex Manager OVA ZIP		PowerFlex_Manager_4.0.0.1-169.zip: • PFMP2-4.0.0.1-169.ova (PFMP_Installer for installing Kubernetes cluster) (22GB) • pfmp-k8s-20220712.ova (Kubernetes OVA running SUSE 15 SP3) (3.5GB)

The original OS layout table from 3.6 matches this file: PowerFlex_4.0.100.103_Complete_Core_SW.zip

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PowerFlex version 4.0

What's new in release 4.0

PowerFlex release 4.0 introduces the following new features:

Feature	Details
New PowerFlex management and orchestration stack for all consumption models	PowerFlex Manager now provides a unified management and orchestration (M&O) interface for Dell PowerFlex rack, Dell PowerFlex appliance, VxFlex Ready Node, PowerFlex custom node, and PowerFlex software only. The management and orchestration stack includes these features: ● An improved user experience with enhanced manageability, scalability, and resilience ● Role and scope-based access control ● Common PowerAPIs to improve ecosystem and customer integration ● PowerFlex storage backend (VMware vSphere independent): a new unified M&O solution that replaces the PowerFlex Gateway, presentation server, and legacy PowerFlex Manager management solutions ● Full lifecycle mode (LCM) services applied based on the deployment model, covering the following: ○ PowerFlex core software deployment and upgrade ○ PowerFlex NAS software upgrade ○ Operating system update and patching for managed nodes (PowerFlex rack or PowerFlex appliance) ○ Firmware and driver updates for managed nodes (PowerFlex rack or PowerFlex appliance) ○ Switch updates (PowerFlex rack or PowerFlex appliance) ● Deployment and expansion for all product offerings ● Single sign-on (SSO) to the system with unified user and role management ● Call Home capability with additional CloudIQ based offering ● Unified maintenance and services experience ● Block services and file services volume operations ● NVMe/TCP support ● Support for Management Data Store (MDS) takeover
Unified management UI	PowerFlex Manager includes a unified management user interface. This interface provides PowerFlex block and NAS file capabilities, along with the life cycle and resource management capabilities offered in earlier releases of PowerFlex Manager. The unified management interface introduces a new set of tabs: ● The **Dashboard** tab allows you to see a summary of performance and inventory data. ● The **Block** tab provides a user interface for managing PowerFlex block storage. ● The **File** tab provides a user interface that incorporates NAS capabilities for managing file storage. ● The **Protection** tab provides a user interface for performing PowerFlex storage protection operations. ● The **Lifecycle** tab allows you to create and clone templates and deploy them as resource groups (previously known as services). ● The **Resources** tab allows you to manage all resource types within the system. ● The **Monitoring** tab allows you to track events, alerts, and jobs running.

Feature	Details
	• The **Settings** tab allows you to configure system settings for PowerFlex Manager. The **Settings** tab has been updated and reorganized. In addition, some of the settings from previous releases that are no longer needed have been removed and a few new settings have been added for security and serviceability. **NOTE:** The presentation server is no longer supported in this release.
File services (NAS)	PowerFlex introduces File service (NAS) support. This support includes: • Tightly integrated file services (NFS/SMB/FTP) to supplement existing block storage ○ A unified infrastructure solution providing block, file and HCI with a common interface set (UI and API) ○ Enabling customer efficiency and manageability for lower TCO • Storage services in PowerFlex • Support for NAS deployments for PowerFlex rack and PowerFlex appliance ○ Ability to deploy new NAS resource groups and import existing NAS resource groups for PowerFlex rack and PowerFlex appliance. This enables the capabilities of the **File** menu. ○ Deployment of compute-only resource groups for NAS. Before deploying a NAS compute-only resource group, you must first have a PowerFlex cluster deployed with a hyperconverged or storage-only resource group that uses the block legacy PowerFlex gateway. ○ New template settings. ○ New **PowerFlex File** sample template, which can be cloned to generate a template that has the correct settings. PowerFlex Manager configures the required network on the node and top-of-rack (access) server-facing ports, as well as the required network on the control plane. When you deploy a NAS template, PowerFlex Manager automatically creates control volumes for the deployment. • Support for new NAS deployments for software management ○ Ability to deploy new NAS resource groups for software management environments. PowerFlex Manager does not support importing existing deployments for software management NAS compute-only environments. ○ New template settings, as well as support for resource group operations such as node expansion, node removal, and maintenance mode. ○ New **PowerFlex File - SW Only** sample template, which can be cloned to generate a template that has the correct settings. • Support for NAS service operations for PowerFlex rack and PowerFlex appliance ○ Ability to expand the number of nodes, remove nodes from the resource group, and put nodes in maintenance mode. If necessary, you can also remove the resource group as a whole. ○ Ability to expand the cluster up to a maximum of 16 nodes. PowerFlex Manager requires a minimum of two nodes in a NAS cluster. ○ Ability to remove NAS nodes from the resource group, but not actually delete them. When you remove a resource, PowerFlex Manager removes the deployment information, but does not actually make configuration changes to the resource itself. You can remove a NAS resource group, which only removes the resource group from the appliance. The cluster still exists and can be added back to the appliance. • Support for NAS serviceability ○ ESE integration for NAS alerts ○ SNMP support for NAS ○ Backup ○ Troubleshooting bundle ○ CloudIQ integration

Feature	Details
NVMe/TCP connectivity	PowerFlex 4.0 introduces support for NVMe/TCP storage protocol for front end connectivity, enabling customers to use NVMe/TCP hosts in addition to the existing SDC. **NOTE:** This is a new protocol and fixes by the OS vendors are being released regularly. Refer to the release notes for known OS issues that PowerFlex has found and is working with the various OS vendors to resolve. If you are planning to implement NVMe-oF with PowerFlex, contact your account team for guidance on the best OS and solution architecture to fit your needs. NVMe/TCP connectivity includes: ● Addition of NVMe/TCP as a frontend storage protocol ○ Native OS connectivity, for customers who like to avoid the use of storage agent on the host. ○ Standardized host connectivity and compatibility across operating systems, including VMware ESXi. **NOTE:** Current SDC (agent-based) access is still supported. ● Support for NVMe/TCP deployments ○ Ability to add NVMe/TCP connectivity to storage-only resource groups. PowerFlex hyperconverged and PowerFlex compute-only are not supported in this release. For compute-only, NVMe/TCP initiators are not automatically configured by PowerFlex Manager in this release. You must configure NVMe/TCP initiators manually. ○ Changes to the network labels within the template. ○ Ability to specify whether the Storage Data Target (SDT) package needs to be installed. ● Support for adding the SDT to a resource group ○ Ability to add the SDT to existing nodes in a storage-only resource group. ● Support for software management deployments with NVMe/TCP ○ End-to-end support for software management deployments that use NVMe/TCP. ○ Template creation, deployment, expansion of nodes, networks, as well as the ability to remove or delete resources or the resource group as a whole. ● Ability to perform non-disruptive updates from PowerFlex version 4.0 to a later release with NVMe/TCP ○ Support for non-disruptive updates from version 4.0 of PowerFlex to a later release with NVMe/TCP enabled. NVMe initiator support is available on ESXi 7.0U3f. Linux support is currently in tech preview status. **NOTE:** Refer to the following support matrix source: https://elabnavigator.dell.com/vault/pdf/Dell_NVMe_TCP_Host_Storage_ESSM.pdf?key=1682953508124.
MDM connection availability (MDM proxy)	In PowerFlex 4.0 SDC to MDM resiliency has been added. In PowerFlex 3.6 path resilience capability was introduced for the SDS (for SDC to SDS connection - also known as SDS proxy).
New license support	PowerFlex 4.0 transitions to an XML-based license for both new systems and upgraded systems. In PowerFlex Manager, you can now upload a Management Data Store (MDS) license and a single production license for the whole PowerFlex system. You can also upload other software licenses (for CloudLink, for example). No license is required for the first 90 days of use. During this period, you are running PowerFlex in trial mode, and all features are enabled. You must deploy the MDM cluster before uploading a PowerFlex license. You must discover an MDS Gateway before uploading an MDS license.

Feature	Details
Data protection	Multiple remote replication systems ● Arbitrary topology of up to five systems ● Target volume auto provisioning
Larger storage node	SDS maximum capacity raised to 160TB on medium granularity data layout ● SSD maximum raised to 15.36 TB (SSD drives) ● Allows higher density systems with larger capacity storage nodes (SDS) ● Initially limited to medium granularity data layout (MG) storage pools only (no expansion)
Serviceability	PowerFlex Manager provides support through integration with the Secure Connect Gateway ensuring better alignment with Dell Technologies services initiatives to enhance the user experience. Registering your device with SupportAssist through the Secure Connect Gateway, enables you to do the following: ● Collect system state information and telemetry. ● Automate issue detection and support case creation. ● Resolve common issue remotely. ● Provide critical information to Dell Technologies remote support to resolve complex issues.
System stability	The MDM proxy ability augments the network stability. In the PowerFlex architecture, each SDC is required to be fully connected to the MDM for control and management of actions. If the SDC is unable to reach the MDM, the SDC will move to the disconnected state, which at some point could lead to complete I/O failure. PowerFlex allows clusters that don't use the MDM virtual IP capability to use the secondary MDMs as a proxy for all control traffic, allowing more redundancy in case of network failures. When such an event occurs, if the MDM proxy is in use, the management APIs raise an appropriate alert.
Support for deployments with PowerFlex 4.0	PowerFlex Manager supports the deployment of resource groups with PowerFlex 4.0 hyperconverged, compute-only, and storage-only nodes. VMware hyperconverged is only supported on the PowerFlex rack and PowerFlex appliance offerings.
Ability to import existing resource groups with PowerFlex 4.0	PowerFlex Manager allows you to import PowerFlex 4.0 hyperconverged, compute-only, and storage-only resource groups.
One time migration of existing PowerFlex environments to PowerFlex Manager 4.0	PowerFlex rack and PowerFlex appliance customers running PowerFlex 3.6.x with or without PowerFlex Manager can use the **Initial Configuration Wizard** to perform a one-time migration to version 4.0. Customers with PowerFlex 3.5.x can upgrade to 3.6.x, and then use the **Initial Configuration Wizard** to perform a one-time migration to PowerFlex Manager 4.0. Customers with PowerFlex Manager must be running PowerFlex Manager 3.8 in order to perform the migration. Customers that do not have an existing PowerFlex environment have the option to deploy a new instance of PowerFlex. In the case where a customer has an existing PowerFlex environment with PowerFlex Manager, the migration process backs up and restores data from the existing PowerFlex Manager 3.8 environment. All existing services will then appear on the new **Resource Groups** page. The migrated resource groups are initially non-compliant, because PowerFlex Manager 4.0 is running a later compliance file (RCM or IC) that includes PowerFlex 4.0. These resource groups must be upgraded to the latest compliance version before they can be expanded or managed with automation operations.

Feature	Details
	In the case where a customer has an existing PowerFlex environment but does not have PowerFlex Manager, the customer must add an existing resource group to import the existing PowerFlex environment. These imported resource groups must also be upgraded to the latest compliance version before other operations may be performed.
Ability to perform updates to PowerFlex 4.0	PowerFlex Manager allows you to perform non-disruptive updates from 3.6.x.x releases of PowerFlex to version 4.0 of PowerFlex. You can upgrade hyperconverged, compute-only, and storage-only nodes. PowerFlex Manager allows you to perform a non-disruptive update from version 4.0 of PowerFlex to a later release with NVMe/TCP enabled. This support applies to storage-only resource groups. PowerFlex Manager also allows you to perform a non-disruptive update on a NAS cluster as part of a PowerFlex upgrade. PowerFlex Manager does not support disruptive updates for NAS.
Increased node scalability	PowerFlex Manager has increased the limits for node scalability and greatly improved the user interface performance for templates and resource groups. For storage-only, PowerFlex Manager supports up to 32 nodes per resource group, with a maximum of 128 nodes in a protection domain. For compute-only Linux, PowerFlex Manager supports up to 32 nodes per resource group. For hyperconverged or compute-only VMware ESXi, PowerFlex Manager supports up to 32 nodes per resource group, with a maximum of 128 nodes in a protection domain. For software management, PowerFlex Manager supports up to 128 nodes per resource group, with a maximum of 128 per protection domain, up to 512 total for four protection domains.
Updated troubleshooting bundle service	The troubleshooting bundle includes access switch logs, VMware ESXi logs, iDRAC syslogs, CloudLink Center logs, management logs, and NAS logs. PowerFlex Manager collects logs from the **Resource Groups** page, and includes deployment logs, as well as the resources used in the resource group. The troubleshooting bundle also includes standard output logs from all pods and Kubernetes logs about pods, services, deployments, secrets, drivers, and volumes.
PowerFlex Manager support for software management deployment	Software management enables the installation and configuration of PowerFlex on a set of nodes where an existing operating system is installed and networking is configured. For a list of operating systems that PowerFlex Manager supports, see the *Dell PowerFlex v4.0 Support Matrix*.
Support for automated upgrades of PowerFlex Manager 4.0	PowerFlex Manager provides a built-in mechanism for upgrading the 4.0 management software to a later release. You can upgrade the software by using Dell SupportAssist or by specifying a local repository path.
Documentation	For more information about this release, see the following documents: ● *PowerFlex Technical Overview* ● *PowerFlex Administration Guide* ● *PowerFlex Manager Online Help* ● *PowerFlex CLI Guide* ● *PowerFlex* API on the Dell Technologies Developer portal

Support deprecation

This section lists features that are no longer supported in PowerFlex 4.0.x.

RFCache

The RFCache feature was removed. Any customer using RFCache with the PowerFlex 3.6.x.x release must first remove the RFCache configuration and packages prior to initiating an upgrade.

VASA support

VASA support is not available in PowerFlex 4.0. VASA support is a future roadmap item.

Customers needing VASA support should continue using PowerFlex 3.6.x.x with the latest patch.

Install and upgrade

You can install and configure PowerFlex version 4.0.x, or you can upgrade from PowerFlex version 3.6.x to version 4.0.x.

PowerFlex Manager allows you to use the **Initial Configuration Wizard** to perform an upgrade to version 4.0.x. Customers with PowerFlex 3.5.x can upgrade to 3.6.x, and then use the **Initial Configuration Wizard** to perform an upgrade to PowerFlex Manager 4.0.x.

Customers with PowerFlex Manager must be running PowerFlex Manager 3.8 in order to perform the upgrade.

Customers that do not have an existing PowerFlex environment have the option to deploy a new instance of PowerFlex.

See the *Dell PowerFlex Install and Upgrade Guide* for more information.

Licensing

PowerFlex use requires a license. Please call your Dell account representative to purchase a license for PowerFlex.

Upgrade

The following table lists the versions from which the upgrade can be performed:

Table 2. Upgrade support matrix

ScaleIO / VxFlex OS / PowerFlex base version	Upgrade path	Supporting documentation
v2.0.1.4, v2.6.x	Two-step upgrade flow: 1. Upgrade to latest version of v3.0.1.x 2. Upgrade to v3.6.0.x	*Upgrade VxFlex OS to v3.0.x Guide* *Upgrade PowerFlex to v3.6 Guide*
v3.0, v3.0.0.x, v3.0.1.x, v3.5.x, v3.6, v3.6.0.1, v3.6.0.2, v3.6.0.3	Single-step upgrade from: • VxFlex OS v3.0/v3.0.0.x/v3.0.1.x • PowerFlex v3.5.x • PowerFlex v3.6/v3.6.0.1/v3.6.0.2/ v3.6.0.3 to PowerFlex v3.6.0.4	*Upgrade PowerFlex to v3.6 Guide*
v3.6.0.x	Two-step upgrade from: 1. Upgrade to latest version of v3.6.0.x 2. Upgrade to v4.0.x.	*Dell PowerFlex Install and Upgrade Guide*

Supported operating systems

The user documentation contains a list of supported operating systems. For the most current information, see the *Dell Simple Support Matrix* (ESSM) at https://elabnavigator.dell.com/eln/modernHomeSSM, **Storage** option.

SDC compatibility

By design, the PowerFlex SDC in this release is compatible with newer and previously released systems that are currently supported (for example, VxFlex OS v3.0.1). In addition, the PowerFlex system (backend) in this release is backward and forward compatible with previously released SDC versions that are currently supported.

Some limitations may apply to mixed-version systems, such as:

- Replication is not supported on SDC releases earlier than PowerFlex v3.5.
- The CHAP component authentication feature cannot be enabled unless every MDM, SDC and SDS in the system is at least version 3.5, and conversely, if the feature is enabled, no pre-v3.5 SDCs can be added.
- Fine Granularity Storage Pool-based volumes cannot be mapped to a ScaleIO/VxFlex OS v2.x SDC.
- Snapshot Policy management is not available with SDC releases earlier than VxFlex OS v3.0.

Known issues in 4.0

The Dell engineering team is working to resolve known issues.

Before getting started with PowerFlex and PowerFlex Manager, review the following known issues in version 4.0:

CLI

Issue ID	Symptom	Recommended action
SCI-67737	If the `generate_login_certificate scli` command is executed without parameters, it will fail.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.

MDM

Issue ID	Symptom	Recommended action
SCI-65059	IP addresses are not rebalanced when MDM disconnects from SDT.	If an SDT is completely unreachable, the SDT_NOT_CONNECTED_TO_HOST alert is raised. You can use connectivity monitoring utilities (such as ping) to monitor the network health.

SDS

Issue ID	Symptom	Recommended action
SCI-67744	All SDS devices get into an error state during and after server reboot. This can happen if you run NVME IO for a long time, then run `reboot -f now`.	Do not use `-f` with the `reboot now` command.
SCI-67359	If a device-at-fault event occurs while an SDS is in protected maintenance mode (PMM), some capacity may become un-writeable, resulting in IO errors until the PMM is aborted. The device-at-fault event occurs when a device is readable but not writeable.	Remove from PMM.
SCI-66916	IPv6 is not supported in version 4.0. Therefore, a non-disruptive upgrade from 3.x to v4.0 with IPv6 addresses is not supported. A non-disruptive upgrade from 3.x to v4.0 when SDS uses IPv6 address will fail in the query phase.	None.

SDC

Issue ID	Symptom	Recommended action
SCI-67107	An SDC has two IP addresses, but the **Block** tab only shows one.	Use the CLI to see other IP addresses.
SCI-66634	An error occurs during the loading of module binary /bin/emc/scaleio/scini.ko. This can happen if a machine restarts or a power outage occurs during a package upgrade. For example, this error might occur if an iDRAC power off is initiated during a zypper upgrade command, which updates packages including the PowerFlex SDC package (EMC-ScaleIO-sdc). This error occurs very infrequently.	Upgrade the SDC package again by using this command: `rpm -U <path_to_sdc_rpm>`

SDT

Issue ID	Symptom	Recommended action
SCI-67009	In some cases where the amount of hosts and active connections is close to utilizing the maximal amount of NVMe controllers permitted by the NVMe standard, adding a new SDT to the system may fail.	Disconnect one or more hosts from the system to allow the system to reclaim NVMe controller IDs. While addition of SDT is blocked, the system can still function properly and serve all currently connected hosts.
SCI-66722	It is possible to remove 2 SDTs that are the only paths of a volume. When removing 2 SDTs, after removing the first one, the MDM will replace it in the host connectivity goal with a port from the other SDT and will update all paths that used it with system-ports from other SDTs. (Some might get a path from the new port in	Dell Technologies recommends not removing or entering into maintenance mode a number of SDTs that is greater than or equal to the number of paths per volume, without reconnecting the hosts.

Issue ID	Symptom	Recommended action
	the goal). When removing the second one, the MDM will do the same as in the first removal. However, now some mapping that got the first new port in the goal after the first removal might also get the second new port in the goal. Those mappings will stay without connection.	
SCI-66645	Incorrect alert definition for SDT_MULTIPLE_DISCONNECTIONS_FROM_STORAGE_BACKEND_COMPONENTS. After bringing an interface down on an SDS, SDTs disconnect from it as expected and an alert is generated and processed by PowerFlex Manager. Upon recovery (NIC up), the events confirm that the SDTs and SDCs are connected back to that interface. However, this alert stays in the user interface. Sometimes, SDT alerts do not get properly cleared.	None.
SCI-66285	SDT performance drops approximately 30% when SDT processes span 2 numas. Deploying SDT and SDR on the same server can lead to reduced performance.	Avoid deploying SDT and SDR on the same server.
SCI-64187	NVMe namespace paths may change following disconnect and reconnect to PowerFlex NVMe Target, such that volumes may be used by different paths. Use by-id paths to maintain a stable path to devices across disconnect operations.	Use /dev/disk/by-id/<unique device path>.

PowerFlex Manager

Issue ID	Symptom	Recommended action
SCI-67737	If the `generate_login_certificate scli` command is executed without parameters, it will fail.	Run the command with a single known parameter (`scli --generate_login_certificate --username` for example) or use the full help request `scli --help --generate_login_certificate`.
VXFM-29803	A service that was imported into version 3.8 of PowerFlex Manager cannot use **Update Resource Group Details** in version 4.0. This could happen if you perform a one-time migration on a 3.8 environment that includes a service that was originally imported through **Add Existing Service**. If you remove a node from the resource group after the migration, and then perform **Update Resource Group Details**, you might see a 500 exception.	Remove the resource group, and re-add it by using **Add Existing Resource Group**. This ensures that the resource group is recreated with the correct data for the nodes.
VXFM-29710	The old REST API for logging out does not work. If you logout with `GET https://<ip>/api/logout`, the API returns a 500 error code. In this release, you need to use both the "access_token" and the "refresh_token" you got when you performed the login. The old API uses only one token, so the API returns an error.	Logout with the new API: `POST https://<ip>/rest/auth/logout`

Issue ID	Symptom	Recommended action
VXFM-29659	The PowerFlex Management Platform (PFMP) upgrade sometimes fails on the first attempt. This can happen if you attempt an upgrade immediately after enabling SupportAssist. The button to perform the upgrade is available in the user interface, but it should not be while SupportAssist configuration is still in progress.	Wait a few minutes and try the upgrade again.
VXFM-29656	The Notifier service fails to handle ALM events with "send_to_srs"=false due to a key error. This results in some low-level information being sent when it should not be.	None.
VXFM-29637	It takes PowerFlex Manager up to 40 minutes to fully load after reboot, crash, or power off.	None.
VXFM-29619	**Delete Resource Group** does not remove nodes from CloudLink Center in the case of resource groups with encryption. The nodes are still present, but they are in a **Disconnected** state.	None.
VXFM- 29602	The PowerFlex gateway still responds to the old alert query API, but this was deprecated. In older versions of Powerflex alerts were queried via the gateway using the following API: `/api/types/Alert/instances` In this release, the alerts are handled by the alert service and queried by using this API: `/rest/v1/alerts` The deprecated API should return an error, but instead it responds normally and returns incorrect data that does not include some alerts.	Do not use the deprecated API. Use the new API instead.
VXFM- 29600	The node deployment status on the **Resources** page is set to **Deployment Cancelled** state after a successful non-disruptive update. This is a temporary state and should be corrected in a few minutes.	Check again in a few minutes to see if the deployment status has been updated.
VXFM-29579	A Security Admin user can perform activities defined for the Security Admin role, as well as activities allowed for the Lifecycle Admin role.	None.
VXFM-29576	Event and alert names are not case consistent. The names for events and alerts should follow the convention of upper snake case, with upper case words separated by underscores. However, some of the names use lower snake case, with lower case words separated by underscores.	If you are performing any type of third party integration, treat the names as case insensitive. This should be necessary only if you are parsing JSON files from the events and alerts system.
VXFM-29543	The management workflow to allow swapping primary and backup nodes has not been added to the restricted operations after a NAS server has entered into an out of service state. This situation might occur in the following scenario: 1. You make a NAS storage pool full. 2. The NAS server on the storage pool becomes read only.	None.

Issue ID	Symptom	Recommended action
	3. You try to move the NAS server (this is a prohibited operation). 4. You try to perform a swap of the NAS server (this is successful). This issue has no visible impact on file service operations.	
VXFM-29530	PowerFlex Manager does not allow you to modify a notification source policy.	Create a new source and delete the old source.
VXFM-29501	An SDT in maintenance mode shows `Disconnected` in the SCLI output.	None.
VXFM-29500	PowerFlex Manager does not provide the ability to place an SDT in maintenance mode within the user interface.	None.
VXFM-29496	If you run the `reset_PFMP.sh` script to remove an old install and re-install PFMP, the script removes pre-requisite software rpms: • jq • nmap • skopeo The install fails if these RPMs are not there already.	Install the missing RPMs again.
VXFM-29469	Whenever an ALM event from PowerFlex is received, PowerFlex Manager sends a message when it is received, then again when the alert is acknowledged or disacknowledged, and again when the alert is cleared. It sends an event whenever the status changes.	Ignore the duplicate alert.
VXFM-29416	PowerFlex Manager does not allow you to modify a syslog policy. When you attempt to modify a syslog policy, the user interface reflects the change, but syslog still behaves the same.	Delete the existing syslog policy and create a new syslog policy instead of modifying the existing policy.
VXFM-29404	A hyperconverged resource group fails if an NFS datastore is already present.	Remove the NFS datastore and retry the resource group.
VXFM-29386	The **Latency** graph on the dashboard does not show any data for NVMe/TCP.	None.
VXFM-29381	The **Summary** page of the **Initial Configuration** wizard remains static for 30 seconds while PowerFlex Manager is enabling SupportAssist.	None.
VXFM-29304	Some event and alert services cannot recover from a configuration or certificate change in the AMQ service. All services may lose connection to AMQ and reconnection attempts may fail.	Restart the pod of services that did not disconnect.
VXFM-29294	The dashboard IO performance numbers do not correlate with IO performance numbers reported by SCLI.	None.
VXFM-29291	On the **File** > **NAS Servers** or **File** > **File Systems** page, the performance details do not get updated on the side pane for a selected NAS server or file system.	Click somewhere else on the page and click back again to see the performance details.

Issue ID	Symptom	Recommended action
VXFM-29272	A hyperconverged deployment fails in a partial network configuration when DNS is not configured in the hypervisor management network.	Configure DNS.
VXFM-29241	Registering SupportAssist from the **Initial Configuration Wizard** is not activating the **Default Policy Configuration** on the **Settings** page.	After you complete the initial configuration, go to the **Settings** page and verify that **SupportAssist** has been turned on. You may need to configure the SupportAssist policy explicitly on the **Settings** > **Events and Alerts** > **Notification Policies** page.
VXFM-29234	When you create a login certificate and it later expires, an attempt to login with the expired certificate gives a misleading error message: **`scli --login --p12_path /root/.scli/admin_cert.p12 --p12_password password Error: Failed to connect to MDM 127.0.0.1:8611`** There should be some indication that the certificate is no longer viable and you need to create a new one.	None.
VXFM-29190	NVMe drive replacement fails with the error "Removal of the operating system failed". The error is confusing because it says it failed, but you can remove the drive without issue and the new drive is detected and can be added back to PowerFlex and CloudLink.	None.
VXFM-29174	The **Edit Management Software Upgrade Settings** page does not have a **Test Connection** button. However, when you click **Save**, PowerFlex Manager displays a message indicating whether the supporting connections required for the upgrade are valid.	None.
VXFM-29095	There is no way to delete a source or destination on the **Notification Policies** page.	Remove or disable all the policies involving this source or destination.
VXFM-28986	PowerFlex Manager allows you to deploy 16 nodes for a PowerFlex File (NAS) compute-only deployment. This is not supported. The limit is 16 nodes. The user interface should block NAS deployments that exceed 16 nodes.	None.
VXFM-28983	When trying to add an additional node to a resource group, you may see the deployment fail with a 500 error, if the storage pool media type is set to "transitional". The "transitional" media type is not supported.	None.
VXFM-28907	On the **Protection** > **RCGs** page, while the RCG is in the failover state, the action **Resume (from pause)** should be disabled, because the RCG is not in pause mode but in switchover/failover state.	None.
VXFM-28844	On the **Generate Troubleshooting Bundle** page, the hover text for **Download locally** indicates that NFS is supported. However, NFS is not supported for the troubleshooting bundle. The user interface correctly shows the failure if	None.

Issue ID	Symptom	Recommended action
	you try to use NFS, but the hover text should not show NFS.	
VXFM-28829	If you click **Add Template** in the **Deploy Resources** section of the **Getting Started** page, the user interface shows the **Create** wizard as expected. After selecting **Create a new template**, entering a **Template Name**, and clicking **Next**, the **Create Template** page shows up correctly. After entering all fields and clicking **Save**, the user interface does not show the template details to allow further editing of the template, as it would if you clicked **Lifecycle** > **Templates** and then clicked **Create**. However, the template is created.	Navigate to **Lifecycle** > **Templates**. You will see the template in the list under **My Templates**. You can then click on the template name to edit the template as expected.
VXFM-28828	The **Compliance Versions** page is not refreshing to show the new state of the compliance file when **Allow Unsigned File** is selected.	Refresh the page or navigate away from the page and come back. The **Compliance Versions** page then shows the state correctly.
VXFM-28774	In some extreme bad path scenarios, when ActiveMQ is not available, some MDM events may not reach PowerFlex Manager.	While there is no way to recover those events into PowerFlex Manager, the MDM events are still available locally on the MDM node, by using show_events.
VXFM-28715	You are unable to save changes in the **Destination Protocol Settings (SupportAssist)** window if you attempt to change the connection type from **Connect via Gateway Server** to **Connect Directly**.	De-register before you try to change from one connection method to another.
VXFM-28693	The error message displayed in PowerFlex Manager is not clear when the logged in user tries to change their password.	None.
VXFM-28532	Resizing the service volumes (vCLS volumes) should be blocked by PowerFlex Manager. Currently, you are able to increase the vCLS volume size. After resizing the volumes, you may also see unexpected behaviors. For example, you may see a message saying that the service is in lifecycle mode because the vCLS volumes are on local storage.	None.
VXFM-28433	When trying to export a template without an encryption password, you may see an error.	Create a password on the **Backup Settings** page and the export works as expected.
VXFM-28291	An ambiguous error appears during node expansion if a reverse DNS lookup for hostname fails to find an IP address. The **Summary** screen for adding the node shows: `An unexpected error occurred on the system. Retry the operation later.`	None.
VXFM-28288	A NilClass exception occurs if you deploy a resource group when you don't have a PowerFlex management network and you have general purpose LAN (GPL) networks that do not have a default gateway.	None.
VXFM-28281	PowerFlex Manager allows multiple instances of an iDRAC password update job to run on the same node at the same time.	None.
VXFM-28194	If there is a larger than expected time difference (over 1 hour) between PowerFlex Manager	Align the time between the management platform and storage nodes and upload the license again.

Issue ID	Symptom	Recommended action
	and the PowerFlex storage nodes, an upload license action may fail with the following error: `License Enforcement - Wrong Command Parameters`. This error is displayed on the **License Management** page in PowerFlex Manager. In the PowerFlex Manager log file, the following error appears: `"error_code":62,"error_message":"License Enforcement - Wrong Command Parameters."` In the primary MDM logs `/opt/emc/scaleio/mdm/logs/trc.z.XX`, the error displays as follows: `Error, license activation date in the future`	
VXFM-28034	If you create volumes on the **Block** page, volume creation is cancelled when you switch away from this page. Volume creation should be continued in background. Instead, volume creation is stopped and there is no progress or completion status when the **Block** page is opened again.	Wait for the operation to finish before navigating to another page.
VXFM-27967	The **View Details** page for Cisco switches shows a blank page and no metrics.	None.
VXFM-27411	Uploading a 4.0 license to an existing service that is still on 3.6 should show a more meaningful error message.	Apply the license through the CLI.
VXFM-27388	If you upload a certificate file you have uploaded before using the same name you have used before, PowerFlex Manager does not display an error message and overrides the existing certificate.	None.
VXFM-26570	An attempt to tear down a storage-only CloudLink resource group does not remove the machine groups from CloudLink Center.	None.
VXFM-26550	If you run the command **`"powerflex_cli events quwery --session_token 98329575cf390107 --limit 2 --order timestamp.asc"`** with wrong command flag **`"quwery"`**, the command fails with error `"Error: unknown flag: --limit"`. This is the incorrect error message.	Run the command with the correct command flag: **`"powerflex_cli events query --session_token 98329575cf390107 --limit 2 --order timestamp.asc"`**
VXFM-26549	When using power_cli to query events with filter "severity in (MAJOR,MINOR)" it fails with a parsing error. This error occurs because the "in" operator does not exist in the CLI.	The same query can be achieved by running the query a few times, once for each value in the "in" clause.
VXFM-26476	When using the REST API to query event history, if there is an error in the filter field, the filter is ignored, and the unfiltered list is returned instead of an error message.	Correct your query and run it again.
VXFM-26333	When using the REST API to query alert templates, the filters are ignored, and the full list of alert templates is returned.	Forward the REST output to a file, and use file viewer and search tools to find the alert template you need.

Issue ID	Symptom	Recommended action
VXFM-25895	PowerFlex Manager is unable to configure a Dell OS10 switch for deployment.	None.
VXFM-25284	While creating a snapshot rule, you need to provide the time of day. By default, the value populated on this field is current time in EST. However, internally, the snapshot rule is created with the value in UTC. This may lead to confusion as snapshots are not getting created as expected.	None.

Known issues and limitations from PowerFlex 3.6.x.x

The following table lists known issues and limitations that exist in the PowerFlex 3.6.x.x releases that also apply to this release.

NOTE: If an issue was reported by customers, the customers' Service Request numbers appear in the "Issue number & SR number" column, and serve to correlate between customer-reported issues and the issue number.

The following table lists known issues and limitations for PowerFlex Custom Node.

Issue number & SR number	Problem summary	Workaround
SCI-62961	Disk slot information for PowerEdge nodes introduced in PowerFlex 3.6 does not work on R650/R6525/R750 nodes with HBA355.	Use iDRAC or perccli to get the disk slot information.

The following table lists known issues and limitations for the PowerFlex Gateway.

Issue number & SR number	Problem summary	Workaround
SCI-57304	PowerFlex Installer (also known as the PowerFlex Gateway) is exposed to issue HSTS Missing From HTTPS Server, CVE: https://nvd.nist.gov/vuln/detail/CVE-2017-7789	None
SCI-55819	During NDU to v3.5.1.x, in rare cases, the PowerFlex Gateway might have high CPU usage and get stuck at the upload phase.	Perform a restart of the gateway service using an SSH session to the PowerFlex Gateway: service scaleio-gateway restart Then, in the PowerFlex Installer user interface, click Retry to continue the upgrade.

The following table lists known issues and limitations for the SDC.

Issue number & SR number	Problem summary	Workaround
SCI-57752	The SDC rpm package for SLES15.X installs without any issues. However, after installation, when checking if the SDC driver's scini service is loaded, the following confusing state is returned: service scini status service: no such service scini This is a false positive status alert.	None

The following table lists known issues and limitations for the SDR.

Issue number & SR number	Problem summary	Workaround
SCI-62047	A replication failover on an RCG that is still in the initial copy phase may not present all volume members at the remote target location.	Wait for all volumes in the RCG to complete their initial copy phase before performing the failover operation for that RCG.

The following table lists known issues and limitations for the SDS.

Issue number & SR number	Problem summary	Workaround
SCI-63216	When adding NVMe devices using the by-id path similar to the following: dev/disk/by-id/nvme-Dell_Express_Flash_NVMe_P4610_1.6TB_SFF_BTLN950106ZB1P6AGN the MDM blocks the addition due to the error: new_device_path too long, MDM supports up to 63 characters.	Use the alternative name path in by-id location with nvme-eui in the path. For example: /dev/disk/by-id/nvme-eui.01000000010000005cd2e448b0b75151

Known issues from PowerFlex Manager 3.8.1

The following table lists known issues from PowerFlex Manager 3.8.1 that also apply to this release.

Issue ID	Symptom	Recommended action
ESRCM-3148	Remediation is available for the Apache Log4j remote code execution vulnerability that could be exploited by malicious users to compromise the affected system. Dell recommends implementing this remediation as soon as possible in light of the critical severity of the vulnerability. The PowerFlex presentation server uses Log4j with version 2.14.1 and is impacted by this issue.	For details, see https://www.dell.com/support/kbdoc/enus/000194548/dsa-2021-272-dellpowerflex-securityupdate-for-apachelog4j-remote-codeexecutionvulnerabilitycve-2021-44228?ref=emcadvisory_000194548_Critical_null
VXFM-24115	A service status is displaying incorrectly on the **Services** page after performing service mode operation.	None.
VXFM-23822	An attempt to perform a service upgrade fails after a backup and restore from an earlier release of PowerFlex Manager. This could be caused by a change in the state of the environment after the backup is performed. For example, this would happen if the hostname of the PowerFlex gateway changed after the backup was performed, since PowerFlex Manager restores the state of objects at the time of the backup, and any subsequent service operations would fail because of the mismatch in the gateway hostname.	Ensure that the state of your system resources does not change after a backup is performed. The backup state must match the state of your production environment once the restore is performed.
VXFM-23802	The SDC nodes are not listed under **View details** on the **Resources** page for PowerFlex Gateway.	None.
VXFM-23775	Disruptive upgrade on the PowerFlex hyperconverged service fails, if upgrading from VMware ESXi 7.0.Ux to VMware ESXi 7.0 Ux. The PowerFlex hyperconverged nodes become unresponsive after deactivating the protection domain and shutting down all the VMs.	Use the non-disruptive upgrade.
VXFM-23756	After upgrading from PowerFlex Manager 3.4 to PowerFlex Manager 3.8, PowerFlex storage-	You need to perform **Update Service Details** after adding the latest compatibility matrix.

Issue ID	Symptom	Recommended action
	only service is showing as **Healthy** instead of **Lifecycle Mode**.	
VXFM-23672	The PowerFlex storage-only service shows as `Incomplete` with a banner message displaying `To add volume` even though two PowerFlex service volumes exist.	Volumes created outside of the service will not display within the service until inventory and **Update Service Details** is performed.
VXFM-23639	New OS10 switches are non-compliant with newer firmware versions as expected version in RCMs and Intelligent Catalogs may be lower and downgrades are not allowed.	None.
VXFM-23636	If you perform an update on a node in a service, you may not be able to update a second node in the service immediately after the first node is upgraded without manually running inventory on the PowerFlex gateway in PowerFlex Manager.	Run the inventory for the PowerFlex gateway and re-initiate the node upgrade.
VXFM-23630	Unable to initiate the scaleup of networks on a PowerFlex management controller 2.0 setup.	None.
VXFM-23231	If the server profile is incorrect, PowerFlex storage-only service scaleup volume completes with an **Error** deployment state.	1. Reapply the server profile so that it displays the model name correctly and retry the service. 2. Set the correct model name for PowerFlex Manager appliance when converting from PowerFlex Ready Node to PowerFlex appliance.
VXFM-23193	Editing a service with a new OS image is not reflected in the **Services** or **Compliance and OS Repositories**. New PowerFlex nodes deployed will reflect the new OS image. Only existing PowerFlex nodes upgraded will have this issue.	1. Confirm the current OS image, go to **Services** and confirm. 2. Re-adding the service will identify the latest OS image available and reflect the selected one during import.
VXFM-23136	Reserving virtual IP address comes from the **SDC role**, if the IP address range is small.	None.
VXFM-23132	In the **Resources** page, filtering resource type **PowerFlex Gateway/Presentation Server(PS)** using the corresponding service name displays `No Devices Found`.	None.
VXFM-23124	Multiple inventory jobs are causing service failures.	Retry the service.
VXFM-23066	Cannot search for volumes starting with **+** in the search text.	Do not use **+** in the search criteria.
VXFM-23061	In resource discovery, cannot select **Next** from **Assign Users** step in **Create Node Pool** wizard due to incorrect validation error.	None.
VXFM-22804	The **Filter by Groups** feature for Active Directory is not working.	None. Scroll through the list of users.
VXFM-22573	On a PowerFlex storage-only template, when changing to the **Add Networks to this port** option from **Mirror**, the selected networks are inversed.	Re-edit and the selected folders will be available.
VXFM-21849	The **Import Networks** feature is not working when adding an existing service with VMware NSX and VMware vCenter network folders.	Do not use the **Import Networks** feature while adding an existing service. Manually define the required networks in PowerFlex Manager

Issue ID	Symptom	Recommended action
		- PowerFlex / hypervisor management and PowerFlex data networks. It is not necessary to define the general purpose LAN networks - they can be left blank.
VXFM-21559	During a deployment, if servers are discovered before switches, the service is returned in lifecycle mode.	Manually run inventory on the switches.
VXFM-21531	CloudLink Center SNMP settings are applied without user input and are not fully functional.	After completing a change configuration, save the configuration again without making any changes.
VXFM-21467	Deployment fails if overlapping data network subnets are used.	Set up the correct network subnets and redeploy. Ensure you have one interface per subnet and redeploy.
VXFM-21353	PowerFlex gateway upgrade does not update the compliance report on the PowerFlex nodes.	Rerun the inventory on all the PowerFlex nodes.
VXFM-21278	VMware vCLS volumes fail to migrate to the shared datastore if VMware ESXi nodes are added using an IP address, but resolve to a hostname in the DNS.	Add the PowerFlex nodes back into the vCSA under the resolved hostname.
VXFM-21256	The iDRAC service module is not in a running state during an NVMe replacement. The NVMe eject operation fails.	Restart the iDRAC network and restart the ISM service `dcism`.
VXFM-21233	The migration of vCLS VMs fails on a PowerFlex hyperconverged service if there is a mix of PowerFlex compute-only nodes and hyperconverged-only nodes in same VMware cluster but in different services that you are trying to upgrade. Migration is failing because one of the vCLS VMs belongs to the PowerFlex compute-only service, but it does not have access to the new utility datastores.	Run the VMware vCLS migration on the PowerFlex hyperconverged service. If that fails, go to PowerFlex compute-only service, and run the vCLS migration on that service.
VXFM-21232	Network scale up fails on a hyperconverged service due to port down when adding service.	1. Reseat the network cable to see if uplink can be restored. 2. Remove the service. 3. Readd the service using the **Add Existing Service** function. 4. Verify the network connectivity displays in the **Port View**.
VXFM-21231 and VXFM-21180	Partial network automation option is only intended to be used with LACP networking. However, there is no validation performed in the **Add Existing Service** wizard to prevent the use of partial network automation without LACP networking. Selecting this combination can result in various failures including: ● The service shows a blank **Teaming And Bonding Configuration** in the service's **View All Settings** dialog under **OS Settings**. ● Rebooting the PowerFlex Manager appliance will cause the service to go into an **Upgrade Service Components** error status. ● Service update operations run on the service fail.	Remove the service and add it again by using **Add Existing Service** with the **Full Network Automation** option. Only choose **Partial Network Automation** if your service is using LACP networking.

Issue ID	Symptom	Recommended action
	• PowerFlex cluster upgrades fail.	
VXFM-21186	If you are upgrading a service, the service may be placed in lifecycle mode due to validation errors. The validation errors are displayed as a part of the reason. If the service prompts for vCLS migration also, you can perform vCLS migration on the service. In this case, the validation reasons are removed, and the banner no longer includes the reasons for lifecycle mode. You no longer see the reasons, but the service is in lifecycle mode and prevents any further actions.	None.
VXFM-21049	The service name displayed in the **Duplicate Node Summary** screen shows special characters in an HTML encoded format.	None.
VXFM-20957	During an import of Active Directory users, if you click **View Details**, PowerFlex Manager displays a broken page.	None.
VXFM-20944	Appliance and Alert Connector state shows different timezone and time stamp.	No functional impact, reboot the appliance and timezone and time stamp align.
VXFM-20940	Dell CloudLink Center deployment fails if you are using different subnet than PowerFlex Manager appliance.	Set the subnet manually.
VXFM-20868	If a service only has a single node, the **Add Existing Service** function works, but the overall service health shows error status once the service add finishes.	None.
VXFM-20877	PowerFlex Manager may show the wrong compliance status on nodes. PowerFlex Manager shows the nodes are compliant on the **Resources** page, even though the nodes are not compliant.	Run a manual inventory, and the correct compliance status displays.
VXFM-20816	Adding an OS Image with the same repository name does not return an error.	None.
VXFM-20735	VLAN 4094 through 4096 are reserved VLANs on the Dell PowerSwitch switches.	Do not use any of the reserved VLANs.
VXFM-20686	VMware vSphere vMotion vmk is missing from the second node in an VMware ESXi compute-only node deployment.	Manually create the VMware vSphere vMotion vmk on the second node.
VXFM-20666	No warning displays when adding an existing encrypted service to a brownfield deployment.	Ensure an encryption tool or CloudLink is discovered first.
VXFM-20655	Drive replacement and MDM reconfiguration are allowed through PowerFlex Manager when a service is in an error state.	Do not attempt a drive replacement or an MDM reconfiguration if a service is in an error state.
VXFM-20593	PowerFlex Manager creates an Element Manager credential via REST API fails with wrong error message if `snmpCommunityString` is omitted.	None.
VXFM-20590	When scaling up of the network on a PowerFlex compute-only service, the new network does not show under the port view for one of the three nodes in the service.	None.
VXFM-20588	PowerFlex Manager displays **Update Resource** when current version is higher than expected version.	None.

Issue ID	Symptom	Recommended action
VXFM-20571	PowerFlex Manager lets you add a static route to a subnet that already exists on the node. This fails as it is not needed - the node already has a route to that subnet.	None.
VXFM-20546	If existing port channels that do not have any interfaces are reused on Dell OS10 as part of the new service, any previous VLANs on those port channels are not cleared. If new port channels are created they will only have the correct VLANs.	None.
VXFM-20533	PowerFlex Manager fails to set the default compliance package version when you add a compliance file and select **Make this the default version for compliance checking**.	Click **Change** to reset the default version.
VXFM-20318	A PowerFlex hyperconverged service deployment using an existing VMware cluster with different storage pools fails.	Remove the nodes and add them back with the same storage pool.
VXFM-20215	In the teardown of a PowerFlex hyperconverged service where the datacenter, cluster, and network information are all moved into their own folders in VMware vCenter, the service resources are re-inventoried, and the service is removed. However, the VMware vCenter cluster and nodes are left and not removed.	Manually remove the vCenter cluster and nodes.
VXFM-20169	MDM reconfiguration fails for certain role transitions.	You can set the role manually, if needed. Retry or rescan refreshes the roles.
VXFM-20154	If you attempt to restart a failed job by clicking **Retry**, PowerFlex Manager displays a warning page with a **View Compliance Report** link. This link is broken. You may encounter this issue after a resize volume or migrate vCLS VMs operation fails.	Click **View Compliance Report** under **Service Actions**.
VXFM-19482	PowerFlex Manager fails to configure the server facing ports due to `switchport mode trunk` returning the following error: `% Error: MODE_L2HYBRID, Configured as a lag interface ToR-B(conf-if-eth1/1/3)#`	Remove `swichport mode trunk` before running through PowerFlex Manager.
VXFM-19010	The wizard for updating resources displays incorrect error messages when trying to update the PowerFlex gateway or presentation server resources from the **Services** page.	Go to the management VM service and click **Update resource**.
VXFM-18857	During an **Add Volume** job, a service will go from the **In Progress** state to the **Completed** state, then suddenly return to **In Progress**. While the job is still **In Progress**, some of the service actions may be enabled.	Allow the job to finish before you try to use any of the service actions.
VXFM-18580	Clusters already in services on the appliance are provided as selection options when importing an existing service.	None.
VXFM-18554	During a PowerFlex hyperconverged replication service deployment, retry does not set the VMware VM version. This can cause the	Delete the VM and retry the service.

Issue ID	Symptom	Recommended action
	deployment to fail when trying to add NVDIMM devices to the VM.	
VXFM-18167	Adding an existing service wizard network mapping shows `%2f` instead of `/` in port group name.	Unsupported special characters cause a rendering issue with the port group name and should not be used. Spaces, underscores, and hyphens are allowed.
VXFM-18005	Retrying a failed PowerFlex upgrade fails to cancel previous upgrade job.	If you start the upgrade, and request PowerFlex Manager to start another upgrade, it fails. If you see this failure, you may must cancel the previous upgrade.
VXFM-17760	Service sees old NIC cabling data resulting in failure.	Contact Dell Support for a workaround.
VXFM-17610	The DHCP setting in PowerFlex Manager does not allow you to disable the DHCP/PXE node if the virtual appliance is not currently on the subnet.	Do not change your network while you are working on it. If you must do this, at least disable DHCP first.
VXFM-17023	Unable to use NetApp NFS share from PowerFlex Manager.	PowerFlex Manager does not support NFS shares set up this way.
VXFM-16623	A node discovery may appear to fail with the following log message: `Failed to collect the system configuration for server`	Check for the following conditions: • Networking issues between iDRAC and PowerFlex Manager • The server is booted into the BIOS screen. In this case, jobs cannot be run.
VXFM-16261	During a restore, the PowerFlex Manager appliance gets stuck at **Starting Restoration** stage when a filename is not provided, that is, if only the path is specified.	Use the full path to the restore file.
VXFM-15653	Download of the ONIE firmware upgrade package fails, but the failure is not displayed in the logs.	None.
ESRCM-2526	VMware ESXi 7.0 requires a security patch that is available in VMware ESXi 7.0 Update 1.	For VMware ESXi 7.0, you must perform a manual workaround. For details, see https://kb.vmware.com/s/article/76372.
VXFM-15308	PowerFlex Manager collects switch performance metrics only if NXAPI is listening on both HTTPS (Port 443) and HTTP (Port 80). If either of these ports are disabled, switch performance metrics are not collected.	Be sure that ports for both HTTP and HTTPS are enabled for Cisco switches.
VXFM-15227	PowerFlex Manager allows you to publish and deploy a CloudLink template containing a password with all lowercase characters.	Be sure to include at least one uppercase letter in the password.
VXFM-15267	If you try to upload an SSL certificate on the **Appliance SSL Certificates** section of the **Settings** page, you may encounter security errors with the Certificate Signing Request (CSR). The CSR does not include the Subject Alternative Name (SAN) field. It only includes the Common Name (CN) field. This problem occurs only with the Chrome browser.	Contact Dell Support for a workaround.

Issue ID	Symptom	Recommended action
VXFM-15188	When deploying a service, you may see an error message stating that a static IP address could not be reserved. The error may include empty brackets ([ ]). If you see this error, the IP addresses may not be pingable.	Check your IP addresses to be sure you have enough pingable IP addresses, and try again.
VXFM-15110	You may see incorrect warning messages when using invalid characters for a hostname in a template. If you try to publish the template, you may see a message indicating that a hostname exceeds the limit of 15 characters. The message should indicate that the hostname has invalid characters.	Be sure to use valid characters and conform to the character limit for the operating system. For Windows, the limit is 15 characters for a hostname. For ESXi and Linux, the limit is 26 characters.
VXFM-15069	A node cannot be removed from a storage-only service because of a failure removing the switch. If the switch removal fails, PowerFlex Manager leaves the node in the service.	Update the service details or remove the node manually from the service.
VXFM-15010	Adding replication networks to a hyperconverged template breaks the service.	None.
VXFM-14898	During drive replacement, if you type **`CONTINUE WITHOUT DRIVE`** with a bad drive, PowerFlex Manager tries to add it back and fails.	None.
VXFM-14872	You may notice a type casing inconsistency, when creating templates with uppercase host and SVM names. VMware ESXi uses lowercase host names. There is no impact to functionality.	None.
VXFM-14773	If you try to download two RCMs simultaneously, the operation fails.	Resynchronize the RCMs or download them again separately.
VXFM-14771	If you deploy a service with only one node in it, the service may show **Critical**. This condition is caused by a warning at the component level.	None.
VXFM-14770	The inventory is not running on the PowerFlex gateway after you use service mode. If you place a node in service mode, then exit service mode, then try to enter service mode again, the PowerFlex gateway is left in an abnormal state.	Manually run the inventory or wait for the next hourly update.
VXFM-14767	PowerFlex Manager does not refresh text boxes in the **Discovery Wizard**. This display problem can happen if you select **Node** as the **Resource Type** and choose **Hostname**, then change the **Resource Type** to any other type other than **Node**.	Click the trash can icon to remove the resource, then add it again.
VXFM-14763	Mellanox ConnectX-4 LC firmware must be on 14.25.80.00 or later to upgrade to 14.27.10.16 or later. This problem is not a PowerFlex Manager issue. PowerFlex Manager does not support step upgrades.	None.
VXFM-14753	If an upgrade of the PowerFlex Manager virtual appliance fails, you may see duplicate email messages and alerts.	None.
VXFM-14724	If you try to delete an RCM that is used on a service, the operation fails with a 500 internal server error.	Remove the service that previously used the RCM, delete the RCM, and add the services back as an existing service.
VXFM-14581	When you add an existing hyperconverged or storage-only service that includes a CloudLink	None.

Issue ID	Symptom	Recommended action
	Center VM in a cluster, the **Inventory Summary** page does not show the **Adding** icon for the CloudLink Center VM.	
VXFM-14535 and VXFM-15309	If you change the password for an SVM, **View All Settings** on the **Services** page still shows the old password. This issue also breaks **Update Service Details** if you try to use a new credential.	If you need to change the OS credential to a newly-defined credential and update to use that credential on a service, you have to remove the service and readd it to select that credential.
VXFM-14496 and VXFM-14254	After upgrading from an earlier version of PowerFlex Manager, you may notice that the inventory might not be updated correctly or that subsequent operations such as **Add Existing Service** may fail.	Run a manual inventory after the upgrade.
VXFM-14494 and VXFM-14762	After changing an iDRAC password, if you select the same credentials again, you may see an error message in the **Update Password** wizard. After you see the error, you cannot proceed with the wizard.	Cancel and restart the wizard.
VXFM-14470	PowerFlex Manager does not perform validation to ensure that you have a minimum of three fault sets for a protection domain and storage pool.	Ensure that your fault set configuration conforms to best practices. Dell recommends that all nodes in a protection domain should be in the same storage pool.
VXFM-14397	PowerFlex Manager displays the wrong progress bar during the resizing of a datastore.	None.
VXFM-14377	After you remove a node from a service, PowerFlex Manager may not update the PowerFlex gateway inventory.	Run a manual inventory after removing the node.
VXFM-14028	The **Services** page does not show lifecycle mode as the deployment state for NSXT services that are in lifecycle mode.	None.
VXFM-13994	If a Dell CloudLink Center is non-compliant, PowerFlex Manager disables node selection for upgrade.	Upgrade the CloudLink Center from the **Resources** page and try the upgrade from the **Services** page.
VXFM-13744	An RCM upgrade to a major version of NSX and ESXi may fail with a VIB error. This failure could be caused by a VIB incompatibility if you are using a nonstandard Dell image. Be aware that NSX VIBs are specific to the version of ESXi that is installed on the host. If you upgrade ESXi, you must install new NSX VIBs appropriate for the new ESXi version.	Install new NSX VIBs for the new ESXi version. For more information, see Upgrade to ESXi 6.5 or 6.7 in an NSX Environment.
VXFM-12990	The state of a discovery job may show as **Running** on the **Jobs** page, even though it has completed.	Restart the PowerFlex Manager virtual appliance.
VXFM-12477	On the **Resources** page, if you select a machine with self-encrypting drives (SEDs) for a CloudLink Center, you might see two entries for each drive. One entry shows the status as "encrypted" with a type of "data_sed". The other entry shows the status as "unencrypted" with a type of "data_sds". The SED drives are encrypted, but PowerFlex Manager should not show duplicate entries.	None.
VXFM-12402	After deleting a node from a storage-only service, you may see an incorrect error message in	None.

Issue ID	Symptom	Recommended action
	Recent Activity. The message indicates a failure removing the Storage Data Client (SDC), even though an SDC is not installed on a storage-only node.	
VXFM-12391	If you put an SDS into Protected Maintenance Mode (PMM) manually, and then run inventory for the PowerFlex gateway within PowerFlex Manager, the service does not exit service mode. When you click Exit Service Mode, PowerFlex Manager shows an error. This error occurs because PowerFlex Manager is trying to exit Instant Maintenance Mode (IMM) instead of PMM.	Manually take the SDS out of PMM outside of PowerFlex Manager.
VXFM-12366	If you cancel a service deployment, and then retry the deployment later, the service succeeds. However, the warning banner that is displayed while PowerFlex Manager is canceling the deployment remains during and after the successful deployment.	Remove the service and add it back again.
VXFM-12096	The deployment for a storage-only service may sometimes fail to start.	Retry the service deployment.
VXFM-12074	On the **View Details** page for CloudLink, PowerFlex Manager shows a hyperlink for the **IP Address** field for the SDS. The hyperlink should not be displayed for an SDS IP address.	None.
VXFM-11998	PowerFlex Manager does not always show the new status when you add an operating system image on the **OS Image Repositories** tab.	Manually refresh the screen to see the new status.
VXFM-11977	If you delete a node that was not deployed successfully in a service, PowerFlex Manager might show the node as deployed.	To remove the failed node from the service, choose the **Remove Resource** option, rather than the **Delete Resource** option.
VXFM-11930	After upgrading from version 3.3 or 3.4, you might notice that a service that has a fault set enabled shows **Deployed** as the **Deployment State**, even though it should show **Lifecycle Mode**.	Click **Update Service Details** to show the correct state.
VXFM-11760	After updating the firmware and software on a node, or during a service mode operation, PowerFlex Manager might show the node as being in service mode, even though it is no longer in service mode. Due to a time lag, the service mode status that is shown on the service does not always match the state of the system exactly. The node eventually comes out-of-service mode, and the update completes successfully.	None.
VXFM-11758	A compute-only service with ESXi displays the **Protected Maintenance Mode** option for a firmware and software update on a PowerFlex 3.5 cluster. This type of service offers only compute resources, and does not have an SDS component. The **Protected Maintenance Mode** option should not be available. The update succeeds even if you choose this option.	None.

Issue ID	Symptom	Recommended action
VXFM-11705	If you attempt to discover a Cisco switch with terminal color configured, the discovery fails.	Disable the terminal color option by running `configure no terminal color persist`.
VXFM-11242	If you click **Update Service Details**, PowerFlex Manager might display an incorrect error indicating that the `device could not be connected through SSH`. The message should state that the iDRAC was not responding, most likely because of bad cabling.	None.
VXFM-10663	PowerFlex Manager has only one firmware component id set for all OS10 switches in PowerFlex Manager. This means that PowerFlex Manager can only include one set of OS10 switch versions for all models of OS10 switch in a given RCM package. This is a problem for management switches.	Do not discover the management switches.
VXFM-3727	If you try to create a network containing IP addresses that are in use by vCenter or the PowerFlex gateway inventory, `Invalid IP address range is entered` errors are thrown. An extra error message states: `The entered IP range overlaps with the IP ranges configured on one or more configured networks`. This situation only happens if you have multiple networks with the same subnet. For example, this situation might happen if you have an SDC only network and a network that supports both SDS and SDC traffic.	Remove your vCenter or gateway from the inventory, create the network ranges, and then rediscover them.
VXFM-3517	If the PowerFlex gateway is deployed onto an operating system that does not have `systemctl` (such as SuSE 11 or Red Hat Enterprise Linux 6), the PowerFlex configuration script fails to restart the gateway after configuring it. However, it does not return any error, so ultimately the deployment fails when trying to talk to the gateway REST API. This configuration is unsupported, but PowerFlex Manager does not block the configuration or provide a clear error message.	None.

Operating system known issues and limitations

This section lists known issues and limitations for specific operating systems.

Issue number & SR number	Problem summary	Workaround
VXFV-360	The device name appears incorrectly as a non-PCI device instead of iDRAC, but the current and baseline versions are accurate (6.00.30.201 to 6.00.30.202, respectively).	Dell recommends upgrading the OME catalog to the latest version.
	PowerFlex nodes running CPLD 1.0.7 stop responding during Power On Self-Test (POST) when upgrading to certain RCM or IC packages that include BIOS 1.8.2 and iDRAC 6.00.30.202.	Before upgrading to the latest version of PowerFlex, ensure that the PowerFlex nodes are running CPLD 1.0.8 or later. For details, see https://www.dell.com/support/kbdoc/en-us/000205925.

Issue number & SR number	Problem summary	Workaround
VXFV-296	Mixing the Mellanox CX-6 100GbE NIC driver with the Broadcom PCIe/OCP 25GbE NIC driver is not supported.	Remove the `bnxt_en` driver and install the Mellanox driver.
VXFV-283	iDRAC support for "Prepare for Removal" on PowerEdge R650/R6525/R750/ nodes fails.	NVMe hot-swap on Linux operating systems is supported. Skip the Prepare for Removal step in iDRAC.
VXFV-265	During execution of the command `create-namespace` with 4k alignment on RHEL 8.4, the following error is encountered: Warning: namespace0.0 unable to validate alignment	This is a false notification. The command result is as expected, and the resulting DAX device can be used with PowerFlex acceleration devices.
VXFV-243	Upgrade of a RedHat-based operating system to RHEL 8.3 fails if the node contains NVDIMMs. This is due to a known issue on RHEL 8.3: https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/8/html-single/8.3_release_notes/index#known-issue_installer-and-image-creation	Follow the procedures described in "RHEL/CentOS upgrade to version 8.3" in the .
VXFV-224	In a RHEL 8.x-based node using Cloudlink encrypted SDS devices, after reboot the device moves to failed state. **NOTE:** This issue is much rarer on RHEL7.x-based nodes, but it might occur.	1. SSH to the relevant SDS's operating system. 2. Delete the SDS service. 3. Clear the device errors using the GUI or CLI. 4. Recreate the SDS service. 5. Verify that devices come back online. Rebuild/Rebalance will occur if instant maintenance mode or protected maintenance mode was not used. 6. If instant maintenance mode or protected maintenance mode was used, exit maintenance mode.
VXFV-157	NVMe disks are not supported on RHEL 8.0 due to the following issue: https://access.redhat.com/solutions/4280341 Failure to observe this limitation might cause RHEL-based SDSs with NVMes to experience kernel panic during a disk removal flow.	None

Fixed issues in 4.0

Fixed issues are defects that exist in the previous release but are resolved in the 4.0 release of PowerFlex and PowerFlex Manager.

Issue ID	Symptom
VXFM-28231	PowerFlex Manager-3.6.1-6493 upgrade to 3.8.1-8198 fails to reboot and the UI gets stuck at initializing.
VXFM-24465	After adding a new hyperconverged or compute-only ESXi node to a service, the new node does not have access to the PowerFlex datastores. This is an intermittent issue, it does not occur every time you add a node to a service.
VXFM-24171	Firmware or software updates or deployments fail due to nightly inventory run. This may impact large deployments, since the nightly inventory can run at most 100 jobs at a time.
VXFM-22133	You cannot set the default catalog if any services have been created.

Issue ID	Symptom
VXFM-19963	Deployment of management VMs into a nested VDS folder fails.

Software packages

Learn about the contents of the PowerFlex 4.0 software packages.

Download location

The PowerFlex complete software package, containing all the packages listed below, can be downloaded from the following location:

https://www.dell.com/support/home/en-us/product-support/product/scaleio/overview

Packages

Package name	Size	Contents
RCM for PowerFlex rack	25GB	Flex_RCM_3_7_0_0_r32.zip
Intelligent catalog for PowerFlex appliance	17GB	Intelligent_Catalog_40_370_00_r21.zip
SW only complete ZIP	~6.5GB	Complete software-only ZIP. Includes the following: ● PowerFlex_4.0.0.1211_Complete_Core_SW.zip (4GB) ● PowerFlex_4.0.0.1211_SDCs_for_manual_install.zip (SDC RPMs) (0.5GB) ● PowerFlex_Software_4.0.0.0_1211_r2.zip (2GB)
GPG file for compatibility management		cm-20220728-01.gpg
Management ZIPs		PowerFlex Manager OVA ZIP (~25GB) contains: ● PFMP2-4.0.0-161.ova (PFMP_Installer for installing Kubernetes cluster) (22GB) ● pfmp-k8s-20220712.ova (Kubernetes OVA running SUSE 15 SP3) (3.5GB)

The original OS layout table from 3.6 matches this file: PowerFlex_4.0.0.1211_Complete_Core_SW.zip (4GB)